Problem 4-1

If **A**, **B**, and **D** are given vectors, prove the distributive law for the vector cross product, i.e., $\mathbf{A}\times(\mathbf{B}+\mathbf{D}) = (\mathbf{A}\times\mathbf{B}) + (\mathbf{A}\times\mathbf{D})$.

Solution:

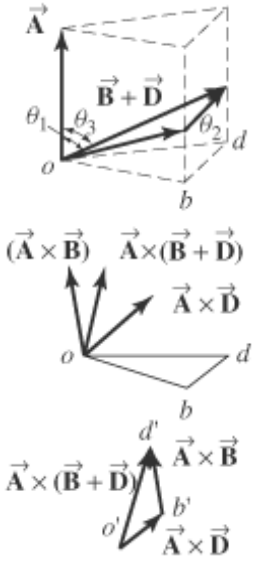
Consider the three vectors; with **A** vertical.

Note triangle *obd* is perpendicular to **A**.

$$od = \left|\mathbf{A}\times(\mathbf{B}+\mathbf{D})\right| = \left|\mathbf{A}\right|\left(\left|\mathbf{B}+\mathbf{D}\right|\right)\sin(\theta_3)$$

$$ob = \left|\mathbf{A}\times\mathbf{B}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right|\sin(\theta_1)$$

$$bd = \left|\mathbf{A}\times\mathbf{D}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right|\sin(\theta_2)$$



Also, these three cross products all lie in the plane *obd* since they are all perpendicular to A. As noted the magnitude of each cross product is proportional to the length of each side of the triangle.

The three vector cross - products also form a closed triangle *o'b'd'* which is similar to triangle *obd*. Thus from the figure,

$$\mathbf{A}\times(\mathbf{B}+\mathbf{D}) = \mathbf{A}\times\mathbf{B} + \mathbf{A}\times\mathbf{D} \qquad \text{(QED)}$$

Note also,

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$$

$$\mathbf{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{K}$$

$$\mathbf{D} = D_x\mathbf{i} + D_y\mathbf{j} + D_z\mathbf{k}$$

$$\mathbf{A}\times(\mathbf{B}+\mathbf{D}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x + D_x & B_y + D_y & B_z + D_z \end{vmatrix}$$

$$= \left[A_y(B_z+D_z) - A_z(B_y+D_y)\right]\mathbf{i} - \left[A_x(B_z+D_z) - A_z(B_x+D_x)\right]\mathbf{j} + \left[A_x(B_y+D_y) - A_y(B_x-D_x)\right]\mathbf{k}$$

$$= \left[(A_yB_z - A_zB_y)\mathbf{i} - (A_xB_z - A_zB_x)\mathbf{j} + (A_xB_y - A_yB_x)\mathbf{k}\right] \ldots$$
$$+ \left[(A_yD_z - A_zD_y)\mathbf{i} - (A_xD_z - A_zD_x)\mathbf{j} + (A_xD_y - A_yD_x)\mathbf{k}\right]$$

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$$= \left|\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{pmatrix}\right| + \left|\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{pmatrix}\right|$$

$$= (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{D}) \qquad \text{(QED)}$$

Problem 4-2

Prove the triple scalar product identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$.

Solution:

As shown in the figure

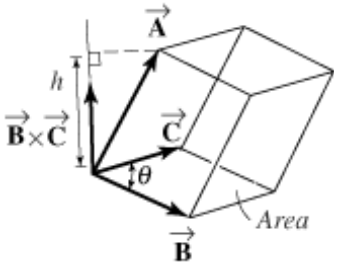
$$Area = B(C\sin(\theta)) = |\mathbf{B} \times \mathbf{C}|$$

Thus,

Volume of parallelopiped is $|B \times C|\ |h|$

But,

$$|h| = \left|\mathbf{A} \cdot \mathbf{u}_{B \times C}\right| = \left|\mathbf{A} \cdot \left(\frac{\mathbf{B} \times \mathbf{C}}{|\mathbf{B} \times \mathbf{C}|}\right)\right|$$



Thus,

$$Volume = |\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})|$$

Since $|\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}|$ represents this same volume then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C} \qquad \text{(QED)}$$

Also,

$$LHS = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

$$= (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \left|\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{pmatrix}\right|$$

$$= A_x(B_yC_z - B_zC_y) - A_y(B_xC_z - B_zC_x) + A_z(B_xC_y - B_yC_x)$$

$$= A_xB_yC_z - A_xB_zC_y - A_yB_xC_z + A_yB_zC_x + A_zB_xC_y - A_zB_yC_x$$

$$RHS = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$

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$$= \left|\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{pmatrix}\right| \left(C_x\mathbf{i} + C_y\mathbf{j} + C_z\mathbf{k}\right)$$

$$= C_x\left(A_yB_z - A_zB_y\right) - C_y\left(A_xB_z - A_zB_x\right) + C_z\left(A_xB_y - A_yB_x\right)$$

$$= A_xB_yC_z - A_xB_zC_y - A_yB_xC_z + A_yB_zC_x + A_zB_xC_y - A_zB_yC_x$$

Thus, $LHS = RHS$

$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C} = \mathbf{A}\times\mathbf{B}\cdot\mathbf{C}$ (QED)

Problem 4-3

Given the three nonzero vectors **A**, **B**, and **C**, show that if $\mathbf{A}\cdot\left(\mathbf{B}\times\mathbf{C}\right) = 0$, the three vectors *must* lie in the same plane.

Solution:

Consider,

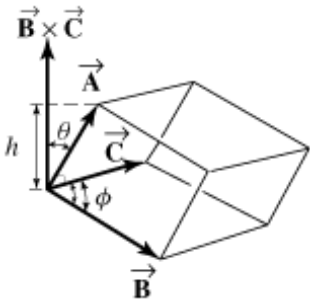
$$\left|\mathbf{A}\cdot\left(\mathbf{B}\times\mathbf{C}\right)\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\times\mathbf{C}\right|\cos\left(\theta\right)$$

$$= \left(\left|\mathbf{A}\right|\cos\left(\theta\right)\right)\left|\mathbf{B}\times\mathbf{C}\right|$$

$$= \left|h\right|\left|\mathbf{B}\times\mathbf{C}\right|$$

$$= BC\left|h\right|\sin\left(\phi\right)$$

= volume of parallelepiped.



If $\mathbf{A}\cdot\left(\mathbf{B}\times\mathbf{C}\right) = 0$, then the volume equals zero, so that **A**, **B**, and **C** are coplanar.

Problem 4-4

Determine the magnitude and directional sense of the resultant moment of the forces at *A* and *B* about point *O*.

Given:

$F_1 = 40$ lb

$F_2 = 60$ lb

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$\theta_1 = 30$ deg

$\theta_2 = 45$ deg

$a = 5$ in

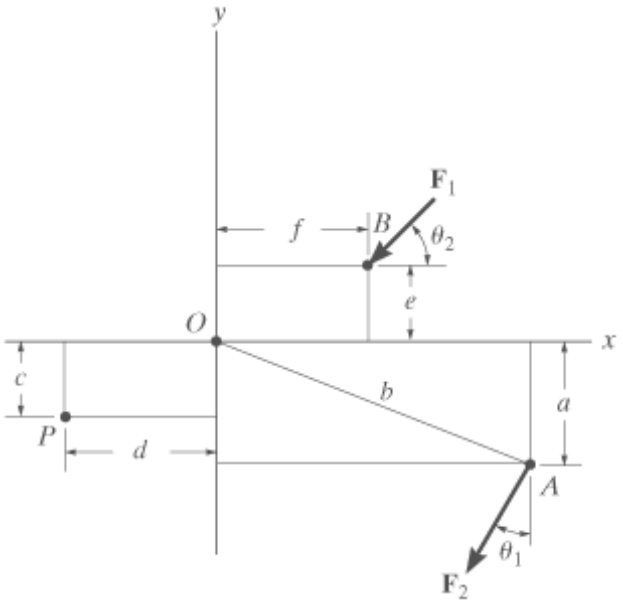
$b = 13$ in

$c = 3$ in

$d = 6$ in

$e = 3$ in

$f = 6$ in



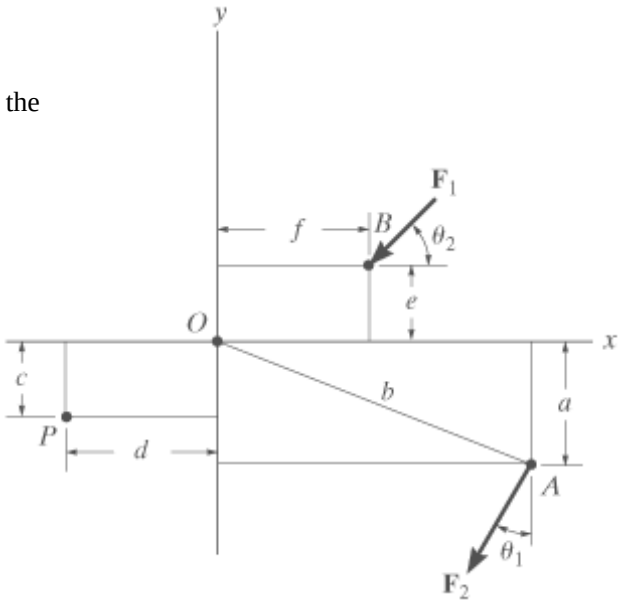
Solution:

$$\curvearrowleft + M_{RO} = \Sigma M_O;\quad M_{RO} = F_1\cos(\theta_2)e - F_1\sin(\theta_2)f - F_2\cos(\theta_1)\sqrt{b^2 - a^2} - F_2\sin(\theta_1)a$$

$$M_{RO} = -858\,\text{lb}\cdot\text{in} \qquad |M_{RO}| = 858\,\text{lb}\cdot\text{in}$$

Problem 4-5

Determine the magnitude and directional sense of the resultant moment of the forces at A and B about point P.



Units Used:

kip = 1000 lb

Given:

$F_1 = 40$ lb $\quad b = 13$ in

$F_2 = 60$ lb $\quad c = 3$ in

$\theta_1 = 30$ deg $\quad d = 6$ in

$\theta_2 = 45$ deg $\quad e = 3$ in

$a = 5$ in $\quad f = 6$ in

Solution:

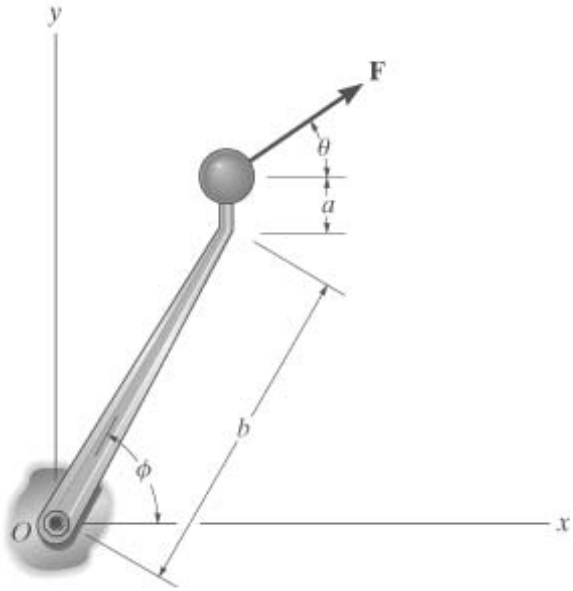
$$\curvearrowleft + M_{RP} = \Sigma M_P;\quad M_{RP} = F_1\cos(\theta_2)(e + c) - F_1\sin(\theta_2)(d + f) - F_2\cos(\theta_1)\left(\sqrt{b^2 - a^2} + d\right) \ldots$$
$$+ -F_2\sin(\theta_1)(a - c)$$

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$M_{RP} = -1165\,\text{lb}\cdot\text{in}$ $\quad |M_{RP}| = 1.17\,\text{kip}\cdot\text{in}$

Problem 4-6

Determine the magnitude of the force **F** that should be applied at the end of the lever such that this force creates a clockwise moment *M* about point *O*.



Given:

$M = 15\ \text{N m}$

$\phi = 60\ \text{deg}$

$\theta = 30\ \text{deg}$

$a = 50\ \text{mm}$

$b = 300\ \text{mm}$

Solution:

$$\circlearrowleft + \quad M = F\cos(\theta)(a + b\sin(\phi)) - F\sin(\theta)(b\cos(\phi))$$

$$F = \frac{M}{\cos(\theta)(a + b\sin(\phi)) - \sin(\theta)(b\cos(\phi))} \qquad F = 77.6\ \text{N}$$

Problem 4-7

Determine the angle θ ($0 <= \theta <= 90$ deg) so that the force **F** develops a clockwise moment *M* about point *O*.

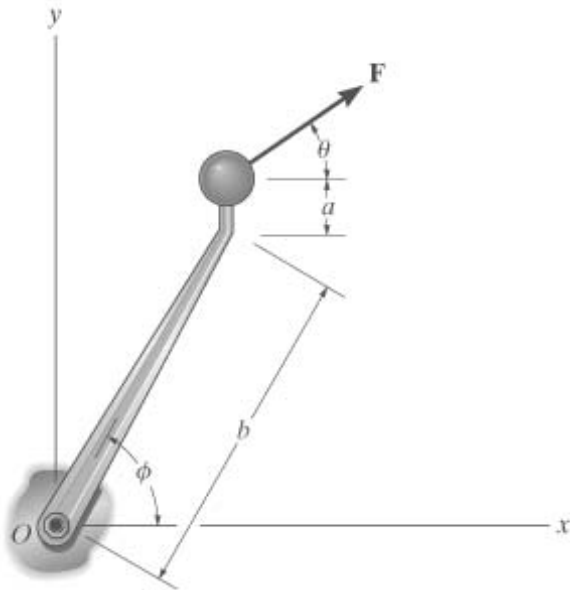
Given:

$F = 100\ \text{N} \qquad \phi = 60\ \text{deg}$

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$M = 20\ \text{N}\cdot\text{m}$ $a = 50\ \text{mm}$

$\theta = 30\ \text{deg}$ $b = 300\ \text{mm}$



Solution:

Initial Guess $\theta = 30\ \text{deg}$

Given

$$M = F\cos(\theta)(a + b\sin(\phi)) - F\sin(\theta)(b\cos(\phi))$$

$\theta = \text{Find}(\theta)$ $\theta = 28.6\,\text{deg}$

Problem 4-8

Determine the magnitude and directional sense of the moment of the forces about point *O*.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

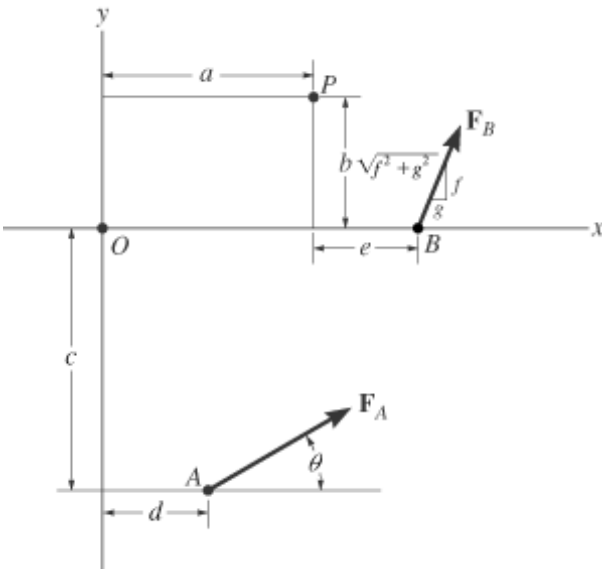
$F_B = 260\ \text{N}$ $e = 2\ \text{m}$

$a = 4\ \text{m}$ $f = 12$

$b = 3\ \text{m}$ $g = 5$

$c = 5\ \text{m}$ $\theta = 30\ \text{deg}$

$d = 2\ \text{m}$ $F_A = 400\ \text{N}$



Solution:

$$\curvearrowleft + M_O = F_A\sin(\theta)d + F_A\cos(\theta)c + F_B\frac{f}{\sqrt{f^2+g^2}}(a+e)$$

$M_O = 3.57\,\text{kN}\cdot\text{m}$ (positive means counterclockwise)

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Problem 4-9

Determine the magnitude and directional sense of the moment of the forces about point P.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_B = 260\ \text{N}$ $e = 2\ \text{m}$

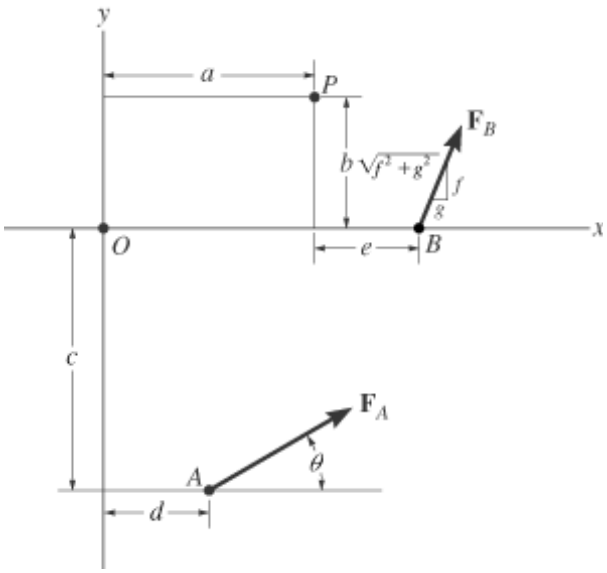
$a = 4\ \text{m}$ $f = 12$

$b = 3\ \text{m}$ $g = 5$

$c = 5\ \text{m}$ $\theta = 30\ \text{deg}$

$d = 2\ \text{m}$

$F_A = 400\ \text{N}$



Solution:

$$\curvearrowleft + \quad M_p = F_B \frac{g}{\sqrt{f^2+g^2}} b + F_B \frac{f}{\sqrt{f^2+g^2}} e - F_A \sin(\theta)(a-d) + F_A \cos(\theta)(b+c)$$

$M_p = 3.15\ \text{kN}\cdot\text{m}$ (positive means counterclockwise)

Problem 4-10

A force **F** is applied to the wrench. Determine the moment of this force about point O. Solve the problem using both a scalar analysis and a vector analysis.

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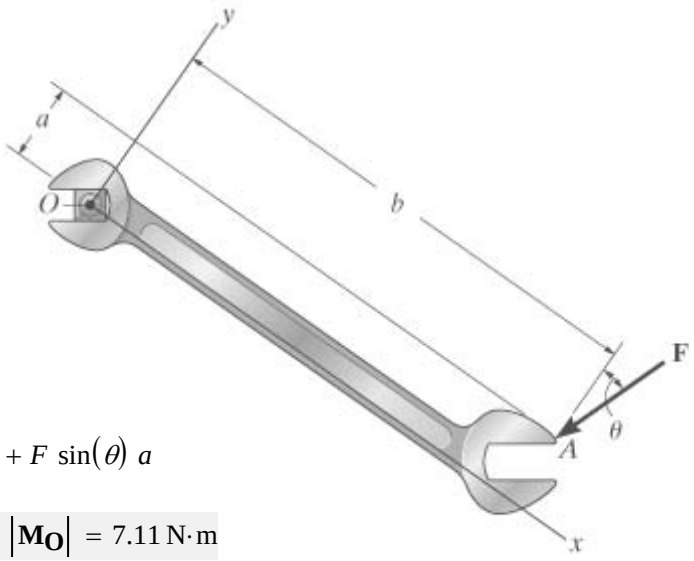
Given:

$F = 40$ N

$\theta = 20$ deg

$a = 30$ mm

$b = 200$ mm



Scalar Solution

$$\left(+\right) \mathbf{M_O} = -F\cos(\theta)\, b + F\sin(\theta)\, a$$

$$\mathbf{M_O} = -7.11\ \text{N}\cdot\text{m} \qquad \left|\mathbf{M_O}\right| = 7.11\ \text{N}\cdot\text{m}$$

Vector Solution

$$\mathbf{M_O} = \begin{pmatrix} b \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} -F\sin(\theta) \\ -F\cos(\theta) \\ 0 \end{pmatrix} \qquad \mathbf{M_O} = \begin{pmatrix} 0 \\ 0 \\ -7.11 \end{pmatrix} \text{N}\cdot\text{m} \qquad \left|\mathbf{M_O}\right| = 7.107\ \text{N}\cdot\text{m}$$

Problem 4-11

Determine the magnitude and directional sense of the resultant moment of the forces about point *O*.

Units Used:

$\text{kip} = 10^3$ lb

Given:

$F_1 = 300$ lb $\quad e = 10$ ft

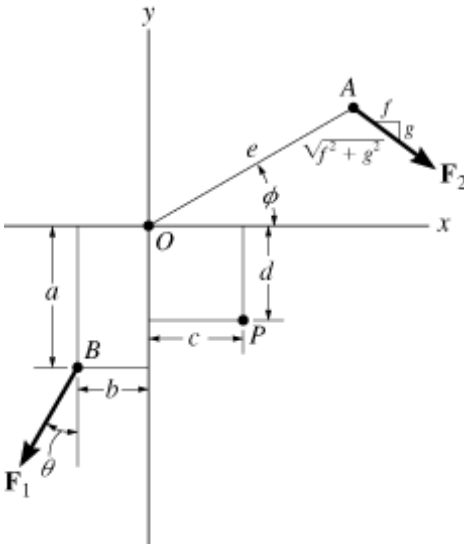
$F_2 = 250$ lb $\quad f = 4$

$a = 6$ ft $\quad g = 3$

$b = 3$ ft $\quad \theta = 30$ deg

$c = 4$ ft $\quad \phi = 30$ deg

$d = 4$ ft



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Solution:

$$\circlearrowleft + \quad M_O = F_2 \frac{f}{\sqrt{f^2 + g^2}} e \sin(\phi) + F_2 \frac{g}{\sqrt{f^2 + g^2}} e \cos(\phi) + F_1 \sin(\theta) a - F_1 \cos(\theta) b$$

$M_O = 2.42\ \text{kip}\cdot\text{ft}$ positive means clockwise

---

Problem 4-12

To correct a birth defect, the tibia of the leg is straightened using three wires that are attached through holes made in the bone and then to an external brace that is worn by the patient. Determine the moment of each wire force about joint *A*.

Given:

$F_1 = 4\ \text{N}$ $d = 0.15\ \text{m}$

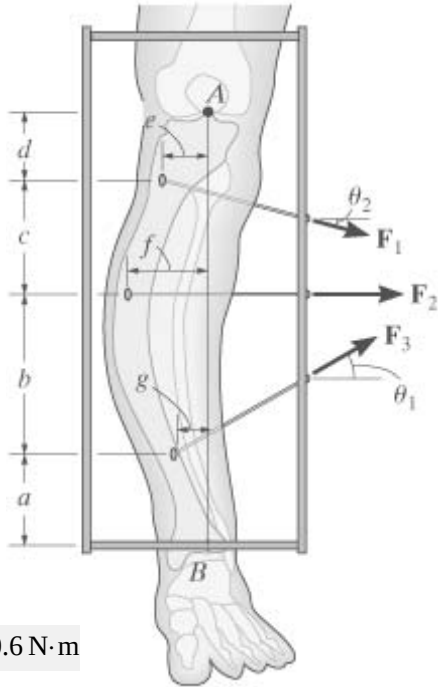
$F_2 = 8\ \text{N}$ $e = 20\ \text{mm}$

$F_3 = 6\ \text{N}$ $f = 35\ \text{mm}$

$a = 0.2\ \text{m}$ $g = 15\ \text{mm}$

$b = 0.35\ \text{m}$ $\theta_1 = 30\ \text{deg}$

$c = 0.25\ \text{m}$ $\theta_2 = 15\ \text{deg}$



Solution: Positive means counterclockwise

$M_{A1} = F_1 \cos(\theta_2) d + F_1 \sin(\theta_2) e$ $M_{A1} = 0.6\ \text{N}\cdot\text{m}$

$M_{A2} = F_2(c + d)$ $M_{A2} = 3.2\ \text{N}\cdot\text{m}$

$M_{A3} = F_3 \cos(\theta_1)(b + c + d) - F_3 \sin(\theta_1) g$ $M_{A3} = 3.852\ \text{N}\cdot\text{m}$

Problem 4-13

To correct a birth defect, the tibia of the leg is straightened using three wires that are attached through holes made in the bone and then to an external brace that is worn by the patient. Determine the moment of each wire force about joint *B*.

Given:

$F_1 = 4\ \text{N}$ $d = 0.15\ \text{m}$

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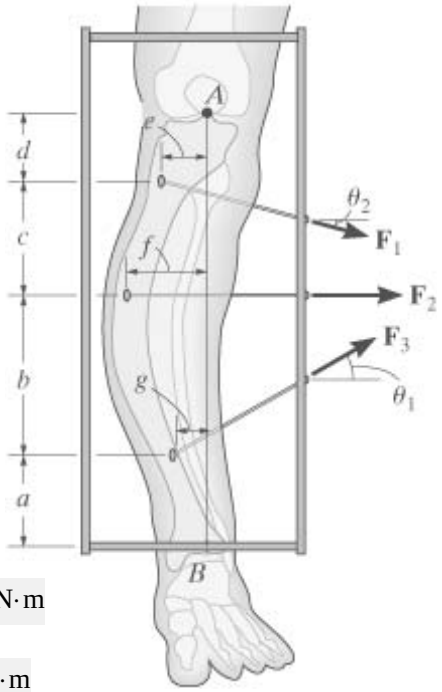
$F_2 = 8$ N $\quad e = 20$ mm

$F_3 = 6$ N $\quad f = 35$ mm

$a = 0.2$ m $\quad g = 15$ mm

$b = 0.35$ m $\quad \theta_1 = 30$ deg

$c = 0.25$ m $\quad \theta_2 = 15$ deg



Solution:

Positive means clockwise

$M_{B1} = F_1 \cos(\theta_2)(a + b + c) - F_1 \sin(\theta_2)e$ $\quad M_{B1} = 3.07\,\text{N}\cdot\text{m}$

$M_{B2} = F_2(a + b)$ $\quad M_{B2} = 4.4\,\text{N}\cdot\text{m}$

$M_{B3} = F_3 \cos(\theta_1)a + F_3 \sin(\theta_1)g$ $\quad M_{B3} = 1.084\,\text{N}\cdot\text{m}$

Problem 4-14

Determine the moment of each force about the bolt located at *A*.

Given:

$F_B = 40$ lb $\quad a = 2.5$ ft $\quad \alpha = 20$ deg $\quad \gamma = 30$ deg

$F_C = 50$ lb $\quad b = 0.75$ ft $\quad \beta = 25$ deg

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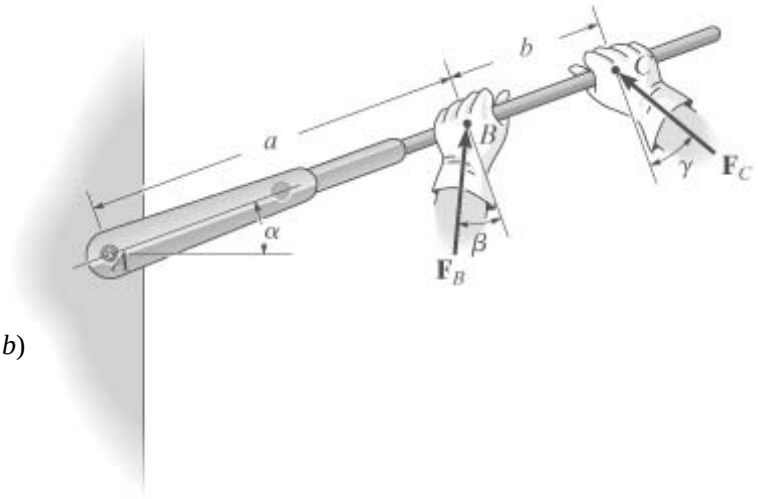
Solution:

$$\curvearrowleft + \quad M_B = F_B \cos(\beta) a$$

$$M_B = 90.6\,\text{lb}\cdot\text{ft}$$

$$\curvearrowleft + \quad M_C = F_C \cos(\gamma)(a + b)$$

$$M_C = 141\,\text{lb}\cdot\text{ft}$$



Problem 4-15

Determine the resultant moment about the bolt located at *A*.

Given:

$F_B = 30$ lb

$F_C = 45$ lb

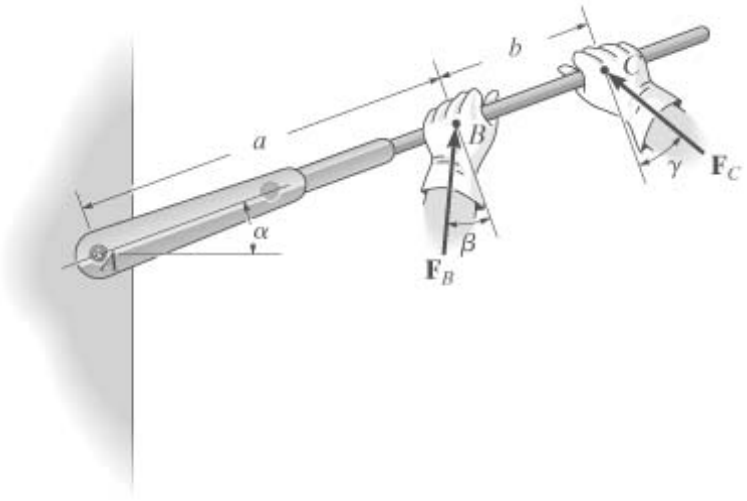
$a = 2.5$ ft

$b = 0.75$ ft

$\alpha = 20$ deg

$\beta = 25$ deg

$\gamma = 30$ deg



Solution:

$$\curvearrowleft + \quad M_A = F_B \cos(\beta) a + F_C \cos(\gamma)(a + b) \qquad M_A = 195\,\text{lb}\cdot\text{ft}$$

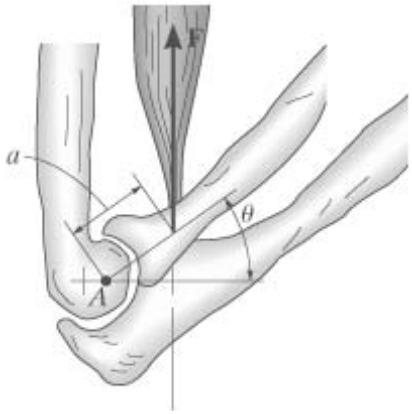
Problem 4-16

The elbow joint is flexed using the biceps brachii muscle, which remains essentially vertical as the arm moves in the vertical plane. If this muscle is located a distance *a* from the pivot point *A* on the humerus, determine the variation of the moment capacity about *A* if the constant force developed by the muscle is *F*. Plot these results of *M* vs. θ for $-60 \le \theta \le 80$.

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Units Used:

$kN = 10^3 N$



Given:

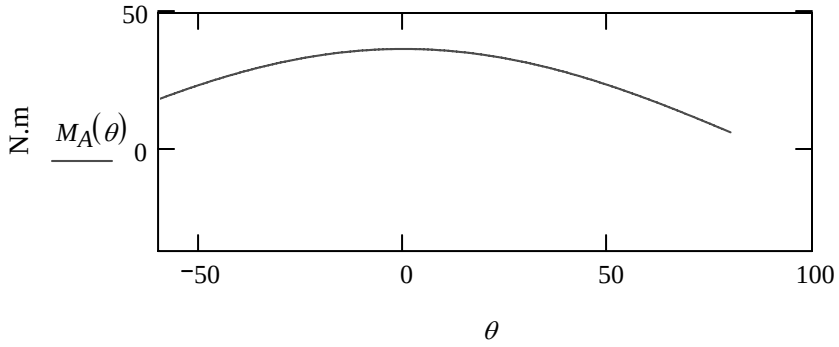
$a = 16\ mm$

$F = 2.30\ kN$

$\theta = (-60\ ..\ 80)$

Solution:

$M_A(\theta) = F\ (a) \cos(\theta\ deg)$



Problem 4-17

The Snorkel Co.produces the articulating boom platform that can support weight W. If the boom is in the position shown, determine the moment of this force about points A, B, and C.

Units Used:

$kip = 10^3 lb$

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Given:

$a = 3$ ft

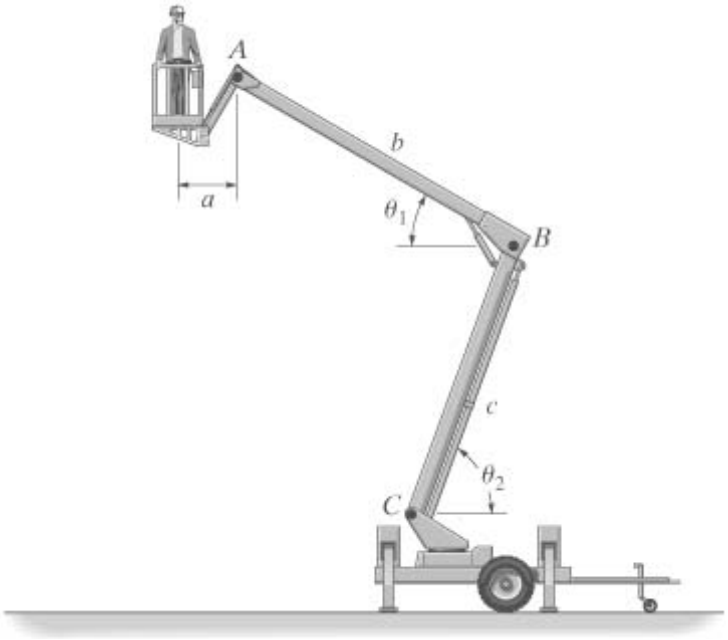
$b = 16$ ft

$c = 15$ ft

$\theta_1 = 30$ deg

$\theta_2 = 70$ deg

$W = 550$ lb



Solution:

$M_A = W a$ $M_A = 1.65\,\text{kip}\cdot\text{ft}$

$M_B = W\left(a + b\,\cos\left(\theta_1\right)\right)$ $M_B = 9.27\,\text{kip}\cdot\text{ft}$

$M_C = W\left(a + b\,\cos\left(\theta_1\right) - c\,\cos\left(\theta_2\right)\right)$ $M_C = 6.45\,\text{kip}\cdot\text{ft}$

Problem 4-18

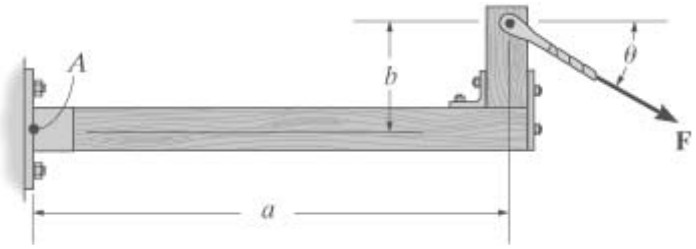
Determine the direction θ ($0° \le \theta \le 180°$) of the force **F** so that it produces **(a)** the maximum moment about point *A* and **(b)** the minimum moment about point *A*. Compute the moment in each case.

Given:

$F = 40$ lb

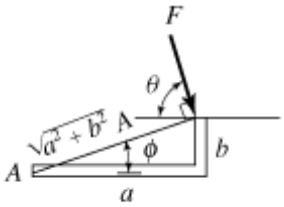
$a = 8$ ft

$b = 2$ ft



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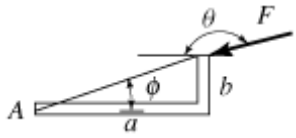
Solution: The maximum occurs when the force is perpendicular to the line between A and the point of application of the force. The minimum occurs when the force is parallel to this line.



(a) $\curvearrowleft + M_{Amax} = F\sqrt{a^2+b^2}$ $\quad M_{Amax} = 329.848\,\text{lb}\cdot\text{ft}$

$\phi_a = \text{atan}\left(\frac{b}{a}\right)$ $\quad \phi_a = 14.04\,\text{deg}$

$\theta_a = 90\ \text{deg} - \phi_a$ $\quad \theta_a = 76.0\,\text{deg}$



(b) $\curvearrowleft + M_{Amin} = 0\ \text{lb}\cdot\text{ft}$ $\quad M_{Amin} = 0\,\text{lb}\cdot\text{ft}$

$\phi_b = \text{atan}\left(\frac{b}{a}\right)$ $\quad \phi_b = 14.04\,\text{deg}$

$\theta_b = 180\ \text{deg} - \phi_b$ $\quad \theta_b = 166\,\text{deg}$

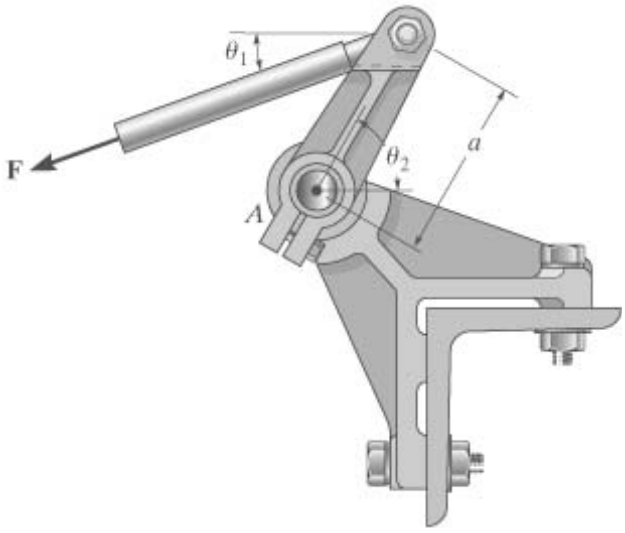
Problem 4-19

The rod on the power control mechanism for a business jet is subjected to force **F**. Determine the moment of this force about the bearing at *A*.

Given:

$F = 80\ \text{N}$ $\quad \theta_1 = 20\ \text{deg}$

$a = 150\ \text{mm}$ $\quad \theta_2 = 60\ \text{deg}$



Solution:

$\circlearrowleft + \quad M_A = F\cos(\theta_1)(a)\sin(\theta_2) - F\sin(\theta_1)(a)\cos(\theta_2)$ $\quad M_A = 7.71\,\text{N}\cdot\text{m}$

Problem 4-20

The boom has length *L*, weight W_b, and mass center at *G*. If the maximum moment that can be developed by the motor at *A* is *M*, determine the maximum load *W*, having a mass center at *G'*, that can be lifted.

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Given:

$L = 30$ ft

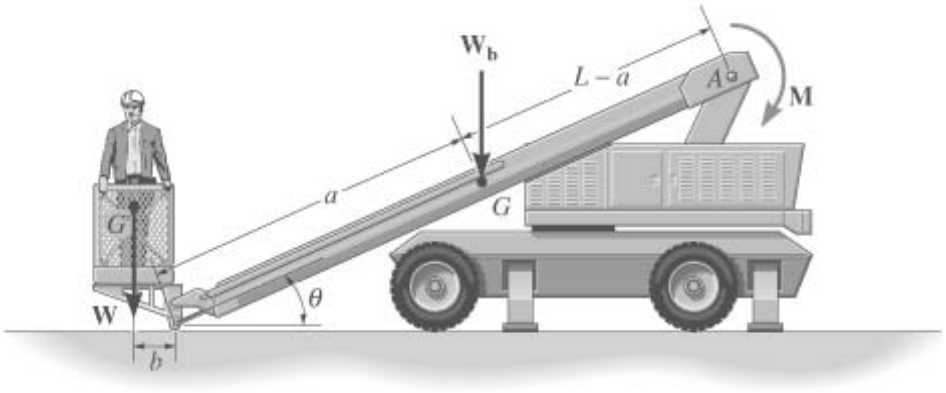
$W_b = 800$ lb

$a = 14$ ft

$b = 2$ ft

$\theta = 30$ deg

$M = 20 \times 10^3$ lb·ft



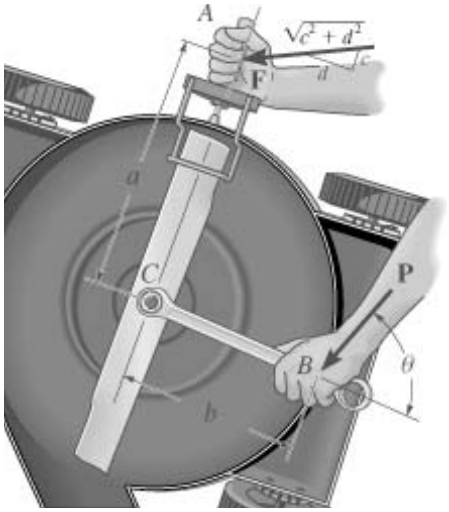
Solution:

$$M = W_b\,(L - a)\cos(\theta) + W\,(L\cos(\theta) + b)$$

$$W = \frac{M - W_b\,(L - a)\,\cos(\theta)}{L\,\cos(\theta) + b} \qquad W = 319\,\text{lb}$$

Problem 4-21

The tool at *A* is used to hold a power lawnmower blade stationary while the nut is being loosened with the wrench. If a force *P* is applied to the wrench at *B* in the direction shown, determine the moment it creates about the nut at *C*. What is the magnitude of force **F** at *A* so that it creates the opposite moment about *C* ?



Given:

$P = 50$ N $\quad b = 300$ mm

$\theta = 60$ deg $\quad c = 5$

$a = 400$ mm $\quad d = 12$

Solution:

(a) $\curvearrowleft + \quad M_A = P\sin(\theta)b$

$M_A = 13.0\,\text{N}\cdot\text{m}$

(b) $\curvearrowleft + \quad M_A - F\,\dfrac{d}{\sqrt{c^2 + d^2}}a = 0$

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$$F = M_A \left(\frac{\sqrt{c^2 + d^2}}{d\ a} \right)$$

$F = 35.2$ N

Problem 4-22

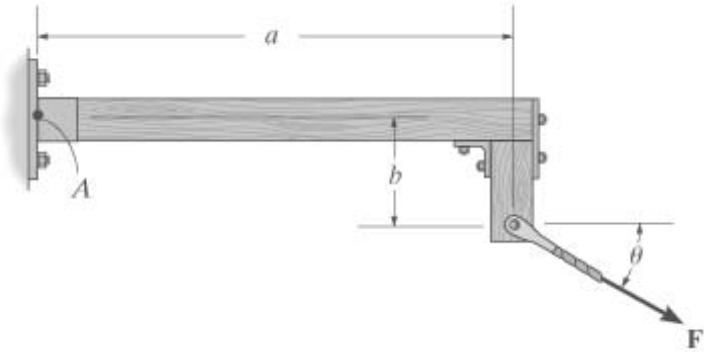
Determine the clockwise direction $\theta\ (0\,\text{deg} \le \theta \le 180\,\text{deg})$ of the force **F** so that it produces (a) the maximum moment about point *A* and (b) no moment about point *A*. Compute the moment in each case.

Given:

$F = 80$ lb

$a = 4$ ft

$b = 1$ ft



Solution:

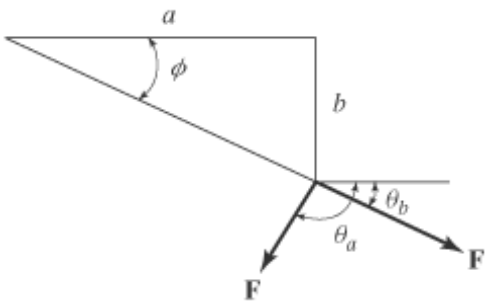
(**a**) $\curvearrowleft +$ $M_{Amax} = F\sqrt{a^2 + b^2}$ $M_{Amax} = 330\,\text{lb}\cdot\text{ft}$

$\phi = \text{atan}\left(\frac{b}{a}\right)$ $\phi = 14.0\,\text{deg}$

$\theta_a = 90\ \text{deg} + \phi$ $\theta_a = 104\,\text{deg}$

(b) $M_{Amin} = 0$

$\theta_b = \text{atan}\left(\frac{b}{a}\right)$ $\theta_b = 14.04\,\text{deg}$



Problem 4-23

The Y-type structure is used to support the high voltage transmission cables. If the supporting cables each exert a force *F* on the structure at *B*, determine the moment of each force about point *A*. Also, by the principle of transmissibility, locate the forces at points *C* and *D* and determine the moments.

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Units Used:

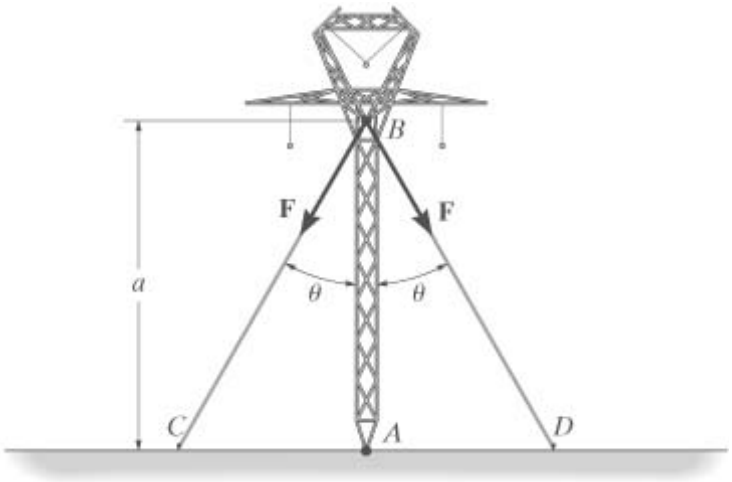
kip = 1000 lb

Given:

$F = 275$ lb

$a = 85$ ft

$\theta = 30$ deg



Solution:

$\curvearrowleft + \quad M_{A1} = F \sin(\theta)\, a \qquad M_{A1} = 11.7\,\text{kip}\cdot\text{ft}$

$\curvearrowright + \quad M_{A2} = F \sin(\theta) a \qquad M_{A2} = 11.7\,\text{kip}\cdot\text{ft}$

Also $b = (a)\tan(\theta)$

$M_{A1} = F \cos(\theta) b \qquad M_{A1} = 11.7\,\text{kip}\cdot\text{ft}$

$M_{A2} = F \cos(\theta) b \qquad M_{A2} = 11.7\,\text{kip}\cdot\text{ft}$

Problem 4-24

The force **F** acts on the end of the pipe at *B*. Determine (a) the moment of this force about point *A*, and (b) the magnitude and direction of a horizantal force, applied at *C*, which produces the same moment.

Given:

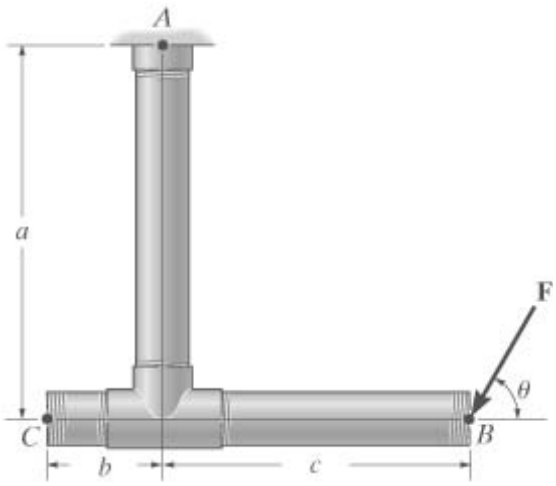
$F = 70$ N

$a = 0.9$ m

$b = 0.3$ m

$c = 0.7$ m

$\theta = 60$ deg



Solution:

(a) $\curvearrowright + \quad M_A = F \sin(\theta)\, c + F \cos(\theta)\, a \qquad M_A = 73.9\,\text{N}\cdot\text{m}$

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(b) $F_C\,(a) = M_A \qquad F_C = \dfrac{M_A}{a} \qquad F_C = 82.2\ \text{N}$

Problem 4-25

The force F acts on the end of the pipe at B. Determine the angles θ ($0° \le \theta \le 180°$) of the force that will produce maximum and minimum moments about point A. What are the magnitudes of these moments?

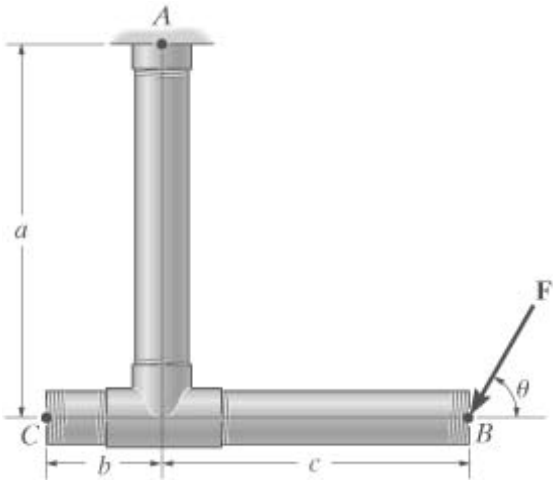
Given:

$F = 70\ \text{N}$

$a = 0.9\ \text{m}$

$b = 0.3\ \text{m}$

$c = 0.7\ \text{m}$



Solution:

$\circlearrowleft_+ \qquad M_A = F\sin(\theta)c + F\cos(\theta)a$

For maximum moment $\dfrac{\mathrm{d}}{\mathrm{d}\theta}M_A = cF\cos(\theta) - aF\sin(\theta) = 0$

$\theta_{max} = \operatorname{atan}\left(\dfrac{c}{a}\right) \qquad \theta_{max} = 37.9\,\text{deg}$

$M_{Amax} = F\sin(\theta_{max})c + F\cos(\theta_{max})a \qquad M_{Amax} = 79.812\,\text{N}\cdot\text{m}$

For minimum moment $M_A = F\sin(\theta)c + F\cos(\theta)a = 0$

$\theta_{min} = 180\ \text{deg} + \operatorname{atan}\left(\dfrac{-a}{c}\right) \qquad \theta_{min} = 128\,\text{deg}$

$M_{Amin} = F\ c\ \sin(\theta_{min}) + F\ (a)\ \cos(\theta_{min}) \qquad M_{Amin} = 0\,\text{N}\cdot\text{m}$

Problem 4-26

The towline exerts force P at the end of the crane boom of length L. Determine the placement

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x of the hook at A so that this force creates a maximum moment about point O. What is this moment?

Unit Used:

$kN = 10^3 N$

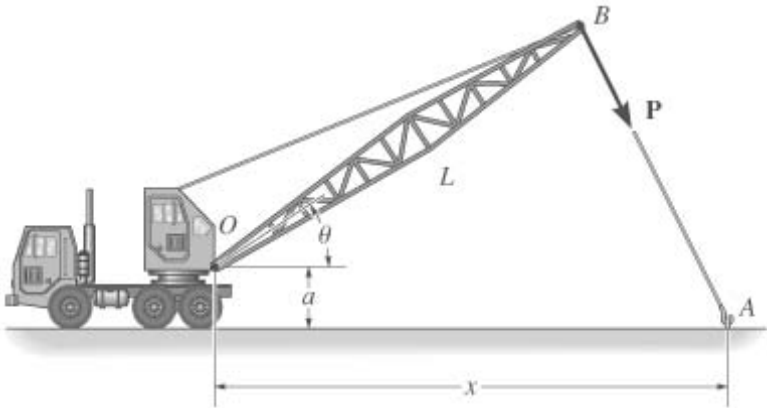
Given:

$P = 4$ kN

$L = 20$ m

$\theta = 30$ deg

$a = 1.5$ m



Solution:

Maximum moment, $OB \perp BA$

Guesses $x = 1$ m $d = 1$ m (Length of the cable from B to A)

Given $L\cos(\theta) + d\sin(\theta) = x$

$a + L\sin(\theta) = d\cos(\theta)$

$\begin{pmatrix} x \\ d \end{pmatrix} = \text{Find}(x, d)$ $x = 23.96$ m

$M_{max} = P\ L$ $M_{max} = 80\,\text{kN}\cdot\text{m}$

Problem 4-27

The towline exerts force P at the end of the crane boom of length L. Determine the position θ of the boom so that this force creates a maximum moment about point O. What is this moment?

Units Used:

$kN = 10^3 N$

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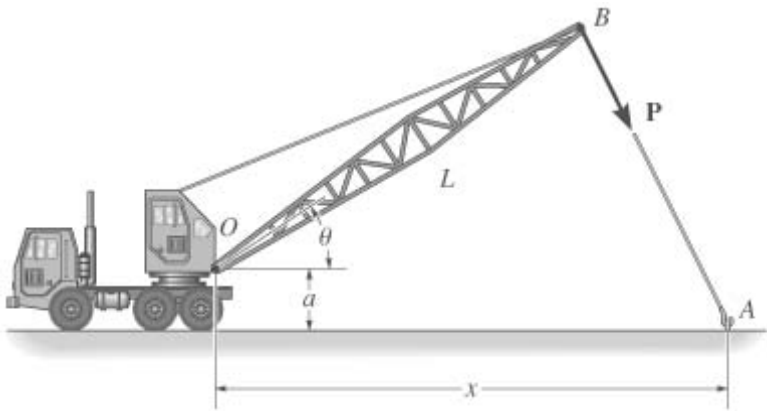
Given:

$P = 4$ kN

$x = 25$ m

$L = 20$ m

$a = 1.5$ m



Solution:

Maximum moment, $OB \perp BA$

Guesses $\theta = 30$ deg $d = 1$ m (length of cable from B to A)

Given $L\cos(\theta) + d\sin(\theta) = x$

$a + L\sin(\theta) = d\cos(\theta)$

$\begin{pmatrix} \theta \\ d \end{pmatrix} = \text{Find}(\theta, d)$ $\theta = 33.573\,\text{deg}$

$M_{max} = P\,L$ $M_{max} = 80\,\text{kN}\cdot\text{m}$

Problem 4-28

Determine the resultant moment of the forces about point A. Solve the problem first by considering each force as a whole, and then by using the principle of moments.

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Units Used:

$kN = 10^3 N$

Given:

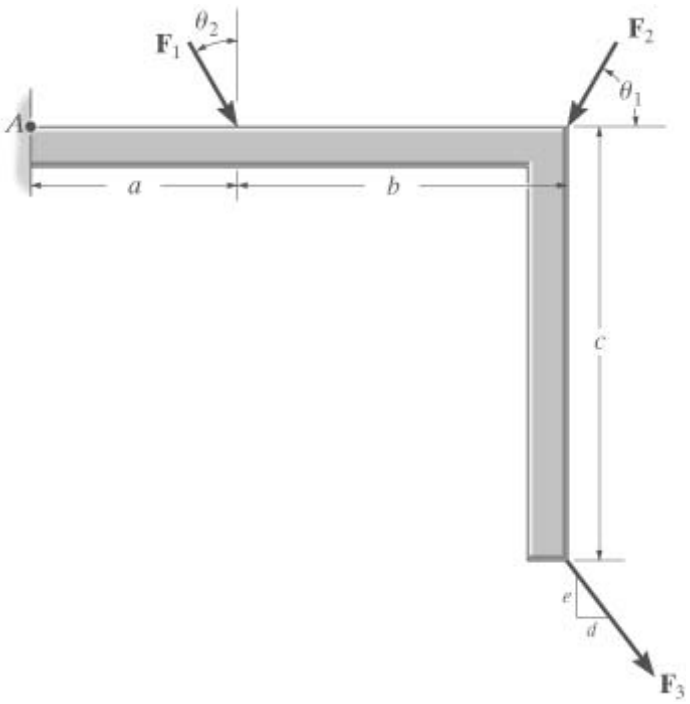
$F_1 = 250\ N \quad a = 2\ m$

$F_2 = 300\ N \quad b = 3\ m$

$F_3 = 500\ N \quad c = 4\ m$

$\theta_1 = 60\ deg \quad d = 3$

$\theta_2 = 30\ deg \quad e = 4$



Solution Using Whole Forces:

Geometry $\alpha = \operatorname{atan}\left(\frac{d}{e}\right) \qquad L = \left(a + b - \frac{d}{e}c\right)\frac{e}{\sqrt{e^2 + d^2}}$

$$M_A = -F_1\left[(a)\cos\left(\theta_2\right)\right] - F_2(a+b)\sin\left(\theta_1\right) - F_3 L \qquad M_A = -2.532\ kN\cdot m$$

Solution Using Principle of Moments:

$$M_A = -F_1\cos\left(\theta_2\right)a - F_2\sin\left(\theta_1\right)(a+b) + F_3\frac{d}{\sqrt{d^2+e^2}}c - F_3\frac{e}{\sqrt{d^2+e^2}}(a+b)$$

$$M_A = -2.532\times 10^3\ N\cdot m$$

Problem 4-29

If the resultant moment about point A is M clockwise, determine the magnitude of $\mathbf{F}_3$.

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Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

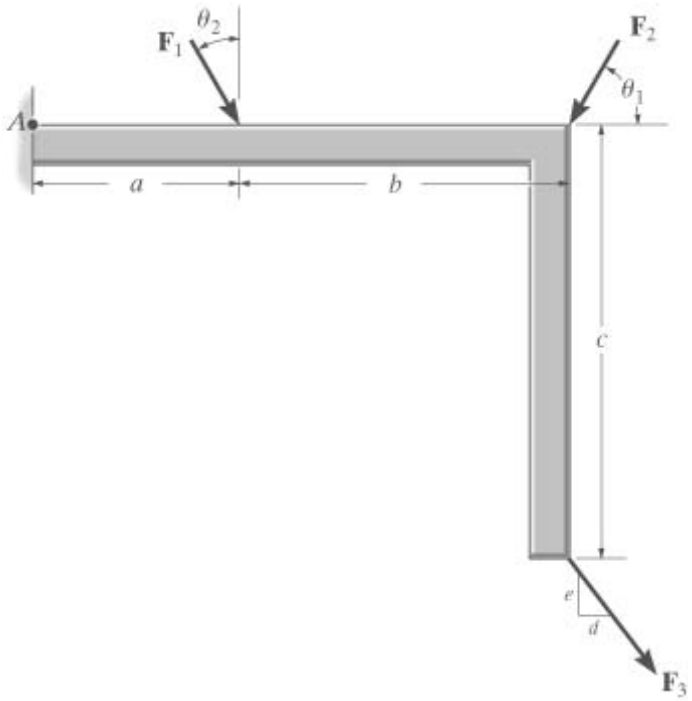
$M = 4.8\ \text{kN·m}$ $a = 2\ \text{m}$

$F_1 = 300\ \text{N}$ $b = 3\ \text{m}$

$F_2 = 400\ \text{N}$ $c = 4\ \text{m}$

$\theta_1 = 60\ \text{deg}$ $d = 3$

$\theta_2 = 30\ \text{deg}$ $e = 4$



Solution:

Initial Guess $F_3 = 1\ \text{N}$

Given

$$-M = -F_1\cos(\theta_2)a - F_2\sin(\theta_1)(a+b) + F_3\left(\frac{d}{\sqrt{d^2+e^2}}\right)c - F_3\left(\frac{e}{\sqrt{d^2+e^2}}\right)(a+b)$$

$F_3 = \text{Find}(F_3)$ $F_3 = 1.593\,\text{kN}$

Problem 4-30

The flat-belt tensioner is manufactured by the Daton Co. and is used with V-belt drives on poultry and livestock fans. If the tension in the belt is **F**, when the pulley is not turning, determine the moment of each of these forces about the pin at *A*.

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Given:

$F = 52$ lb

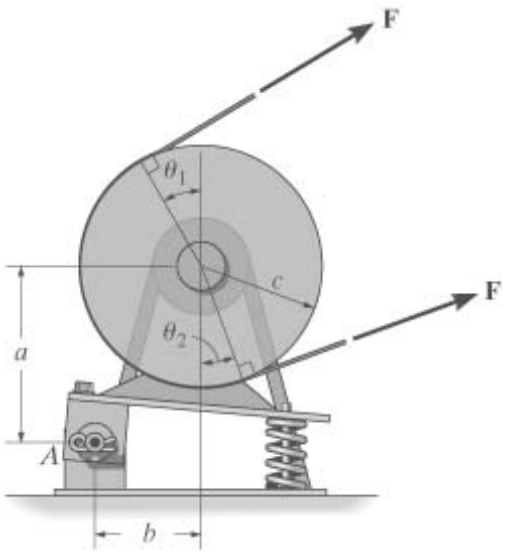
$a = 8$ in

$b = 5$ in

$c = 6$ in

$\theta_1 = 30$ deg

$\theta_2 = 20$ deg



Solution:

$$\circlearrowleft_+ \quad M_{A1} = F\cos(\theta_1)(a + c\cos(\theta_1)) - F\sin(\theta_1)(b - c\sin(\theta_1))$$

$$M_{A1} = 542\,\text{lb}\cdot\text{in}$$

$$\circlearrowleft_+ \quad M_{A2} = F\cos(\theta_2)(a - c\cos(\theta_2)) - F(\sin(\theta_2))(b + c\sin(\theta_2))$$

$$M_{A2} = -10.01\,\text{lb}\cdot\text{in}$$

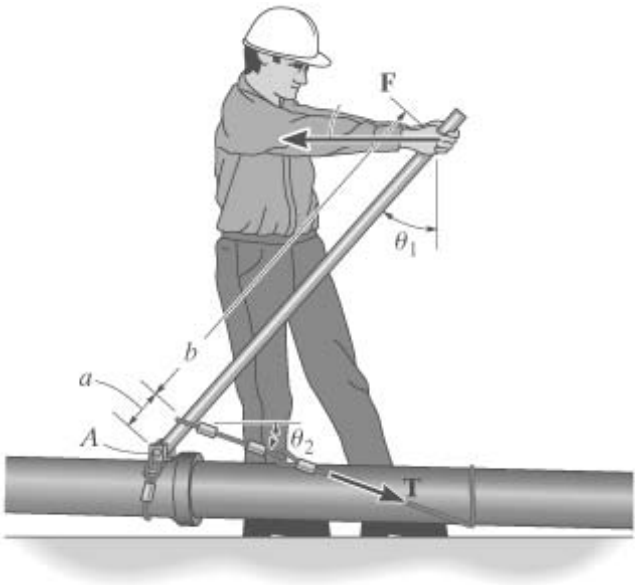
Problem 4-31

The worker is using the bar to pull two pipes together in order to complete the connection. If he applies a horizantal force **F** to the handle of the lever, determine the moment of this force about the end *A*. What would be the tension *T* in the cable needed to cause the opposite moment about point *A*.

Given:

$F = 80$ lb $\quad \theta_1 = 40$ deg $\quad \theta_2 = 20$ deg $\quad a = 0.5$ ft $\quad b = 4.5$ ft

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Solution:

$$\curvearrowleft M_A = F(a+b)\cos(\theta_1)$$

$$M_A = 306\,\text{lb}\cdot\text{ft}$$

Require $M_A = T\cos(\theta_2)(a)\cos(\theta_1) + T\sin(\theta_2)(a)\sin(\theta_1)$

$$T = \frac{M_A}{(a)\left(\cos(\theta_2)\cos(\theta_1) + \sin(\theta_2)\sin(\theta_1)\right)} \qquad T = 652\,\text{lb}$$

Problem 4-32

If it takes a force **F** to pull the nail out, determine the smallest vertical force **P** that must be applied to the handle of the crowbar. *Hint:* This requires the moment of **F** about point *A* to be equal to the moment of **P** about *A*. Why?

Given:

$F = 125$ lb

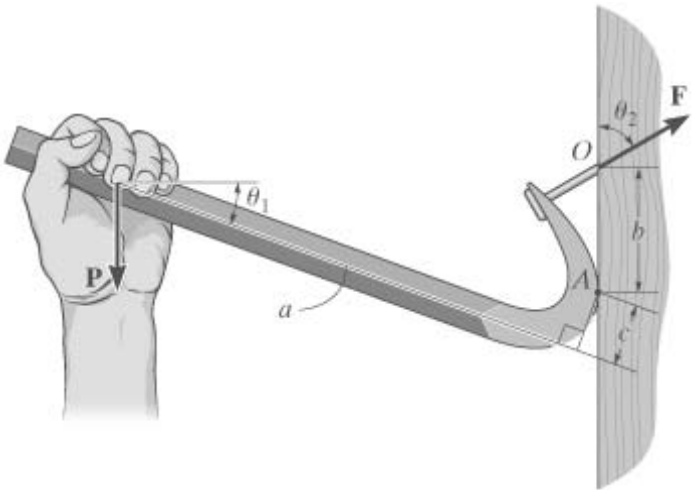
$a = 14$ in

$b = 3$ in

$c = 1.5$ in

$\theta_1 = 20$ deg

$\theta_2 = 60$ deg



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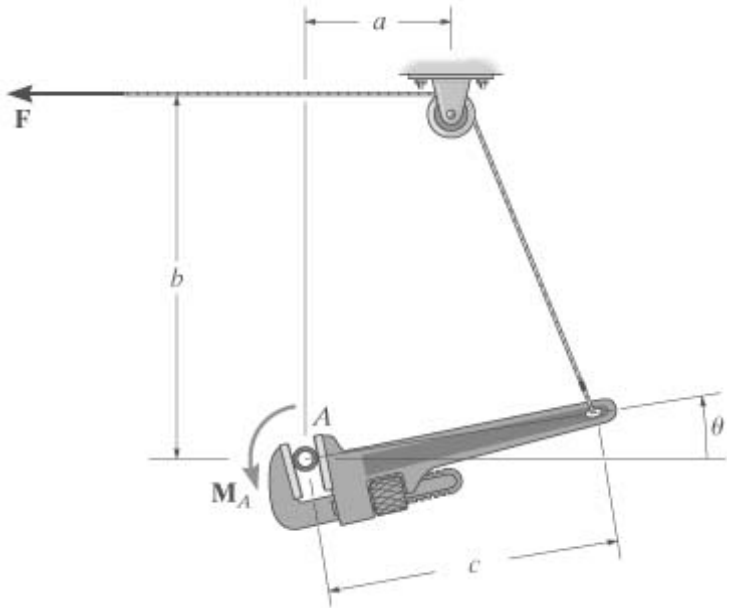
Solution:

$\circlearrowleft_+ \quad M_F = F\sin(\theta_2)(b) \qquad M_F = 325\,\text{lb}\cdot\text{in}$

$\circlearrowleft_+ \quad P\left[(a)\cos(\theta_1) + (c)\sin(\theta_1)\right] = M_F \qquad P = \dfrac{M_F}{(a)\,\cos(\theta_1) + (c)\,\sin(\theta_1)} \qquad P = 23.8\,\text{lb}$

Problem 4-33

The pipe wrench is activated by pulling on the cable segment with a horizantal force $\mathbf{F}$. Determine the moment M_A produced by the wrench on the pipe at θ. Neglect the size of the pulley.



Given:

$F = 500$ N

$a = 0.2$ m

$b = 0.5$ m

$c = 0.4$ m

$\theta = 20$ deg

Solution:

Initial Guesses

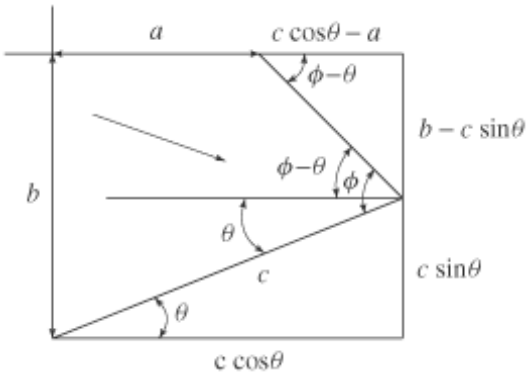
$\phi = 20$ deg

$M_A = 1$ N·m

Given

$\dfrac{b - c\sin(\theta)}{c\cos(\theta) - a} = \tan(\phi - \theta)$

$M_A = F\,c\sin(\phi)$



$\begin{pmatrix} \phi \\ M_A \end{pmatrix} = \text{Find}(\phi, M_A) \qquad \phi = 84.161\,\text{deg} \qquad M_A = 199\,\text{N}\cdot\text{m}$

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Problem 4-34

Determine the moment of the force at *A* about point *O*. Express the result as a Cartesian vector.

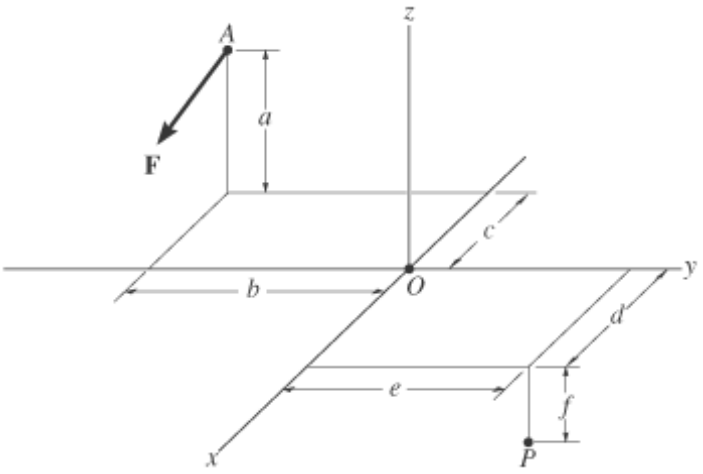
Given:

$$\mathbf{F} = \begin{pmatrix} 60 \\ -30 \\ -20 \end{pmatrix} \text{ N}$$

$a = 4$ m $\quad d = 4$ m

$b = 7$ m $\quad e = 6$ m

$c = 3$ m $\quad f = 2$ m



Solution:

$$\mathbf{r_{OA}} = \begin{pmatrix} -c \\ -b \\ a \end{pmatrix} \qquad \mathbf{M_O} = \mathbf{r_{OA}} \times \mathbf{F} \qquad \mathbf{M_O} = \begin{pmatrix} 260 \\ 180 \\ 510 \end{pmatrix} \text{ N}\cdot\text{m}$$

Problem 4-35

Determine the moment of the force at *A* about point *P*. Express the result as a Cartesian vector.

Given:

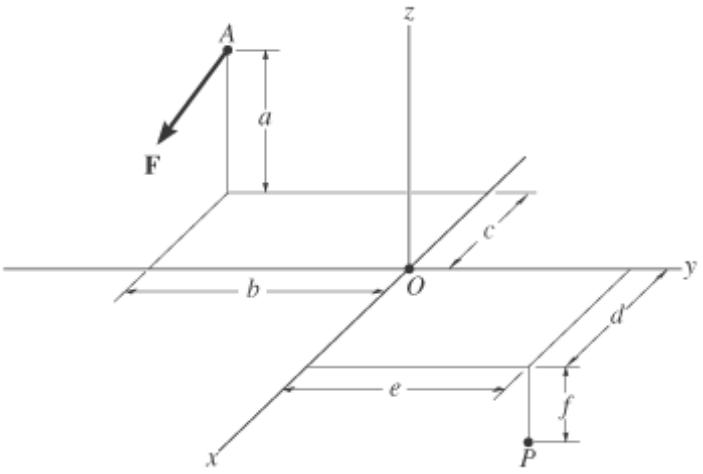
$a = 4$ m $\quad b = 7$ m $\quad c = 3$ m $\quad d = 4$ m $\quad e = 6$ m $\quad f = 2$ m

$$\mathbf{F} = \begin{pmatrix} 60 \\ -30 \\ -20 \end{pmatrix} \text{ N}$$

Solution:

$$\mathbf{r_{PA}} = \begin{pmatrix} -c - d \\ -b - e \\ a + f \end{pmatrix} \qquad \mathbf{M_P} = \mathbf{r_{PA}} \times \mathbf{F}$$

$$\mathbf{M_P} = \begin{pmatrix} 440 \\ 220 \\ 990 \end{pmatrix} \text{ N}\cdot\text{m}$$



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Problem 4-36

Determine the moment of the force **F** at *A* about point *O*. Express the result as a cartesian vector.

Units Used:

$kN = 10^3 N$

Given:

$F = 13$ kN

$a = 6$ m

$b = 2.5$ m

$c = 3$ m

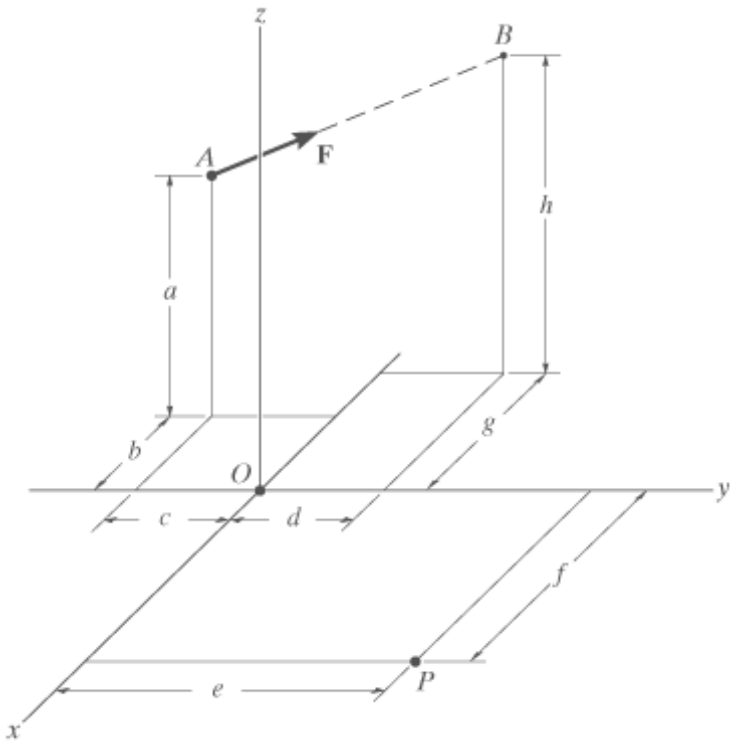
$d = 3$ m

$e = 8$ m

$f = 6$ m

$g = 4$ m

$h = 8$ m



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} b-g \\ c+d \\ h-a \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} -b \\ -c \\ a \end{pmatrix} \qquad \mathbf{F_1} = F\,\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|}$$

$$\mathbf{M_O} = \mathbf{r_{OA}} \times \mathbf{F_1} \qquad \mathbf{M_O} = \begin{pmatrix} -84 \\ -8 \\ -39 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-37

Determine the moment of the force **F** at *A* about point *P*. Express the result as a Cartesian vector.

Units Used: $kN = 10^3 N$

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Given:

$F = 13$ kN

$a = 6$ m

$b = 2.5$ m

$c = 3$ m

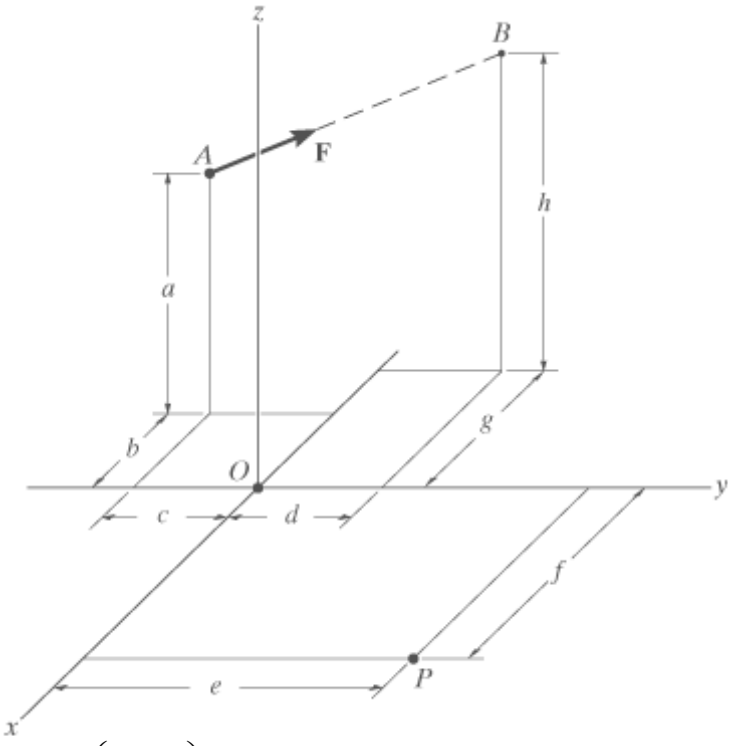
$d = 3$ m

$e = 8$ m

$f = 6$ m

$g = 4$ m

$h = 8$ m



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} b-g \\ c+d \\ h-a \end{pmatrix} \qquad \mathbf{r_{PA}} = \begin{pmatrix} -b-f \\ -c-e \\ a \end{pmatrix} \qquad \mathbf{F_1} = F\,\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|}$$

$$\mathbf{M_O} = \mathbf{r_{PA}} \times \mathbf{F_1} \qquad \mathbf{M_O} = \begin{pmatrix} -116 \\ 16 \\ -135 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-38

The curved rod lies in the *x-y* plane and has radius *r*. If a force **F** acts at its end as shown, determine the moment of this force about point *O*.

Given:

$r = 3$ m $\quad a = 1$ m $\quad \theta = 45$ deg

$F = 80$ N $\quad b = 2$ m

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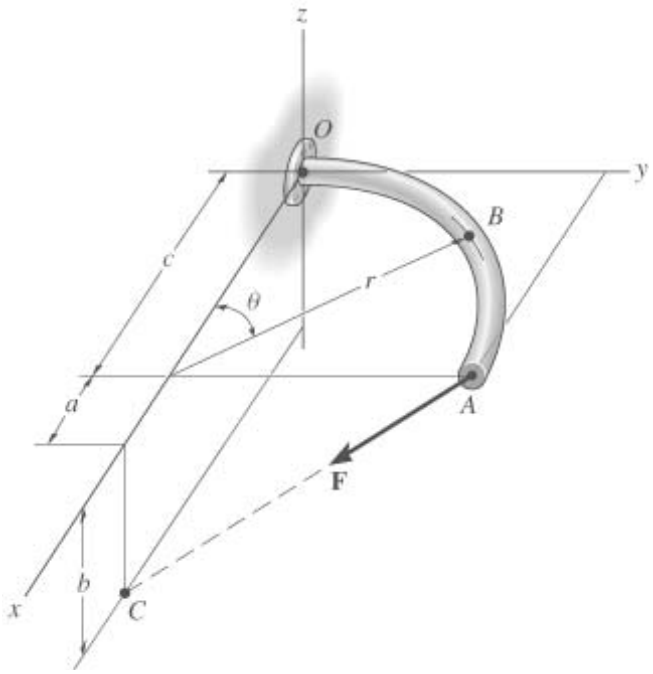
Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} a \\ -r \\ -b \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} 1 \\ -3 \\ -2 \end{pmatrix} \text{ m}$$

$$\mathbf{F_v} = F\frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{F_v} = \begin{pmatrix} 21.381 \\ -64.143 \\ -42.762 \end{pmatrix} \text{ N}$$

$$\mathbf{r_{OA}} = \begin{pmatrix} r \\ r \\ 0 \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} \text{ m}$$

$$\mathbf{M_O} = \mathbf{r_{OA}} \times \mathbf{F_v} \qquad \mathbf{M_O} = \begin{pmatrix} -128.285 \\ 128.285 \\ -256.571 \end{pmatrix} \text{N}\cdot\text{m}$$



Problem 4-39

The curved rod lies in the *x-y* plane and has a radius *r*. If a force **F** acts at its end as shown, determine the moment of this force about point *B*.

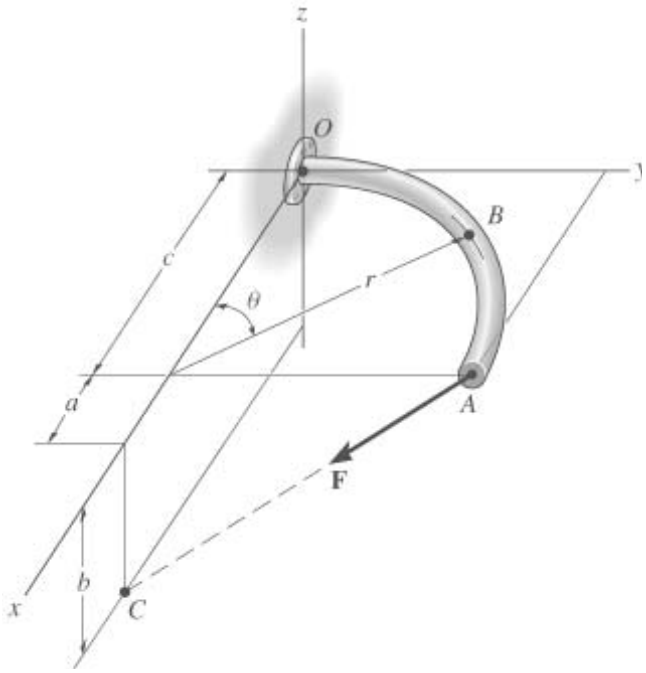
Given:

$F = 80$ N $\quad c = 3$ m

$a = 1$ m $\quad r = 3$ m

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$b = 2\ \text{m} \qquad \theta = 45\ \text{deg}$

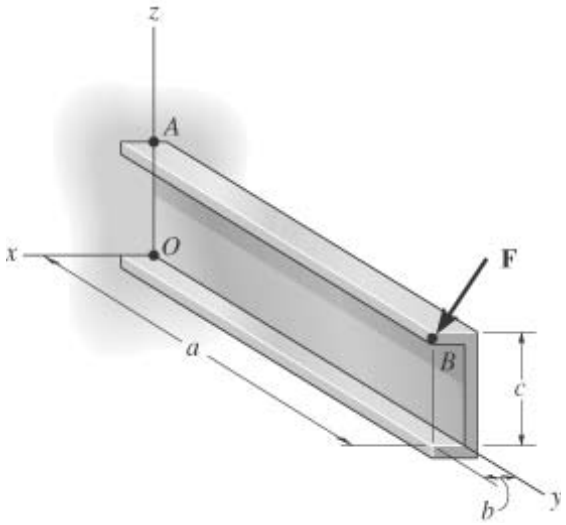


Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} a \\ -c \\ -b \end{pmatrix} \qquad \mathbf{F_v} = F\,\frac{\mathbf{r_{AC}}}{\left|\mathbf{r_{AC}}\right|} \qquad \mathbf{r_{BA}} = \begin{pmatrix} r\cos(\theta) \\ r - r\sin(\theta) \\ 0 \end{pmatrix}$$

$$\mathbf{M_B} = \mathbf{r_{BA}} \times \mathbf{F_v} \qquad \mathbf{M_B} = \begin{pmatrix} -37.6 \\ 90.7 \\ -154.9 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-40



The force **F** acts at the end of the beam. Determine the moment of the force about point *A*.

Given:

$$\mathbf{F} = \begin{pmatrix} 600 \\ 300 \\ -600 \end{pmatrix} \text{N}$$

$a = 1.2\ \text{m}$

$b = 0.2\ \text{m}$

$c = 0.4\ \text{m}$

Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} b \\ a \\ 0 \end{pmatrix} \qquad \mathbf{M_A} = \mathbf{r_{AB}} \times \mathbf{F} \qquad \mathbf{M_A} = \begin{pmatrix} -720 \\ 120 \\ -660 \end{pmatrix} \text{N}\cdot\text{m}$$

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Problem 4-41

The pole supports a traffic light of weight W. Using Cartesian vectors, determine the moment of the weight of the traffic light about the base of the pole at A.

Given:

$W = 22$ lb $\quad a = 12$ ft $\quad \theta = 30$ deg

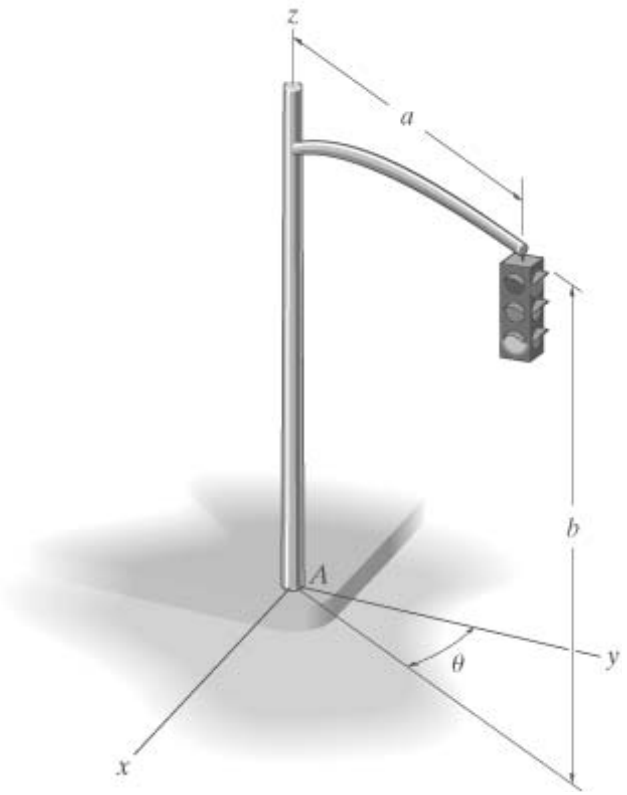
Solution:

$$\mathbf{r} = \begin{bmatrix} (a)\sin(\theta) \\ (a)\cos(\theta) \\ 0 \end{bmatrix}$$

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ -W \end{pmatrix}$$

$$\mathbf{M_A} = \mathbf{r} \times \mathbf{F}$$

$$\mathbf{M_A} = \begin{pmatrix} -229 \\ 132 \\ 0 \end{pmatrix} \text{lb}\cdot\text{ft}$$



Problem 4-42

The man pulls on the rope with a force **F.** Determine the moment that this force exerts about the base of the pole at O. Solve the problem two ways, i.e., by using a position vector from O to A, then O to B.

Given:

$F = 20$ N

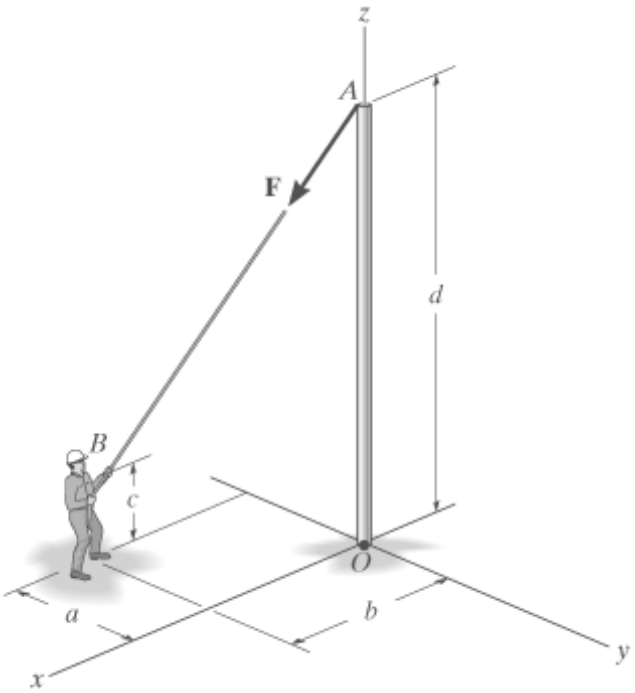
$a = 3$ m

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$b = 4$ m

$c = 1.5$ m

$d = 10.5$ m



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} b \\ -a \\ c-d \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} 0 \\ 0 \\ d \end{pmatrix}$$

$$\mathbf{r_{OB}} = \begin{pmatrix} b \\ -a \\ c \end{pmatrix} \qquad \mathbf{F_v} = F\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|}$$

$$\mathbf{M_{O1}} = \mathbf{r_{OA}} \times \mathbf{F_v} \qquad \mathbf{M_{O1}} = \begin{pmatrix} 61.2 \\ 81.6 \\ 0 \end{pmatrix} \text{N}\cdot\text{m}$$

$$\mathbf{M_{O2}} = \mathbf{r_{OB}} \times \mathbf{F_v} \qquad \mathbf{M_{O2}} = \begin{pmatrix} 61.2 \\ 81.6 \\ -0 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-43

Determine the smallest force *F* that must be applied along the rope in order to cause the curved rod, which has radius *r*, to fail at the support *C*. This requires a moment to be developed at *C* of magnitude *M*.

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Given:

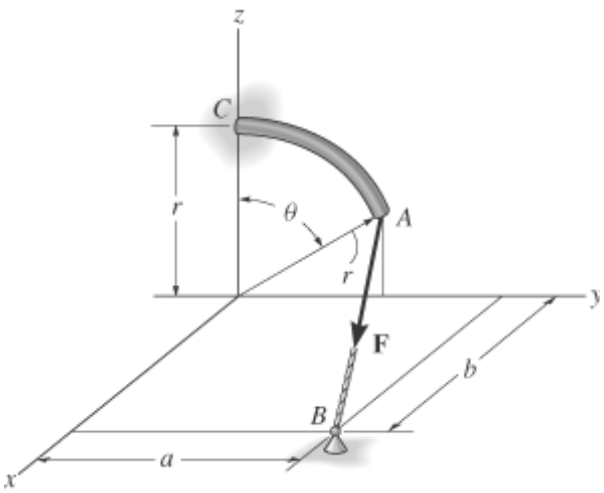
$r = 5$ ft

$M = 80$ lb·ft

$\theta = 60$ deg

$a = 7$ ft

$b = 6$ ft



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} b \\ a - r\sin(\theta) \\ -r\cos(\theta) \end{pmatrix} \qquad \mathbf{u_{AB}} = \frac{\mathbf{r_{AB}}}{\left|\mathbf{r_{AB}}\right|} \qquad \mathbf{r_{CB}} = \begin{pmatrix} b \\ a \\ -r \end{pmatrix}$$

Guess $F = 1$ lb

Given $\left|\mathbf{r_{CB}} \times \left(F\ \mathbf{u_{AB}}\right)\right| = M \qquad F = \text{Find}(F) \qquad F = 18.6\,\text{lb}$

Problem 4-44

The pipe assembly is subjected to the force $\mathbf{F}$. Determine the moment of this force about point A.

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Given:

$F = 80$ N

$a = 400$ mm

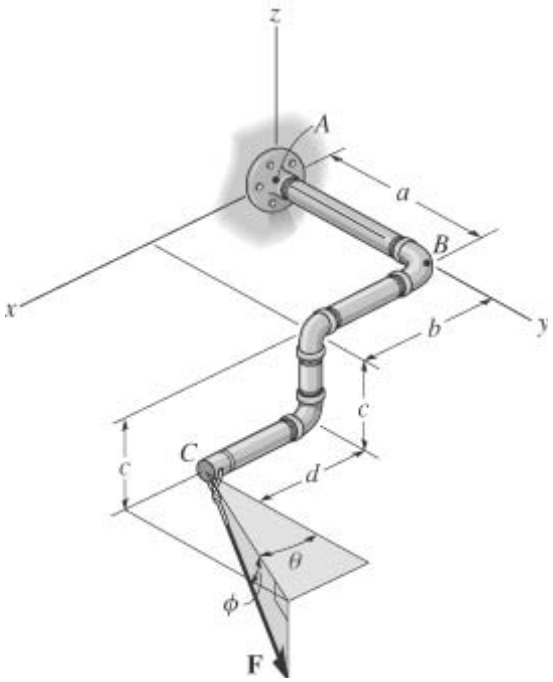
$b = 300$ mm

$c = 200$ mm

$d = 250$ mm

$\theta = 40$ deg

$\phi = 30$ deg



Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} b+d \\ a \\ -c \end{pmatrix} \qquad \mathbf{F_v} = F \begin{pmatrix} \cos(\phi)\sin(\theta) \\ \cos(\phi)\cos(\theta) \\ -\sin(\phi) \end{pmatrix}$$

$$\mathbf{M_A} = \mathbf{r_{AC}} \times \mathbf{F_v} \qquad \mathbf{M_A} = \begin{pmatrix} -5.385 \\ 13.093 \\ 11.377 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-45

The pipe assembly is subjected to the force **F**. Determine the moment of this force about point *B*.

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Given:

$F = 80\ \text{N}$

$a = 400\ \text{mm}$

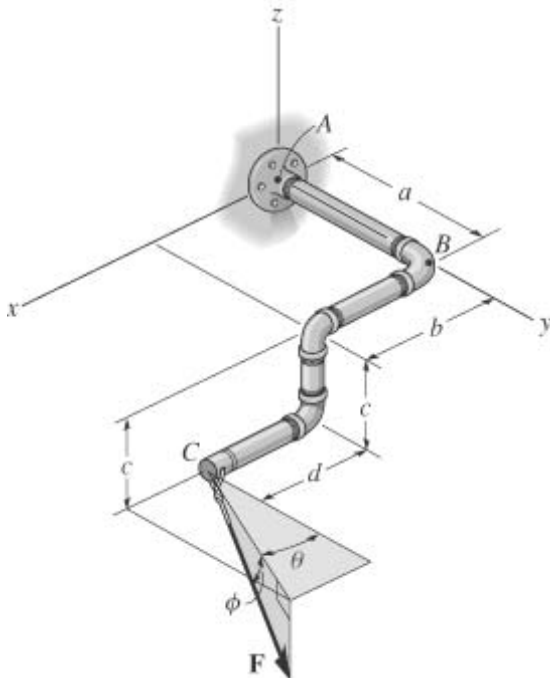
$b = 300\ \text{mm}$

$c = 200\ \text{mm}$

$d = 250\ \text{mm}$

$\theta = 40\ \text{deg}$

$\phi = 30\ \text{deg}$



Solution:

$$\mathbf{r_{BC}} = \begin{pmatrix} b+d \\ 0 \\ -c \end{pmatrix} \qquad \mathbf{r_{BC}} = \begin{pmatrix} 550 \\ 0 \\ -200 \end{pmatrix} \text{mm}$$

$$\mathbf{F_V} = F \begin{pmatrix} \cos(\phi)\sin(\theta) \\ \cos(\phi)\cos(\theta) \\ -\sin(\phi) \end{pmatrix} \qquad \mathbf{F_V} = \begin{pmatrix} 44.534 \\ 53.073 \\ -40 \end{pmatrix} \text{N}$$

$$\mathbf{M_B} = \mathbf{r_{BC}} \times \mathbf{F_V} \qquad \mathbf{M_B} = \begin{pmatrix} 10.615 \\ 13.093 \\ 29.19 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-46

The *x*-ray machine is used for medical diagnosis. If the camera and housing at *C* have mass *M* and a mass center at *G*, determine the moment of its weight about point *O* when it is in the position shown.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

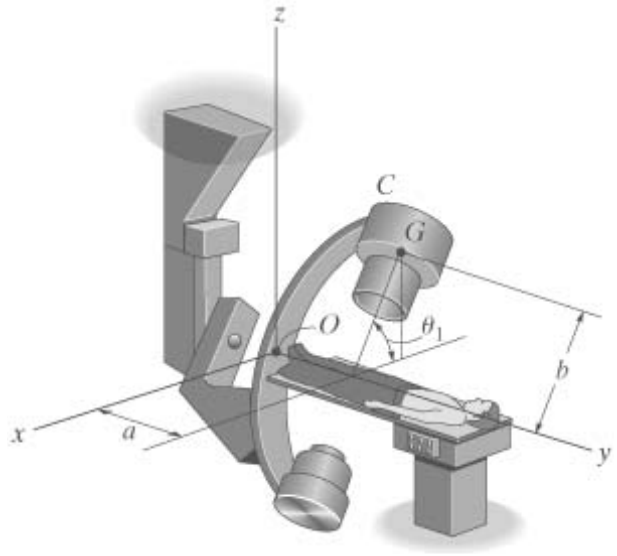
$M = 150\ \text{kg}$

$a = 1.2\ \text{m}$

$b = 1.5\ \text{m}$

$\theta = 60\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:

$$\mathbf{M_O} = \begin{bmatrix} -(b)\cos(\theta) \\ a \\ (b)\sin(\theta) \end{bmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -Mg \end{pmatrix} \qquad \mathbf{M_O} = \begin{pmatrix} -1.77 \\ -1.1 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-47

Using Cartesian vector analysis, determine the resultant moment of the three forces about the base of the column at *A*.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

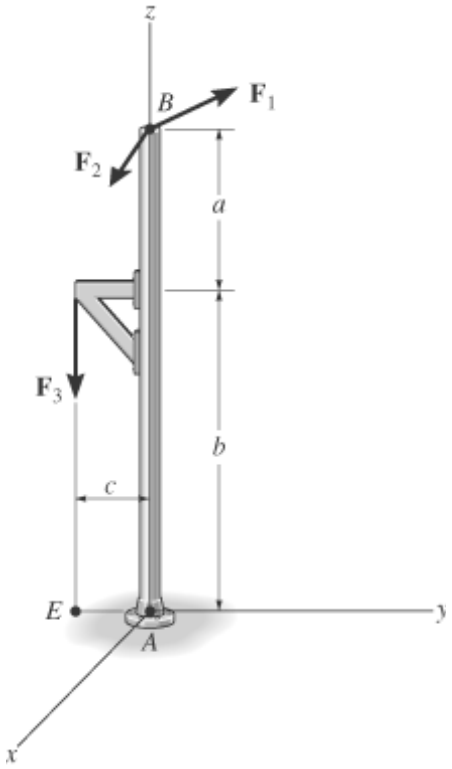
$$\mathbf{F_1} = \begin{pmatrix} 400 \\ 300 \\ 120 \end{pmatrix} \text{N}$$

$$\mathbf{F_2} = \begin{pmatrix} 100 \\ -100 \\ -60 \end{pmatrix} \text{N}$$

$$\mathbf{F_3} = \begin{pmatrix} 0 \\ 0 \\ -500 \end{pmatrix} \text{N}$$

$a = 4\ \text{m}$

$b = 8\ \text{m}$



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$c = 1\ \text{m}$

Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} 0 \\ 0 \\ a+b \end{pmatrix} \qquad \mathbf{r_{A3}} = \begin{pmatrix} 0 \\ -c \\ b \end{pmatrix}$$

The individual moments

$$\mathbf{M_{A1}} = \mathbf{r_{AB}} \times \mathbf{F_1} \qquad \mathbf{M_{A2}} = \mathbf{r_{AB}} \times \mathbf{F_2} \qquad \mathbf{M_{A3}} = \mathbf{r_{A3}} \times \mathbf{F_3}$$

$$\mathbf{M_{A1}} = \begin{pmatrix} -3.6 \\ 4.8 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m} \qquad \mathbf{M_{A2}} = \begin{pmatrix} 1.2 \\ 1.2 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m} \qquad \mathbf{M_{A3}} = \begin{pmatrix} 0.5 \\ 0 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m}$$

The total moment

$$\mathbf{M_A} = \mathbf{M_{A1}} + \mathbf{M_{A2}} + \mathbf{M_{A3}} \qquad \mathbf{M_A} = \begin{pmatrix} -1.9 \\ 6 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-48

A force $\mathbf{F}$ produces a moment $\mathbf{M}_O$ about the origin of coordinates, point O. If the force acts at a point having the given x coordinate, determine the y and z coordinates.

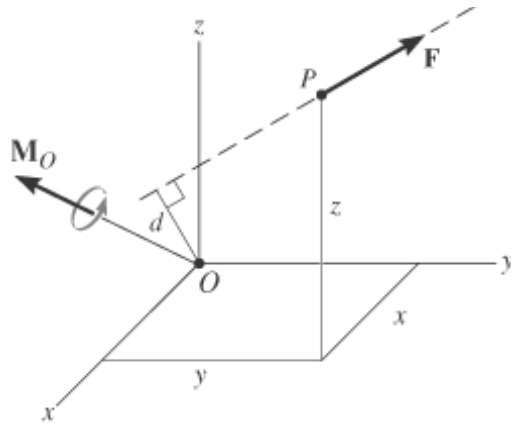
Units Used: $\text{kN} = 10^3\ \text{N}$

Given:

$$\mathbf{F} = \begin{pmatrix} 6 \\ -2 \\ 1 \end{pmatrix} \text{kN}$$

$$\mathbf{M_O} = \begin{pmatrix} 4 \\ 5 \\ -14 \end{pmatrix} \text{kN}\cdot\text{m}$$

$x = 1\ \text{m}$



Solution:

The initial guesses:

$y = 1\ \text{m} \qquad z = 1\ \text{m}$

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Given

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \mathbf{F} = \mathbf{M_O} \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \text{Find}(y, z) \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{m}$$

Problem 4-49

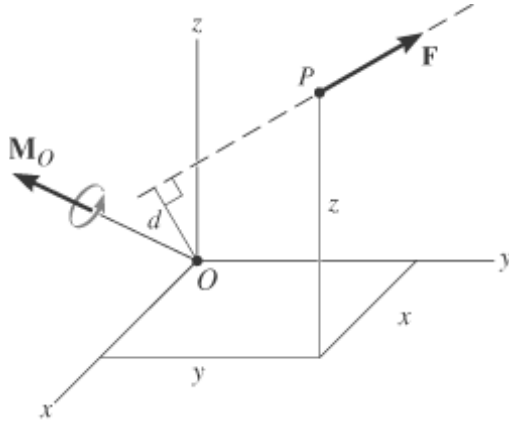
The force $\mathbf{F}$ creates a moment about point O of $\mathbf{M}_O$. If the force passes through a point having the given x coordinate, determine the y and z coordinates of the point. Also, realizing that $M_O = Fd$, determine the perpendicular distance d from point O to the line of action of $\mathbf{F}$.

Given:

$$\mathbf{F} = \begin{pmatrix} 6 \\ 8 \\ 10 \end{pmatrix} \text{N}$$

$$\mathbf{M_O} = \begin{pmatrix} -14 \\ 8 \\ 2 \end{pmatrix} \text{N}\cdot\text{m}$$

$x = 1$ m



Solution:

The initial guesses: $y = 1$ m $z = 1$ m

Given

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \mathbf{F} = \mathbf{M_O} \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \text{Find}(y, z) \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \text{m}$$

$$d = \frac{|\mathbf{M_O}|}{|\mathbf{F}|} \qquad d = 1.149 \text{ m}$$

Problem 4-50

The force $\mathbf{F}$ produces a moment $\mathbf{M}_O$ about the origin of coordinates, point O. If the force acts at a point having the given x-coordinate, determine the y and z coordinates.

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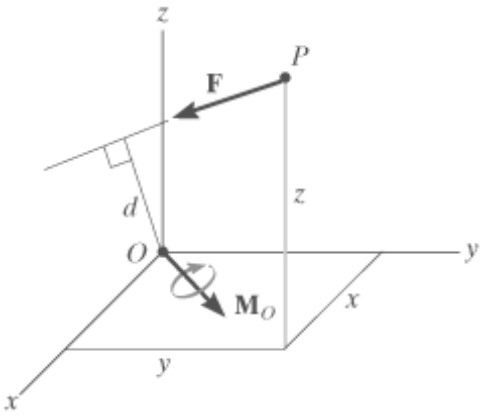
Units Used:

$kN = 10^3 N$

Given:

$x = 1\ m$

$\mathbf{F} = \begin{pmatrix} 6 \\ -2 \\ 1 \end{pmatrix} kN$



$\mathbf{M_O} = \begin{pmatrix} 4 \\ 5 \\ -14 \end{pmatrix} kN \cdot m$

Solution:

Initial Guesses: $y = 1\ m \qquad z = 1\ m$

Given $\qquad \mathbf{M_O} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \mathbf{F} \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \text{Find}(y, z) \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} m$

Problem 4-51

Determine the moment of the force **F** about the *Oa* axis. Express the result as a Cartesian vector.

Given:

$\mathbf{F} = \begin{pmatrix} 50 \\ -20 \\ 20 \end{pmatrix} N$

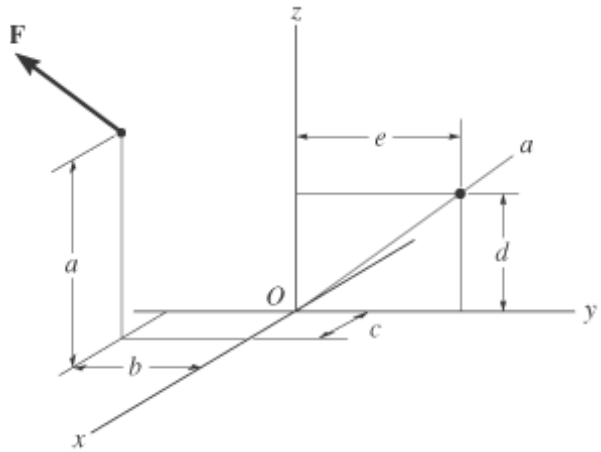
$a = 6\ m$

$b = 2\ m$

$c = 1\ m$

$d = 3\ m$

$e = 4\ m$



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Solution:

$$\mathbf{r_{OF}} = \begin{pmatrix} c \\ -b \\ a \end{pmatrix} \qquad \mathbf{r_{Oa}} = \begin{pmatrix} 0 \\ e \\ d \end{pmatrix} \qquad \mathbf{u_{Oa}} = \frac{\mathbf{r_{Oa}}}{|\mathbf{r_{Oa}}|}$$

$$\mathbf{M_{Oa}} = \left[\left(\mathbf{r_{OF}} \times \mathbf{F}\right) \cdot \mathbf{u_{Oa}}\right]\mathbf{u_{Oa}} \qquad \mathbf{M_{Oa}} = \begin{pmatrix} 0 \\ 217.6 \\ 163.2 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-52

Determine the moment of the force **F** about the *aa* axis. Express the result as a Cartesian vector.

Given:

$F = 600$ lb

$a = 6$ ft

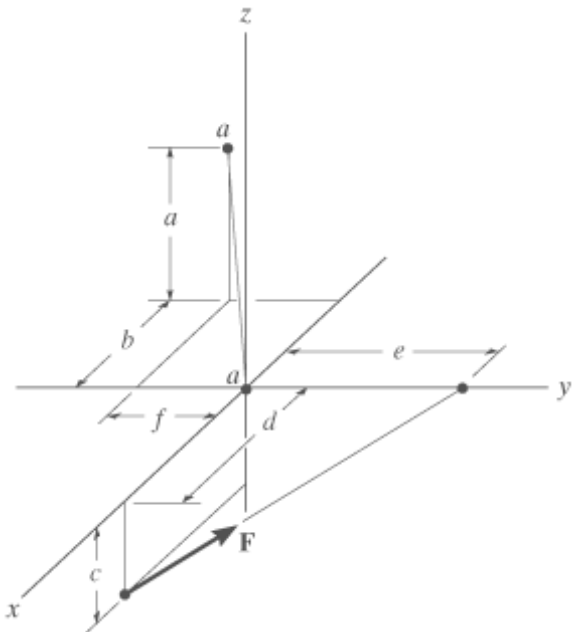
$b = 3$ ft

$c = 2$ ft

$d = 4$ ft

$e = 4$ ft

$f = 2$ ft



Solution:

$$\mathbf{F_v} = \frac{F}{\sqrt{c^2 + d^2 + e^2}}\begin{pmatrix} -d \\ e \\ c \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} d \\ 0 \\ -c \end{pmatrix} \qquad \mathbf{u_{aa}} = \frac{1}{\sqrt{a^2 + b^2 + f^2}}\begin{pmatrix} -b \\ -f \\ a \end{pmatrix}$$

$$\mathbf{M_{aa}} = \left[\left(\mathbf{r} \times \mathbf{F_v}\right) \cdot \mathbf{u_{aa}}\right]\mathbf{u_{aa}} \qquad \mathbf{M_{aa}} = \begin{pmatrix} -441 \\ -294 \\ 882 \end{pmatrix} \text{lb}\cdot\text{ft}$$

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Problem 4-53

Determine the resultant moment of the two forces about the *Oa* axis. Express the result as a Cartesian vector.

Given:

$F_1 = 80$ lb

$F_2 = 50$ lb

$\alpha = 120$ deg

$\beta = 60$ deg

$\gamma = 45$ deg

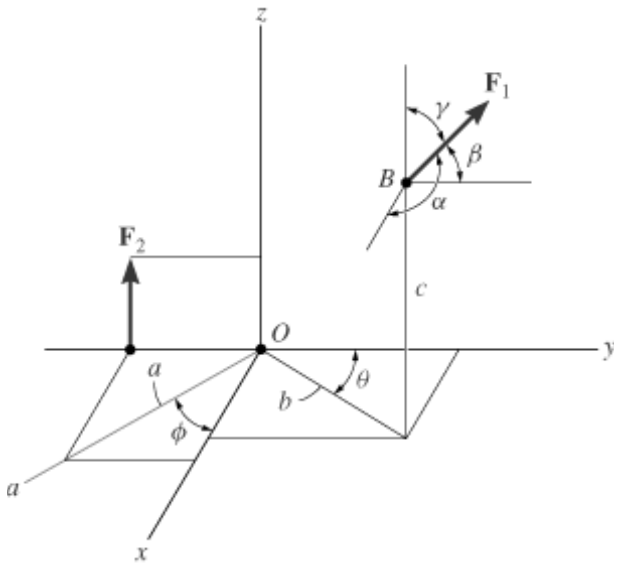
$a = 5$ ft

$b = 4$ ft

$c = 6$ ft

$\theta = 30$ deg

$\phi = 30$ deg



Solution:

$$\mathbf{F_{1v}} = F_1 \begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ 0 \\ F_2 \end{pmatrix}$$

$$\mathbf{r_1} = \begin{bmatrix} (b)\sin(\theta) \\ (b)\cos(\theta) \\ c \end{bmatrix} \qquad \mathbf{r_2} = \begin{bmatrix} 0 \\ -(a)\sin(\phi) \\ 0 \end{bmatrix}$$

$$\mathbf{u_{aa}} = \begin{pmatrix} \cos(\phi) \\ -\sin(\phi) \\ 0 \end{pmatrix} \qquad \mathbf{M_{aa}} = \left[\left(\mathbf{r_1} \times \mathbf{F_{1v}} + \mathbf{r_2} \times \mathbf{F_{2v}}\right)\mathbf{u_{aa}}\right]\mathbf{u_{aa}}$$

$$\mathbf{M_{aa}} = \begin{pmatrix} 26.132 \\ -15.087 \\ 0 \end{pmatrix} \text{lb}\cdot\text{ft}$$

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Problem 4-54

The force **F** is applied to the handle of the box wrench. Determine the component of the moment of this force about the *z* axis which is effective in loosening the bolt.

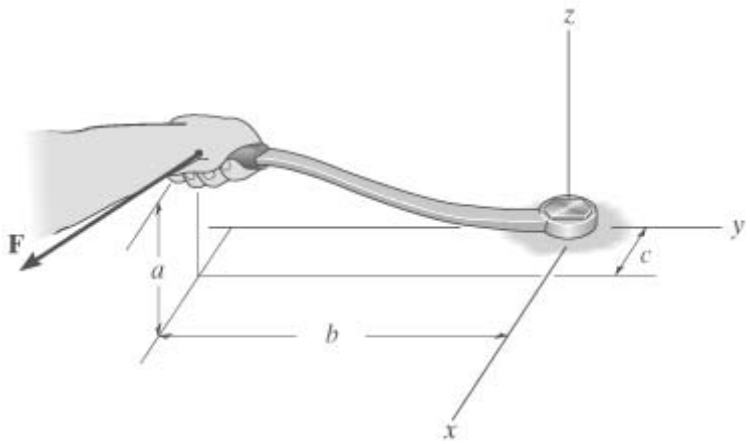
Given:

$a = 3$ in

$b = 8$ in

$c = 2$ in

$$\mathbf{F} = \begin{pmatrix} 8 \\ -1 \\ 1 \end{pmatrix} \text{ lb}$$



Solution:

$$\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} c \\ -b \\ a \end{pmatrix} \qquad M_Z = (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{k} \qquad M_Z = 62 \,\text{lb} \cdot \text{in}$$

Problem 4-55

The force **F** acts on the gear in the direction shown. Determine the moment of this force about the *y* axis.

Given:

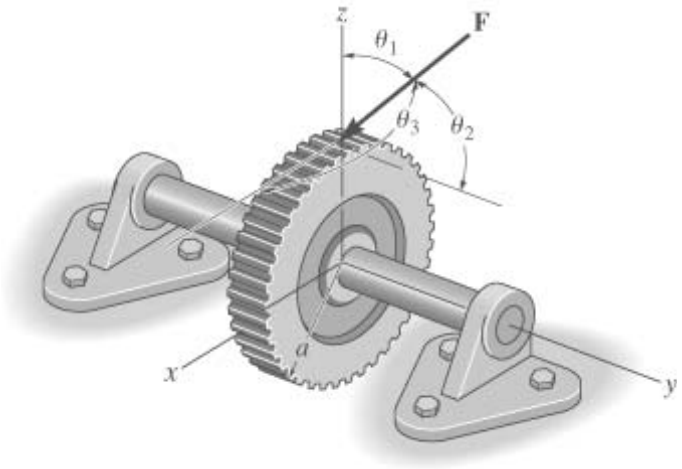
$F = 50$ lb

$a = 3$ in

$\theta_1 = 60$ deg

$\theta_2 = 45$ deg

$\theta_3 = 120$ deg



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Solution:

$$\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix} \quad \mathbf{F_v} = F \begin{pmatrix} -\cos(\theta_3) \\ -\cos(\theta_2) \\ -\cos(\theta_1) \end{pmatrix} \quad M_y = (\mathbf{r} \times \mathbf{F_v}) \cdot \mathbf{j} \qquad M_y = 75\,\text{lb}\cdot\text{in}$$

Problem 4-56

The RollerBall skate is an in-line tandem skate that uses two large spherical wheels on each skate, rather than traditional wafer-shape wheels. During skating the two forces acting on the wheel of one skate consist of a normal force $\mathbf{F}_2$ and a friction force $\mathbf{F}_1$. Determine the moment of both of these forces about the axle *AB* of the wheel.

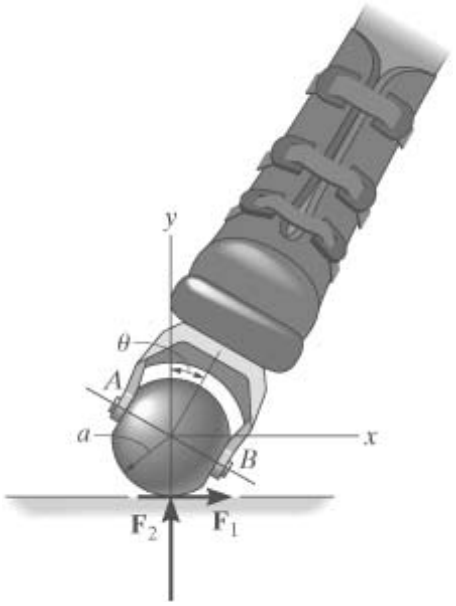
Given:

$\theta = 30$ deg

$F_1 = 13$ lb

$F_2 = 78$ lb

$a = 1.25$ in



Solution:

$$\mathbf{F} = \begin{pmatrix} F_1 \\ F_2 \\ 0 \end{pmatrix} \quad \mathbf{r} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix}$$

$$\mathbf{ab} = \begin{pmatrix} \cos(\theta) \\ -\sin(\theta) \\ 0 \end{pmatrix} \qquad M_{ab} = (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{ab} \qquad M_{ab} = 0\,\text{lb}\cdot\text{in}$$

Problem 4-57

The cutting tool on the lathe exerts a force $\mathbf{F}$ on the shaft in the direction shown. Determine the moment of this force about the *y* axis of the shaft.

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Units Used:

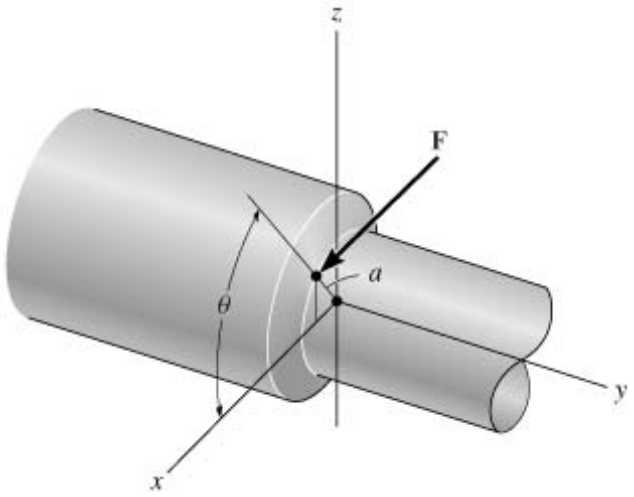
$kN = 10^3 N$

Given:

$\mathbf{F} = \begin{pmatrix} 6 \\ -4 \\ -7 \end{pmatrix} kN$

$a = 30 \text{ mm}$

$\theta = 40 \text{ deg}$



Solution:

$\mathbf{r} = a \begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix}$ $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $M_y = (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{j}$ $M_y = 0.277 \text{ kN} \cdot \text{m}$

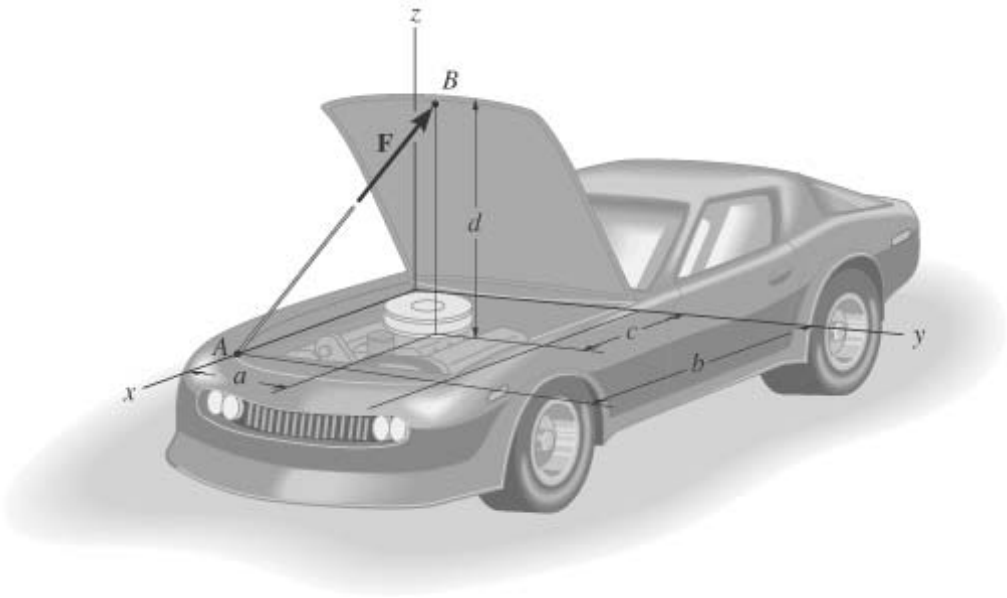
Problem 4-58

The hood of the automobile is supported by the strut *AB*, which exerts a force **F** on the hood. Determine the moment of this force about the hinged axis *y*.

Given:

$F = 24 \text{ lb}$ $a = 2 \text{ ft}$ $b = 4 \text{ ft}$ $c = 2 \text{ ft}$ $d = 4 \text{ ft}$

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Solution:

$$\mathbf{r_A} = \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -b + c \\ a \\ d \end{pmatrix} \qquad \mathbf{F_v} = F \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_v} = \begin{pmatrix} -9.798 \\ 9.798 \\ 19.596 \end{pmatrix} \text{lb}$$

$$\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad M_y = (\mathbf{r_A} \times \mathbf{F_v}) \cdot \mathbf{j} \qquad M_y = -78.384 \,\text{lb} \cdot \text{ft}$$

Problem 4-59

The lug nut on the wheel of the automobile is to be removed using the wrench and applying the vertical force $\mathbf{F}$ at A. Determine if this force is adequate, provided a torque M about the x axis is initially required to turn the nut. If the force $\mathbf{F}$ can be applied at A in any other direction, will it be possible to turn the nut?

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Given:

$F = 30$ N

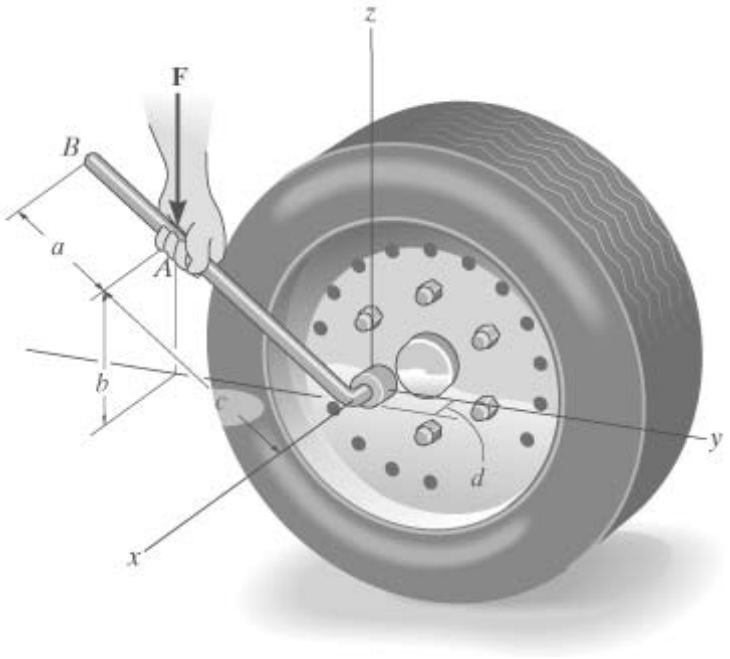
$M = 14$ N·m

$a = 0.25$ m

$b = 0.3$ m

$c = 0.5$ m

$d = 0.1$ m



Solution:

$$M_X = F\sqrt{c^2 - b^2}$$

$$M_X = 12\,\text{N}\cdot\text{m}$$

$M_X < M$ No

For M_{xmax}, apply force perpendicular to the handle and the x-axis.

$$M_{xmax} = F\,c$$

$$M_{xmax} = 15\,\text{N}\cdot\text{m}$$

$M_{xmax} > M$ Yes

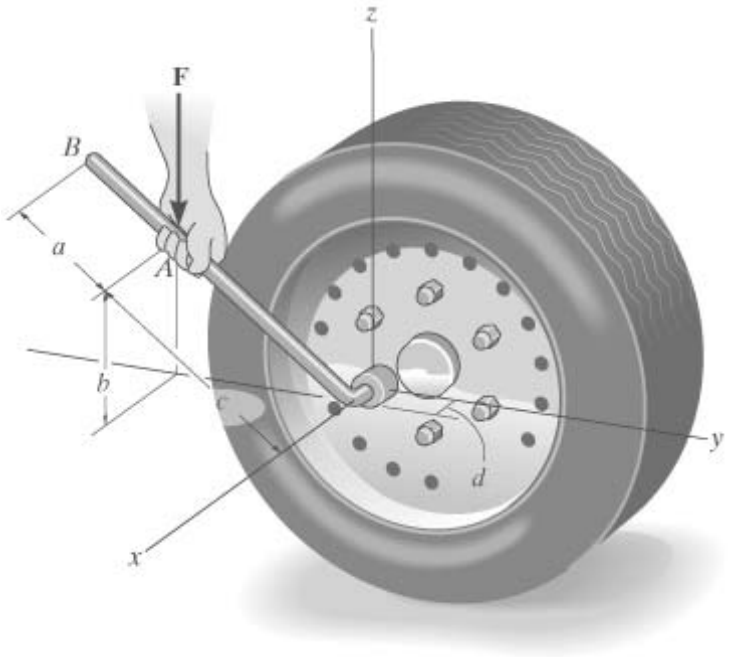
Problem 4-60

The lug nut on the wheel of the automobile is to be removed using the wrench and applying the vertical force **F**. Assume that the cheater pipe *AB* is slipped over the handle of the wrench and the **F** force can be applied at any point and in any direction on the assembly. Determine if this force is adequate, provided a torque *M* about the *x* axis is initially required to turn the nut.

Given:

$F_1 = 30$ N $M = 14$ N·m $a = 0.25$ m $b = 0.3$ m $c = 0.5$ m $d = 0.1$ m

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Solution:

$$M_x = F_1 \frac{a+c}{c}\sqrt{c^2 - b^2}$$

$$M_x = 18\,\text{N}\cdot\text{m}$$

$M_x > M$ Yes

M_{xmax} occurs when force is applied perpendicular to both the handle and the x-axis.

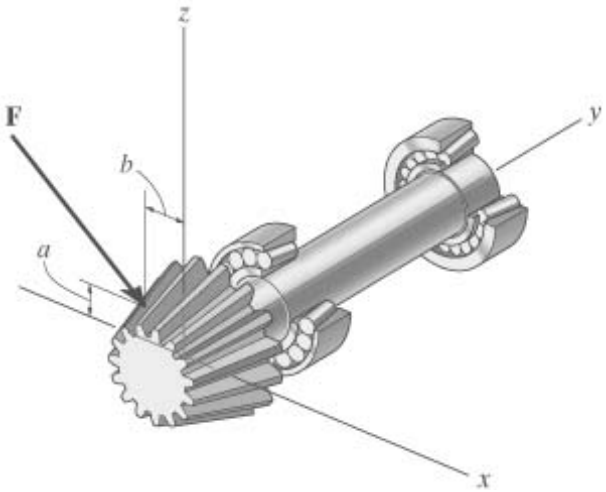
$$M_{xmax} = F_1(a+c)$$

$$M_{xmax} = 22.5\,\text{N}\cdot\text{m}$$

$M_{xmax} > M$ Yes

Problem 4-61

The bevel gear is subjected to the force **F** which is caused from contact with another gear. Determine the moment of this force about the y axis of the gear shaft.



Given:

$a = 30$ mm

$b = 40$ mm

$$\mathbf{F} = \begin{pmatrix} 20 \\ 8 \\ -15 \end{pmatrix} \text{N}$$

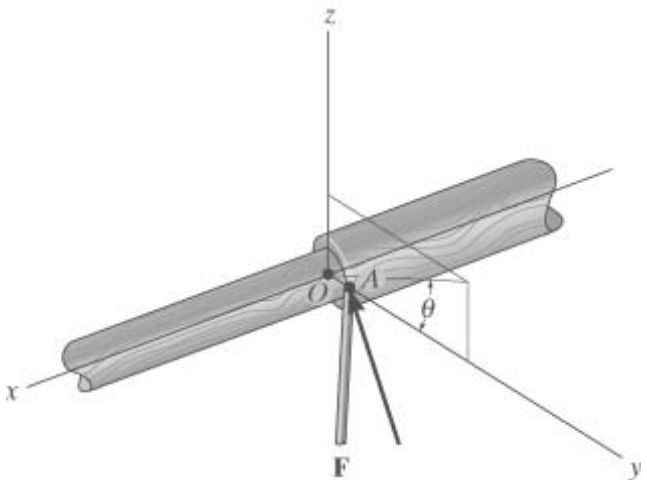
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Solution:

$$\mathbf{r} = \begin{pmatrix} -b \\ 0 \\ a \end{pmatrix} \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad M_y = (\mathbf{r} \times \mathbf{F}) \cdot \mathbf{j} \qquad M_y = 0\,\text{N}\cdot\text{m}$$

Problem 4-62

The wooden shaft is held in a lathe. The cutting tool exerts force **F** on the shaft in the direction shown. Determine the moment of this force about the *x* axis of the shaft. Express the result as a Cartesian vector. The distance *OA* is *a*.



Given:

$a = 25$ mm

$\theta = 30$ deg

$$\mathbf{F} = \begin{pmatrix} -5 \\ -3 \\ 8 \end{pmatrix} \text{N}$$

Solution:

$$\mathbf{r} = \begin{bmatrix} 0 \\ (a)\cos(\theta) \\ (a)\sin(\theta) \end{bmatrix} \qquad \mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{M_x} = \left[(\mathbf{r} \times \mathbf{F}) \cdot \mathbf{i}\right]\mathbf{i} \qquad \mathbf{M_x} = \begin{pmatrix} 0.211 \\ 0 \\ 0 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-63

Determine the magnitude of the moment of the force **F** about the base line *CA* of the tripod.

Given:

$$\mathbf{F} = \begin{pmatrix} 50 \\ -20 \\ -80 \end{pmatrix} \text{N}$$

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$a = 4$ m

$b = 2.5$ m

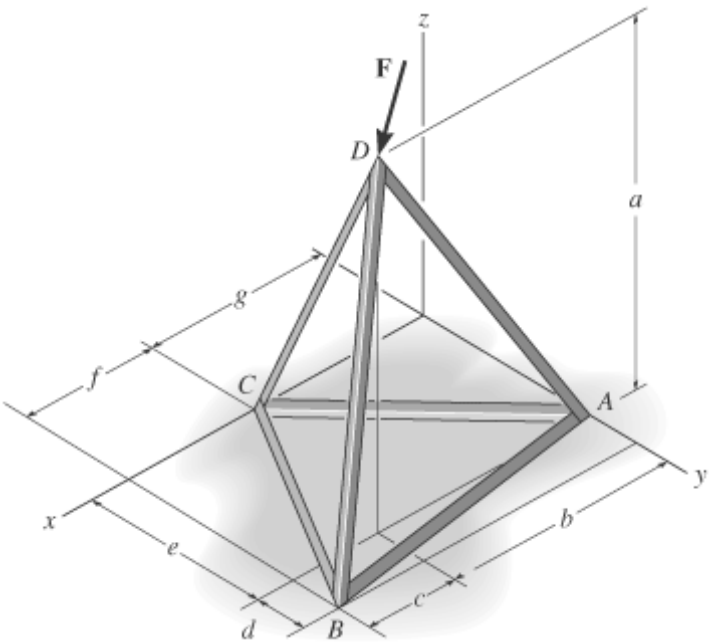
$c = 1$ m

$d = 0.5$ m

$e = 2$ m

$f = 1.5$ m

$g = 2$ m

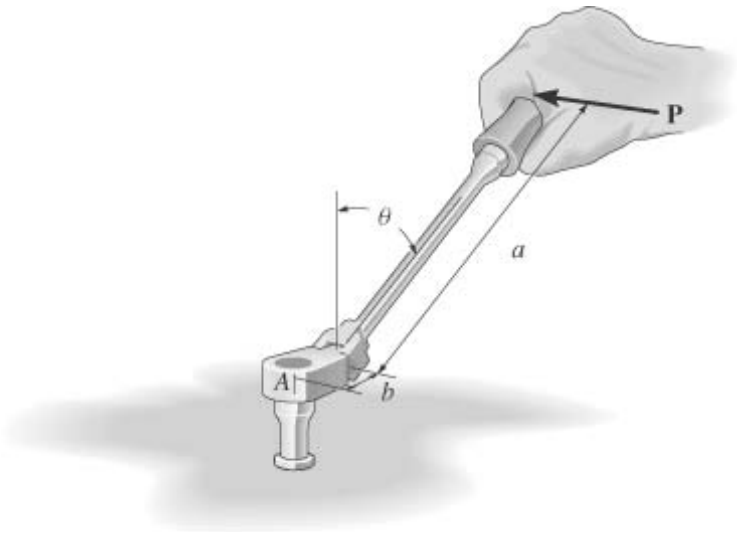


Solution:

$$\mathbf{r_{CA}} = \begin{pmatrix} -g \\ e \\ 0 \end{pmatrix} \quad \mathbf{u_{CA}} = \frac{\mathbf{r_{CA}}}{|\mathbf{r_{CA}}|} \quad \mathbf{r_{CD}} = \begin{pmatrix} b-g \\ e \\ a \end{pmatrix} \quad M_{CA} = (\mathbf{r_{CD}} \times \mathbf{F}) \cdot \mathbf{u_{CA}} \qquad M_{CA} = 226\,\text{N}\cdot\text{m}$$

Problem 4-64

The flex-headed ratchet wrench is subjected to force **P**, applied perpendicular to the handle as shown. Determine the moment or torque this imparts along the vertical axis of the bolt at *A*.



Given:

$P = 16$ lb $\quad a = 10$ in

$\theta = 60$ deg $\quad b = 0.75$ in

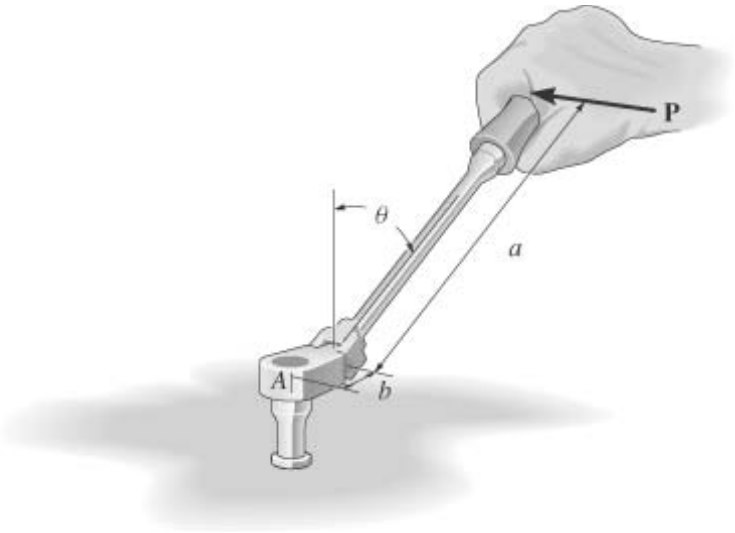
Solution:

$$M = P\left[b + (a)\sin(\theta)\right] \qquad M = 150.564\,\text{lb}\cdot\text{in}$$

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Problem 4-65

If a torque or moment M is required to loosen the bolt at A, determine the force **P** that must be applied perpendicular to the handle of the flex-headed ratchet wrench.



Given:

$M = 80\ \text{lb}\cdot\text{in}$

$\theta = 60\ \text{deg}$

$a = 10\ \text{in}$

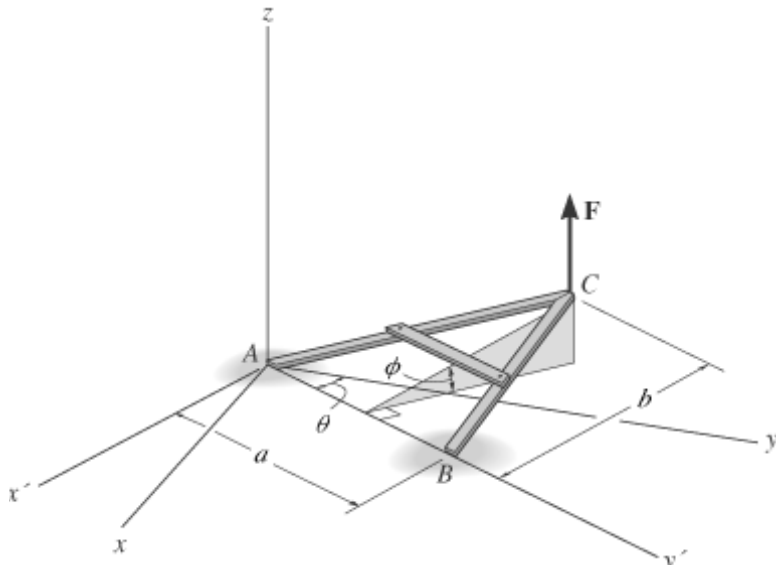
$b = 0.75\ \text{in}$

Solution:

$$M = P\left[b + (a)\sin(\theta)\right] \qquad P = \frac{M}{b + (a)\sin(\theta)} \qquad P = 8.50\,\text{lb}$$

Problem 4-66

The A-frame is being hoisted into an upright position by the vertical force **F**. Determine the moment of this force about the y axis when the frame is in the position shown.



Given:

$F = 80\ \text{lb}$

$a = 6\ \text{ft}$

$b = 6\ \text{ft}$

$\theta = 30\ \text{deg}$

$\phi = 15\ \text{deg}$

Solution:

Using the primed coordinates we have

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$$\mathbf{j} = \begin{pmatrix} -\sin(\theta) \\ \cos(\theta) \\ 0 \end{pmatrix} \qquad \mathbf{F_v} = F \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -b\cos(\phi) \\ \frac{a}{2} \\ b\sin(\phi) \end{pmatrix}$$

$$M_y = (\mathbf{r_{AC}} \times \mathbf{F_v}) \cdot \mathbf{j} \qquad M_y = 281.528\,\text{lb}\cdot\text{ft}$$

Problem 4-67

Determine the moment of each force acting on the handle of the wrench about the a axis.

Given:

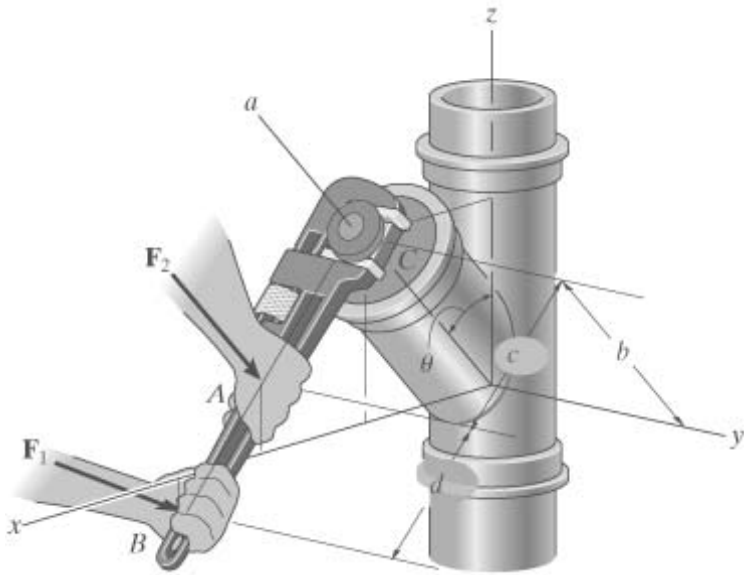
$$\mathbf{F_1} = \begin{pmatrix} -2 \\ 4 \\ -8 \end{pmatrix} \text{lb} \qquad \mathbf{F_2} = \begin{pmatrix} 3 \\ 2 \\ -6 \end{pmatrix} \text{lb}$$

$b = 6$ in

$c = 4$ in

$d = 3.5$ in

$\theta = 45$ deg



Solution:

$$\mathbf{u_a} = \begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix}$$

$$\mathbf{r_1} = b\begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix} + (c+d)\begin{pmatrix} \sin(\theta) \\ 0 \\ -\cos(\theta) \end{pmatrix} \qquad \mathbf{r_2} = b\begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix} + c\begin{pmatrix} \sin(\theta) \\ 0 \\ -\cos(\theta) \end{pmatrix}$$

$$M_{1a} = (\mathbf{r_1} \times \mathbf{F_1}) \cdot \mathbf{u_a} \qquad M_{1a} = 30\,\text{lb}\cdot\text{in}$$

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$M_{2a} = (\mathbf{r_2} \times \mathbf{F_2}) \cdot \mathbf{u_a}$ $M_{2a} = 8\,\text{lb}\cdot\text{in}$

Problem 4-68

Determine the moment of each force acting on the handle of the wrench about the z axis.

Given:

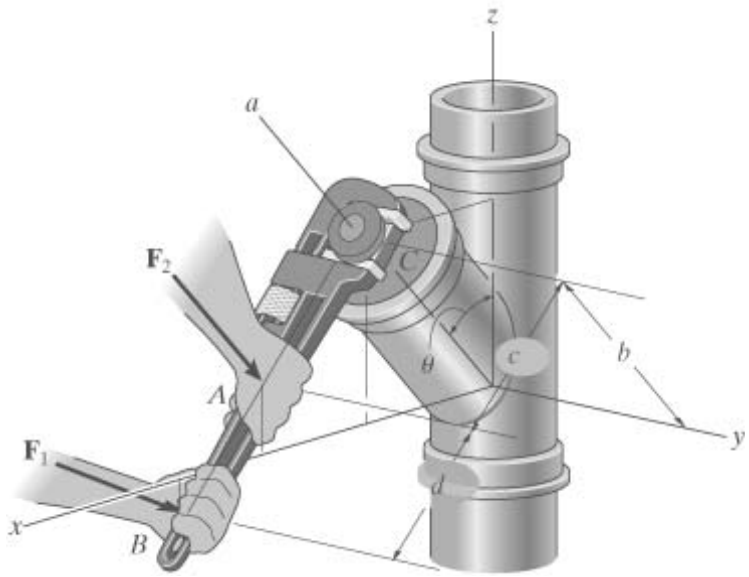
$$\mathbf{F_1} = \begin{pmatrix} -2 \\ 4 \\ -8 \end{pmatrix} \text{lb} \qquad \mathbf{F_2} = \begin{pmatrix} 3 \\ 2 \\ -6 \end{pmatrix} \text{lb}$$

$b = 6$ in

$c = 4$ in

$d = 3.5$ in

$\theta = 45$ deg



Solution:

$$\mathbf{r_1} = b\begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix} + (c+d)\begin{pmatrix} \sin(\theta) \\ 0 \\ -\cos(\theta) \end{pmatrix} \qquad \mathbf{r_2} = b\begin{pmatrix} \cos(\theta) \\ 0 \\ \sin(\theta) \end{pmatrix} + c\begin{pmatrix} \sin(\theta) \\ 0 \\ -\cos(\theta) \end{pmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$M_{1z} = (\mathbf{r_1} \times \mathbf{F_1}) \cdot \mathbf{k}$ $M_{1z} = 38.2\,\text{lb}\cdot\text{in}$

$M_{2z} = (\mathbf{r_2} \times \mathbf{F_2}) \cdot \mathbf{k}$ $M_{2z} = 14.1\,\text{lb}\cdot\text{in}$

Problem 4-69

Determine the magnitude and sense of the couple moment.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

$F = 5$ kN

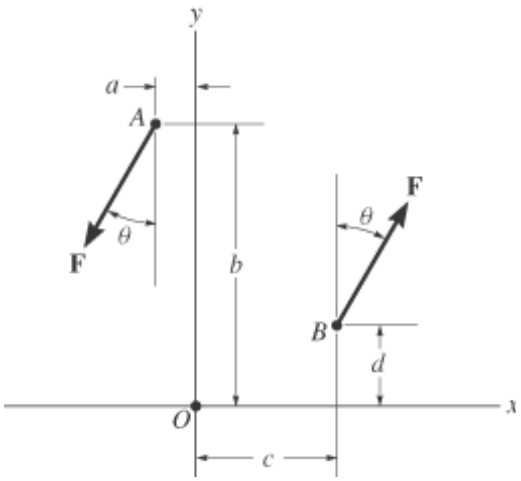
$\theta = 30$ deg

$a = 0.5$ m

$b = 4$ m

$c = 2$ m

$d = 1$ m



Solution:

$$\curvearrowright M_C = F\cos(\theta)(a + c) + F\sin(\theta)(b - d)$$

$$M_C = 18.325\,\text{kN}\cdot\text{m}$$

Problem 4-70

Determine the magnitude and sense of the couple moment. Each force has a magnitude F.

Given:

$F = 65$ lb

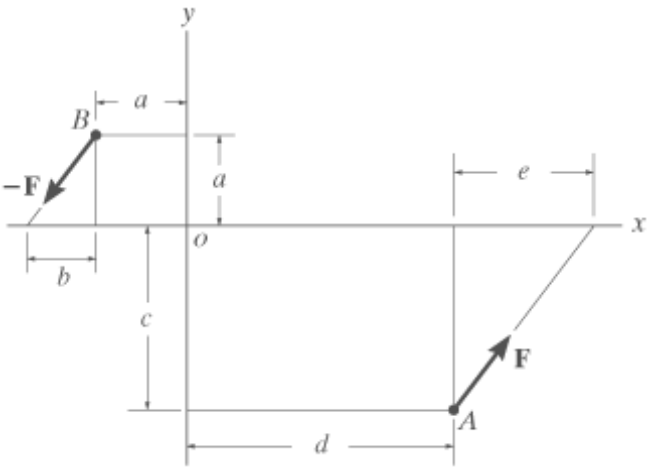
$a = 2$ ft

$b = 1.5$ ft

$c = 4$ ft

$d = 6$ ft

$e = 3$ ft



Solution:

$$M_c = \Sigma M_{B;} \qquad M_C = \left[F\left(\frac{c}{\sqrt{c^2 + e^2}}\right)(d + a)\right] + \left[F\left(\frac{e}{\sqrt{c^2 + e^2}}\right)(c + a)\right]$$

$$M_C = 650\,\text{lb}\cdot\text{ft} \quad \text{(Counterclockwise)}$$

Problem 4-71

Determine the magnitude and sense of the couple moment.

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Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F = 150 \text{ lb}$

$a = 8 \text{ ft}$

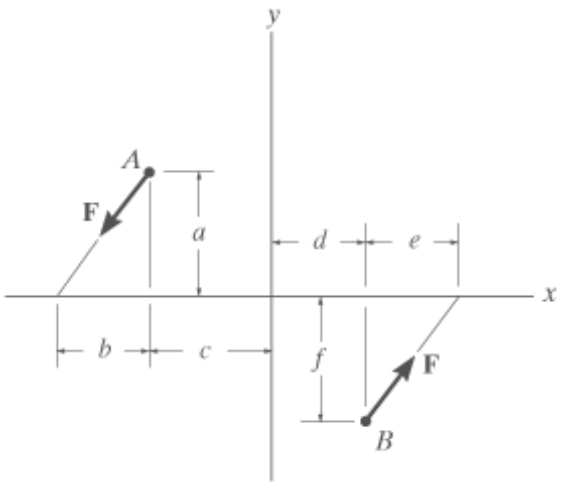
$b = 6 \text{ ft}$

$c = 8 \text{ ft}$

$d = 6 \text{ ft}$

$e = 6 \text{ ft}$

$f = 8 \text{ ft}$



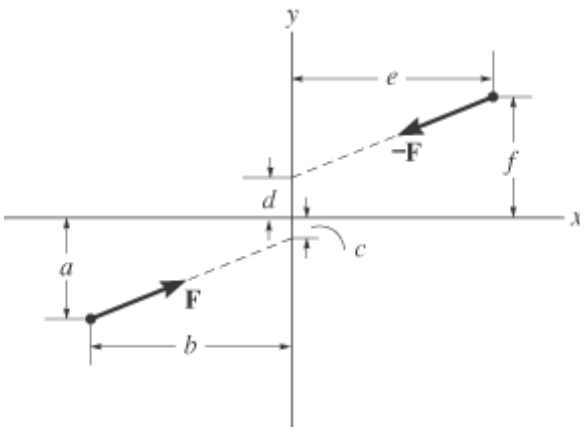
Solution:

$$\circlearrowleft + M_C = \Sigma M_A; \qquad M_C = F\frac{d}{\sqrt{d^2 + f^2}}(a + f) + F\frac{f}{\sqrt{d^2 + f^2}}(c + d)$$

$$M_C = 3120 \,\text{lb}\cdot\text{ft} \qquad M_C = 3.120 \,\text{kip}\cdot\text{ft}$$

Problem 4-72

If the couple moment has magnitude M, determine the magnitude F of the couple forces.



Given:

$M = 300 \text{ lb}\cdot\text{ft}$

$a = 6 \text{ ft}$

$b = 12 \text{ ft}$

$c = 1 \text{ ft}$

$d = 2 \text{ ft}$

$e = 12 \text{ ft}$

$f = 7 \text{ ft}$

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Solution:

$$M = F\left[\frac{e(f+a)}{\sqrt{(f-d)^2+e^2}} - \frac{(f-d)(b+e)}{\sqrt{(f-d)^2+e^2}}\right]$$

$$F = \frac{M}{\dfrac{e(f+a)}{\sqrt{(f-d)^2+e^2}} - \dfrac{(f-d)(b+e)}{\sqrt{(f-d)^2+e^2}}} \qquad F = 108\,\text{lb}$$

Problem 4-73

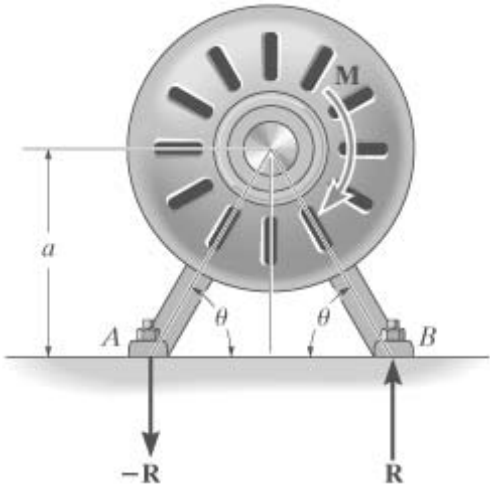
A clockwise couple M is resisted by the shaft of the electric motor. Determine the magnitude of the reactive forces $-\mathbf{R}$ and $\mathbf{R}$ which act at supports A and B so that the resultant of the two couples is zero.

Given:

$a = 150$ mm

$\theta = 60$ deg

$M = 5$ N·m



Solution:

$$\circlearrowleft + \quad M_C = -M + \frac{2R\,a}{\tan(\theta)} = 0 \qquad R = \frac{M}{2}\,\frac{\tan(\theta)}{a} \qquad R = 28.9\,\text{N}$$

Problem 4-74

The resultant couple moment created by the two couples acting on the disk is $\mathbf{M_R}$. Determine the magnitude of force $\mathbf{T}$.

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Units Used:

$kip = 10^3\ lb$

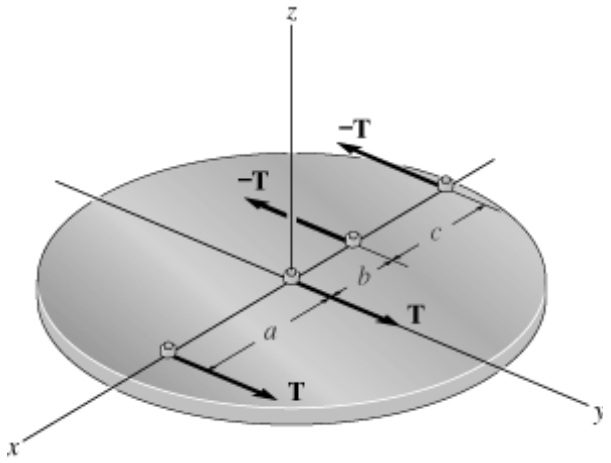
Given:

$$\mathbf{M_R} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} kip \cdot in$$

$a = 4\ in$

$b = 2\ in$

$c = 3\ in$



Solution:

Initial Guess $T = 1\ kip$

Given $$\begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ T \\ 0 \end{pmatrix} + \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -T \\ 0 \end{pmatrix} + \begin{pmatrix} -b-c \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -T \\ 0 \end{pmatrix} = \mathbf{M_R}$$

$T = \text{Find}(T)$ $T = 0.909\,kip$

Problem 4-75

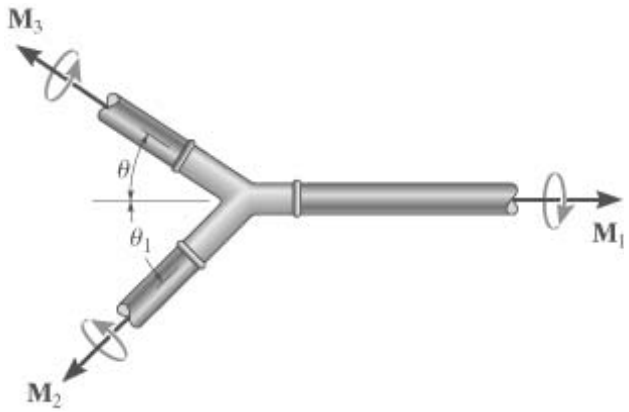
Three couple moments act on the pipe assembly. Determine the magnitude of M_3 and the bend angle θ so that the resultant couple moment is zero.

Given:

$\theta_1 = 45\ deg$

$M_1 = 900\ N \cdot m$

$M_2 = 500\ N \cdot m$



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Solution:

Initial guesses: $\theta = 10 \text{ deg}$ $M_3 = 10 \text{ N·m}$

Given

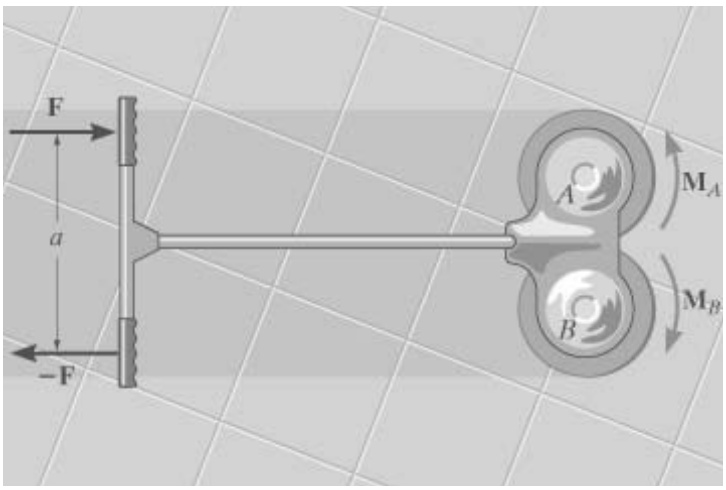
$\overset{+}{\rightarrow} \Sigma M_x = 0;$ $M_1 - M_3 \cos(\theta) - M_2 \cos(\theta_1) = 0$

$+\uparrow \Sigma M_y = 0;$ $M_3 \sin(\theta) - M_2 \sin(\theta_1) = 0$

$\begin{pmatrix} \theta \\ M_3 \end{pmatrix} = \text{Find}(\theta, M_3)$ $\theta = 32.9 \text{ deg}$ $M_3 = 651 \text{ N·m}$

Problem 4-76

The floor causes couple moments M_A and M_B on the brushes of the polishing machine. Determine the magnitude of the couple forces that must be developed by the operator on the handles so that the resultant couple moment on the polisher is zero. What is the magnitude of these forces if the brush at B suddenly stops so that $M_B = 0$?



Given:

$a = 0.3 \text{ m}$

$M_A = 40 \text{ N·m}$

$M_B = 30 \text{ N·m}$

Solution:

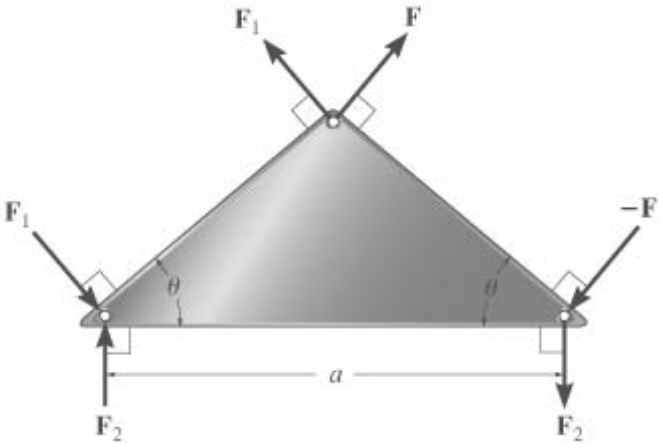
$M_A - M_B - F_1 a = 0$ $F_1 = \dfrac{M_A - M_B}{a}$ $F_1 = 33.3 \text{ N}$

$M_A - F_2 a = 0$ $F_2 = \dfrac{M_A}{a}$ $F_2 = 133 \text{ N}$

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Problem 4-77

The ends of the triangular plate are subjected to three couples. Determine the magnitude of the force **F** so that the resultant couple moment is *M* clockwise.



Given:

$F_1 = 600$ N

$F_2 = 250$ N

$a = 1$ m

$\theta = 40$ deg

$M = 400$ N·m

Solution:

Initial Guess $F = 1$ N

Given $F_1\left(\frac{a}{2\cos(\theta)}\right) - F_2 a - F\left(\frac{a}{2\cos(\theta)}\right) = -M$ $F = \text{Find}(F)$ $F = 830$ N

Problem 4-78

Two couples act on the beam. Determine the magnitude of **F** so that the resultant couple moment is *M* counterclockwise. Where on the beam does the resultant couple moment act?

Given:

$M = 450$ lb·ft

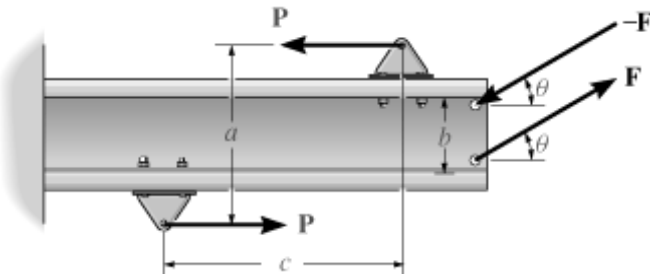
$P = 200$ lb

$a = 1.5$ ft

$b = 1.25$ ft

$c = 2$ ft

$\theta = 30$ deg



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Solution:

$$\curvearrowleft \quad M_R = \Sigma M \qquad M = F\,b\cos(\theta) + P\,a \qquad F = \frac{M - P\,a}{b\cos(\theta)} \qquad F = 139\,\text{lb}$$

The resultant couple moment is a free vector. It can act at any point on the beam.

Problem 4-79

Express the moment of the couple acting on the pipe assembly in Cartesian vector form. Solve the problem **(a)** using Eq. 4-13, and **(b)** summing the moment of each force about point *O*.

Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ 25 \end{pmatrix}\ \text{N}$$

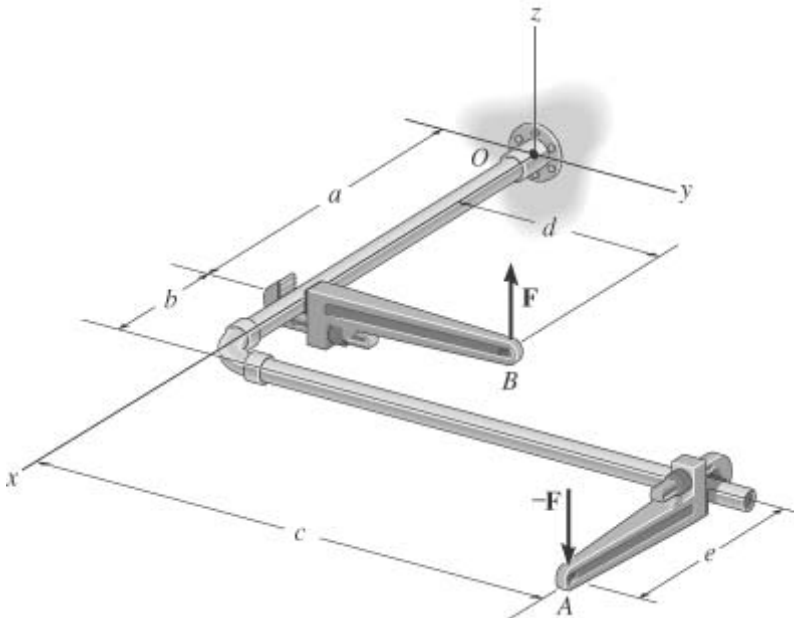
$a = 300$ mm

$b = 150$ mm

$c = 400$ mm

$d = 200$ mm

$e = 200$ mm



Solution:

(a) $\mathbf{r_{AB}} = \begin{pmatrix} -e - b \\ -c + d \\ 0 \end{pmatrix} \qquad \mathbf{M} = \mathbf{r_{AB}} \times \mathbf{F} \qquad \mathbf{M} = \begin{pmatrix} -5 \\ 8.75 \\ 0 \end{pmatrix}\ \text{N}\cdot\text{m}$

(b) $\mathbf{r_{OB}} = \begin{pmatrix} a \\ d \\ 0 \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} a + b + e \\ c \\ 0 \end{pmatrix}$

$$\mathbf{M} = \mathbf{r_{OB}} \times \mathbf{F} + \mathbf{r_{OA}} \times (-\mathbf{F}) \qquad \mathbf{M} = \begin{pmatrix} -5 \\ 8.75 \\ 0 \end{pmatrix}\ \text{N}\cdot\text{m}$$

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Problem 4-80

If the couple moment acting on the pipe has magnitude M, determine the magnitude F of the vertical force applied to each wrench.

Given:

$M = 400\ \text{N}\cdot\text{m}$

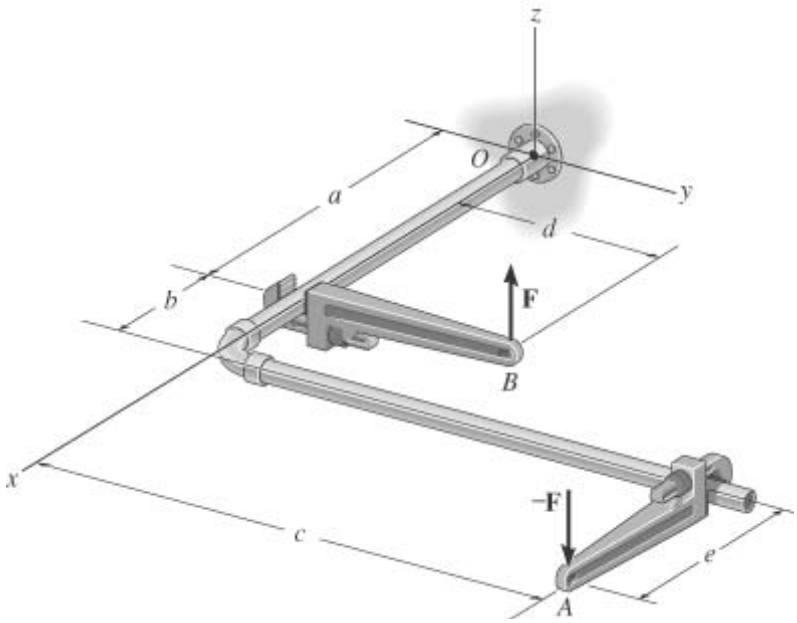
$a = 300\ \text{mm}$

$b = 150\ \text{mm}$

$c = 400\ \text{mm}$

$d = 200\ \text{mm}$

$e = 200\ \text{mm}$



Solution:

$$\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{r_{AB}} = \begin{pmatrix} -e - b \\ -c + d \\ 0 \end{pmatrix}$$

Guesss $F = 1\ \text{N}$

Given $\left|\mathbf{r_{AB}} \times (F\mathbf{k})\right| = M$ $F = \text{Find}(F)$ $F = 992.278\ \text{N}$

Problem 4-81

Determine the resultant couple moment acting on the beam. Solve the problem two ways: (a) sum moments about point *O*; and (b) sum moments about point *A.*

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

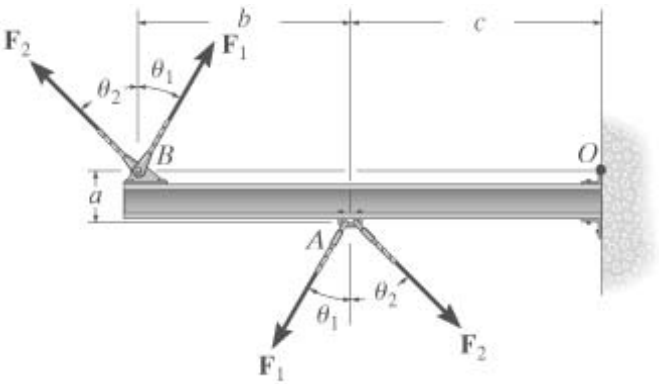
$F_1 = 2$ kN $\quad \theta_1 = 30$ deg

$F_2 = 8$ kN $\quad \theta_2 = 45$ deg

$a = 0.3$ m

$b = 1.5$ m

$c = 1.8$ m



Solution:

(a) $\curvearrowleft M_R = \Sigma M_O;$

$$M_{Ra} = \left(F_1\cos\left(\theta_1\right) + F_2\cos\left(\theta_2\right)\right)c + \left(F_2\cos\left(\theta_2\right) - F_1\sin\left(\theta_1\right)\right)a \ldots$$
$$+ -\left(F_2\cos\left(\theta_2\right) + F_1\cos\left(\theta_1\right)\right)(b + c)$$

$$M_{Ra} = -9.69\,\text{kN}\cdot\text{m}$$

(b) $\curvearrowleft M_R = \Sigma M_A;$

$$M_{Rb} = \left(F_2\sin\left(\theta_2\right) - F_1\sin\left(\theta_1\right)\right)a - \left(F_2\cos\left(\theta_2\right) + F_1\cos\left(\theta_1\right)\right)b$$

$$M_{Rb} = -9.69\,\text{kN}\cdot\text{m}$$

Problem 4-82

Two couples act on the beam as shown. Determine the magnitude of **F** so that the resultant couple moment is M counterclockwise. Where on the beam does the resultant couple act?

Given:

$M = 300$ lb·ft

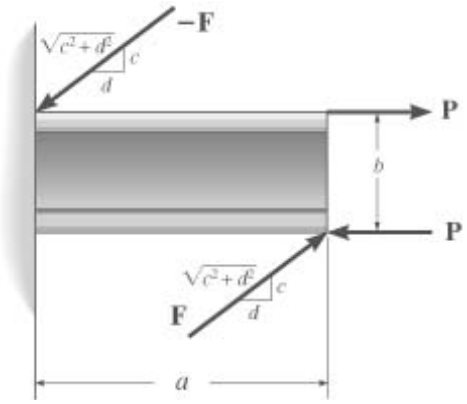
$a = 4$ ft

$b = 1.5$ ft

$P = 200$ lb

$c = 3$

$d = 4$



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Solution:

$$M = \frac{c}{\sqrt{c^2 + d^2}} F\,a + \frac{d}{\sqrt{c^2 + d^2}} F\,b - P\,b$$

$$F = \sqrt{c^2 + d^2}\left(\frac{M + P\,b}{c\,a + d\,b}\right)$$

$F = 167\,\text{lb}$ Resultant couple can act anywhere.

Problem 4-83

Two couples act on the frame. If the resultant couple moment is to be zero, determine the distance d between the couple forces $\mathbf{F_1}$.

Given:

$F_1 = 80$ lb

$F_2 = 50$ lb

$a = 1$ ft

$b = 3$ ft

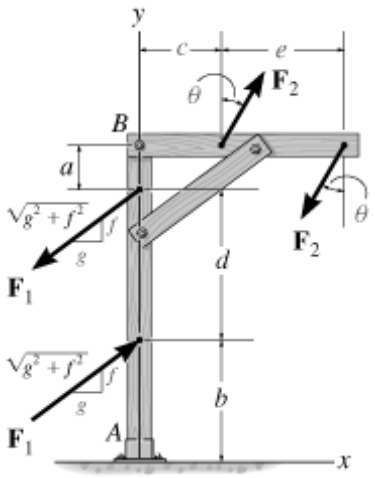
$c = 2$ ft

$e = 3$ ft

$f = 3$

$g = 4$

$\theta = 30$ deg



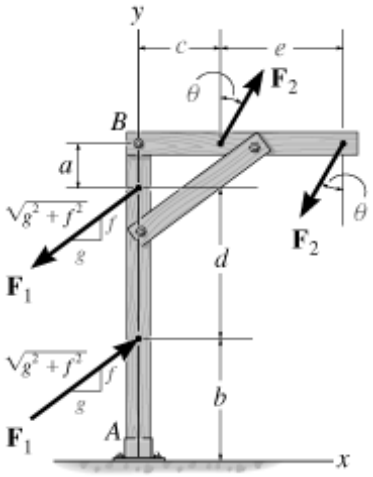
Solution:

$$\left[-F_2\cos(\theta)e + \left(\frac{g}{\sqrt{g^2 + f^2}}\right)F_1\,d\right] = 0 \qquad d = \frac{F_2}{F_1}\cos(\theta)\,e\left(\frac{\sqrt{g^2 + f^2}}{g}\right) \qquad d = 2.03\,\text{ft}$$

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Problem 4-84

Two couples act on the frame. Determine the resultant couple moment. Compute the result by resolving each force into *x* and *y* components and (a) finding the moment of each couple (Eq. 4-13) and (b) summing the moments of all the force components about point *A*.



Given:

$F_1 = 80$ lb $\quad c = 2$ ft $\quad g = 4$

$F_2 = 50$ lb $\quad d = 4$ ft $\quad \theta = 30$ deg

$a = 1$ ft $\quad e = 3$ ft

$b = 3$ ft $\quad f = 3$

Solution:

(a) $\mathbf{M} = \Sigma(\mathbf{r} \times \mathbf{F})$

$$\mathbf{M} = \begin{pmatrix} e \\ 0 \\ 0 \end{pmatrix} \times \left[F_2 \begin{pmatrix} -\sin(\theta) \\ -\cos(\theta) \\ 0 \end{pmatrix} \right] + \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} \times \left[\frac{F_1}{\sqrt{f^2 + g^2}} \begin{pmatrix} -g \\ -f \\ 0 \end{pmatrix} \right] \qquad \mathbf{M} = \begin{pmatrix} 0 \\ 0 \\ 126.096 \end{pmatrix} \text{lb}\cdot\text{ft}$$

(b) Summing the moments of all force components aboout point *A*.

$$M_1 = \left(\frac{-g}{\sqrt{f^2 + g^2}} \right) F_1 b + \left(\frac{g}{\sqrt{f^2 + g^2}} \right) F_1 (b + d)$$

$$M_2 = F_2 \cos(\theta) c - F_2 \sin(\theta)(a + b + d) - F_2 \cos(\theta)(c + e) + F_2 \sin(\theta)(a + b + d)$$

$M = M_1 + M_2 \qquad M = 126.096 \text{ lb}\cdot\text{ft}$

Problem 4-85

Two couples act on the frame. Determine the resultant couple moment. Compute the result by resolving each force into *x* and *y* components and **(a)** finding the moment of each couple (Eq. 4 -13) and **(b)** summing the moments of all the force components about point *B*.

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Given:

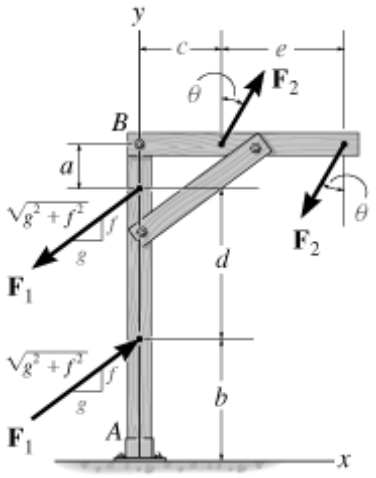
$F_1 = 80$ lb $d = 4$ft

$F_2 = 50$ lb $e = 3$ ft

$a = 1$ ft $f = 3$ ft

$b = 3$ ft $g = 4$ ft

$c = 2$ ft $\theta = 30$ deg



Solution:

(a) $\mathbf{M} = \Sigma(\mathbf{r} \times \mathbf{F})$

$$\mathbf{M} = \begin{pmatrix} e \\ 0 \\ 0 \end{pmatrix} \times \left[F_2 \begin{pmatrix} -\sin(\theta) \\ -\cos(\theta) \\ 0 \end{pmatrix} \right] + \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} \times \left[\frac{F_1}{\sqrt{f^2 + g^2}} \begin{pmatrix} -g \\ -f \\ 0 \end{pmatrix} \right] \qquad \mathbf{M} = \begin{pmatrix} 0 \\ 0 \\ 126.096 \end{pmatrix} \text{lb}\cdot\text{ft}$$

(b) Summing the moments of all force components about point B.

$$M_1 = \left(\frac{g}{\sqrt{f^2 + g^2}} \right) F_1 (a + d) - \left(\frac{g}{\sqrt{f^2 + g^2}} \right) F_1 a$$

$$M_2 = F_2 \cos(\theta) c - F_2 \cos(\theta)(c + e)$$

$$M = M_1 + M_2 \qquad M = 126.096 \text{ lb}\cdot\text{ft}$$

Problem 4-86

Determine the couple moment. Express the result as a Cartesian vector.

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Given:

$$\mathbf{F} = \begin{pmatrix} 8 \\ -4 \\ 10 \end{pmatrix} \text{ N}$$

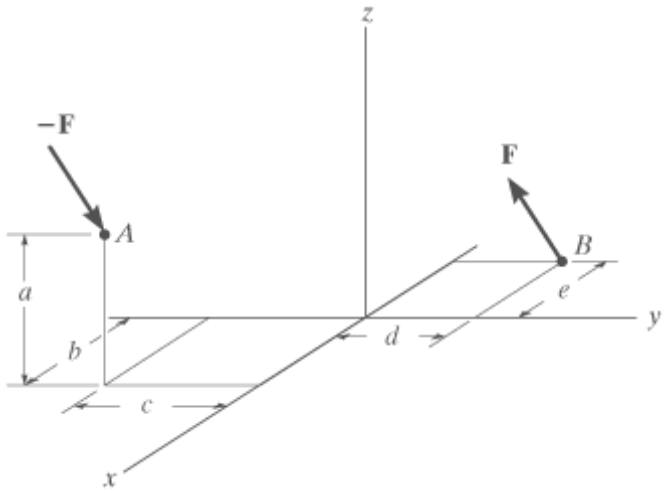
$a = 5$ m

$b = 3$ m

$c = 4$ m

$d = 2$ m

$e = 3$ m



Solution:

$$\mathbf{r} = \begin{pmatrix} -b - e \\ c + d \\ -a \end{pmatrix} \qquad \mathbf{M} = \mathbf{r} \times \mathbf{F} \qquad \mathbf{M} = \begin{pmatrix} 40 \\ 20 \\ -24 \end{pmatrix} \text{ N}\cdot\text{m}$$

Problem 4-87

Determine the couple moment. Express the result as a Cartesian vector.

Given:

$F = 80$ N

$a = 6$ m

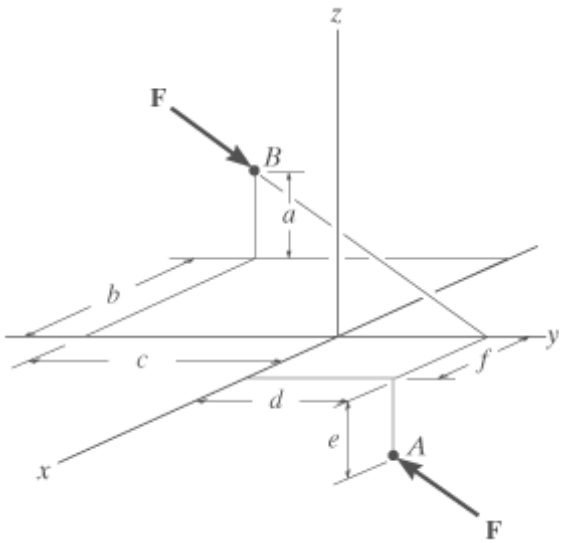
$b = 10$ m

$c = 10$ m

$d = 5$ m

$e = 4$ m

$f = 4$ m



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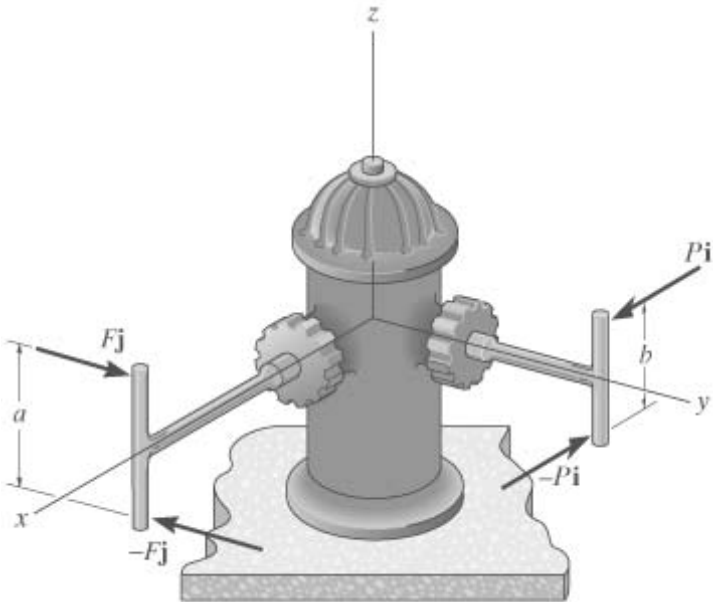
Solution:

$$\mathbf{u} = \frac{1}{\sqrt{a^2 + b^2 + (c+d)^2}} \begin{pmatrix} b \\ c+d \\ -a \end{pmatrix} \qquad \mathbf{F_v} = F\mathbf{u} \qquad \mathbf{r} = \begin{pmatrix} -f-b \\ -d-c \\ e+a \end{pmatrix}$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F_v} \qquad \mathbf{M} = \begin{pmatrix} -252.6 \\ 67.4 \\ -252.6 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-88

If the resultant couple of the two couples acting on the fire hydrant is $\mathbf{M_R} = \{-15\mathbf{i} + 30\mathbf{j}\}$ N·m, determine the force magnitude P.



Given:

$a = 0.2$ m

$b = 0.150$ m

$$\mathbf{M} = \begin{pmatrix} -15 \\ 30 \\ 0 \end{pmatrix} \text{N}\cdot\text{m}$$

$F = 75$ N

Solution:

Initial guess $P = 1$ N

Given

$$\mathbf{M} = \begin{pmatrix} -F\,a \\ P\,b \\ 0 \end{pmatrix} \qquad P = \text{Find}(P) \qquad P = 200 \text{ N}$$

Problem 4-89

If the resultant couple of the three couples acting on the triangular block is to be zero, determine the magnitude of forces **F** and **P**.

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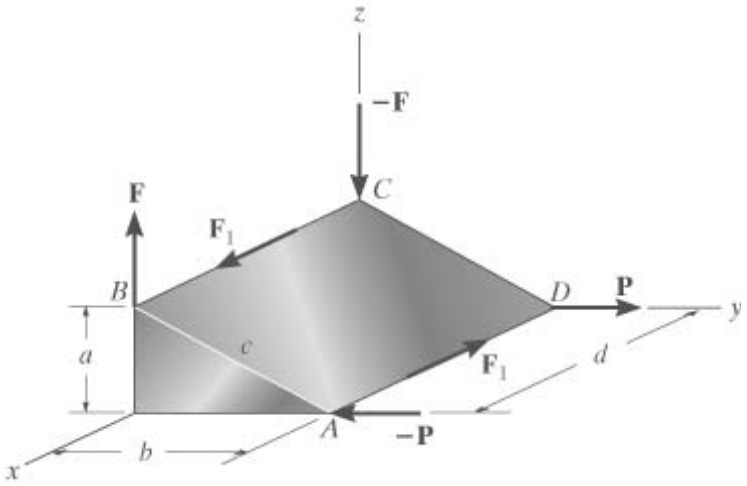
Given:

$F_1 = 150$ N

$a = 300$ mm

$b = 400$ mm

$d = 600$ mm



Solution:

Initial guesses: $F = 1$ N $P = 1$ N

Given $\begin{pmatrix} d \\ 0 \\ a \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ F \end{pmatrix} + \begin{pmatrix} d \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -P \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} -F_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix} \times \begin{pmatrix} F_1 \\ 0 \\ 0 \end{pmatrix} = 0$

$\begin{pmatrix} F \\ P \end{pmatrix} = \text{Find}(F, P)$ $\begin{pmatrix} F \\ P \end{pmatrix} = \begin{pmatrix} 75 \\ 100 \end{pmatrix}$ N

Problem 4-90

Determine the couple moment that acts on the assembly. Express the result as a Cartesian vector. Member *BA* lies in the *x-y* plane.

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Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ 100 \end{pmatrix} \text{N}$$

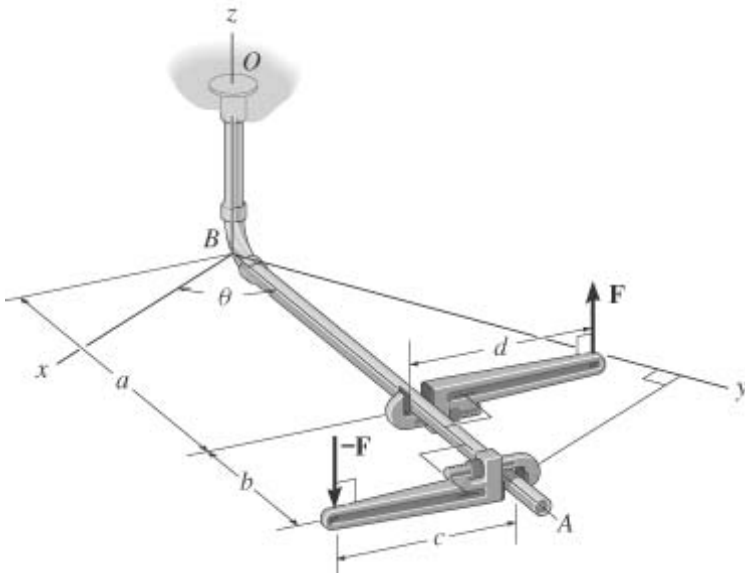
$a = 300$ mm

$b = 150$ mm

$c = 200$ mm

$d = 200$ mm

$\theta = 60$ deg



Solution:

$$\mathbf{r} = \begin{bmatrix} -(c+d)\sin(\theta) - b\cos(\theta) \\ -b\sin(\theta) + (c+d)\cos(\theta) \\ 0 \end{bmatrix} \qquad \mathbf{M} = \mathbf{r} \times \mathbf{F} \qquad \mathbf{M} = \begin{pmatrix} 7.01 \\ 42.14 \\ 0.00 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-91

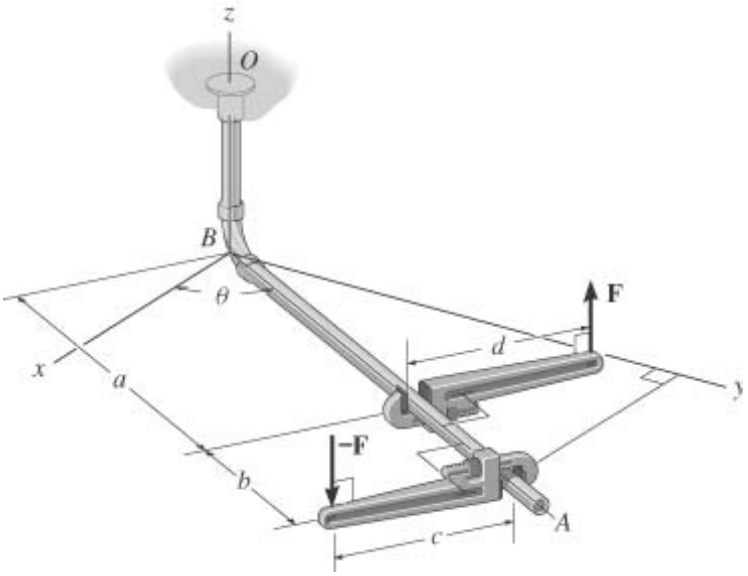
If the magnitude of the resultant couple moment is M, determine the magnitude F of the forces applied to the wrenches.

Given:

$M = 15$ N·m $\quad c = 200$ mm

$a = 300$ mm $\quad d = 200$ mm

$b = 150$ mm $\quad \theta = 60$ deg



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Solution:

$$\mathbf{r} = \begin{bmatrix} -(c+d)\sin(\theta) - b\cos(\theta) \\ -b\sin(\theta) + (c+d)\cos(\theta) \\ 0 \end{bmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Guess $F = 1$ N

Given $\left|\mathbf{r} \times (F\mathbf{k})\right| = M$ $F = \text{Find}(F)$ $F = 35.112$ N

Problem 4-92

The gears are subjected to the couple moments shown. Determine the magnitude and coordinate direction angles of the resultant couple moment.

Given:

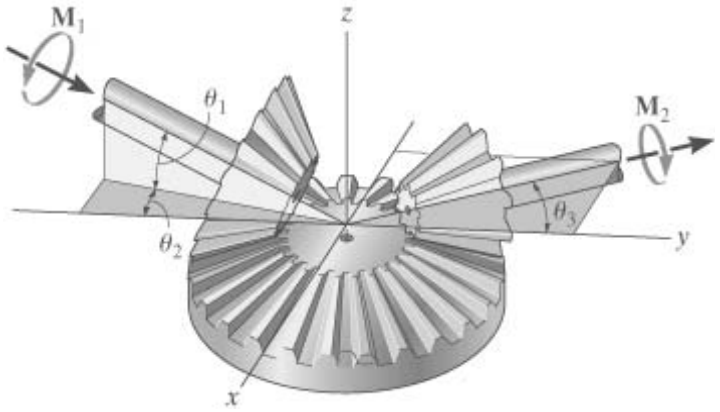
$M_1 = 40$ lb·ft

$M_2 = 30$ lb·ft

$\theta_1 = 20$ deg

$\theta_2 = 15$ deg

$\theta_3 = 30$ deg



Solution:

$$\mathbf{M_1} = \begin{pmatrix} M_1\cos(\theta_1)\sin(\theta_2) \\ M_1\cos(\theta_1)\cos(\theta_2) \\ -M_1\sin(\theta_1) \end{pmatrix} \qquad \mathbf{M_2} = \begin{pmatrix} -M_2\sin(\theta_3) \\ M_2\cos(\theta_3) \\ 0 \end{pmatrix}$$

$\mathbf{M_R} = \mathbf{M_1} + \mathbf{M_2}$ $\left|\mathbf{M_R}\right| = 64.0$ lb·ft

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{M_R}}{\left|\mathbf{M_R}\right|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 94.7 \\ 13.2 \\ 102.3 \end{pmatrix} \text{deg}$$

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Problem 4-93

Express the moment of the couple acting on the rod in Cartesian vector form. What is the magnitude of the couple moment?

Given:

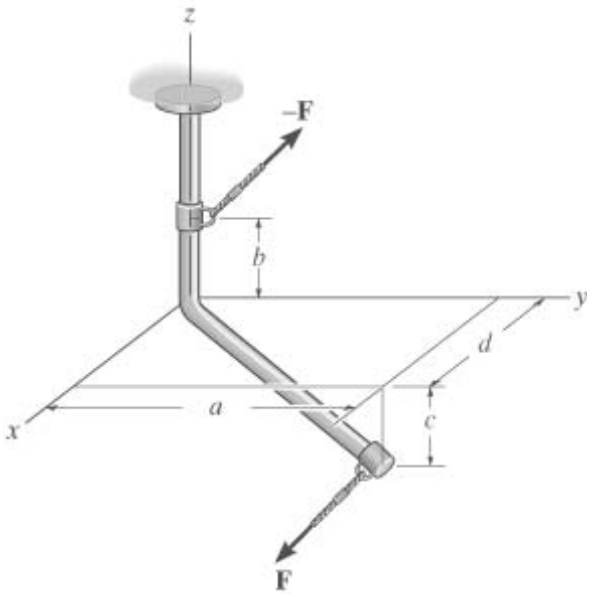
$$\mathbf{F} = \begin{pmatrix} 14 \\ -8 \\ -6 \end{pmatrix} \text{ N}$$

$a = 1.5$ m

$b = 0.5$ m

$c = 0.5$ m

$d = 0.8$ m



Solution:

$$\mathbf{M} = \begin{pmatrix} d \\ a \\ -c \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} 0 \\ 0 \\ b \end{pmatrix} \times (-\mathbf{F}) \qquad \mathbf{M} = \begin{pmatrix} -17 \\ -9.2 \\ -27.4 \end{pmatrix} \text{N}\cdot\text{m} \qquad |\mathbf{M}| = 33.532\,\text{N}\cdot\text{m}$$

Problem 4-94

Express the moment of the couple acting on the pipe assembly in Cartesian vector form. Solve the problem (a) using Eq. 4-13, and (b) summing the moment of each force about point *O*.

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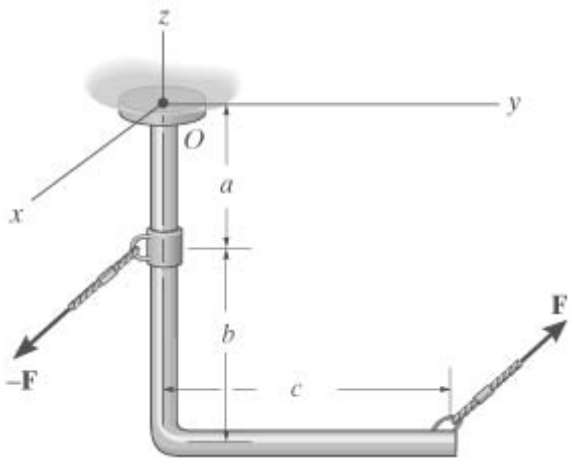
Given:

$a = 0.3$ m

$b = 0.4$ m

$c = 0.6$ m

$\mathbf{F} = \begin{pmatrix} -6 \\ 2 \\ 3 \end{pmatrix}$ N



Solution:

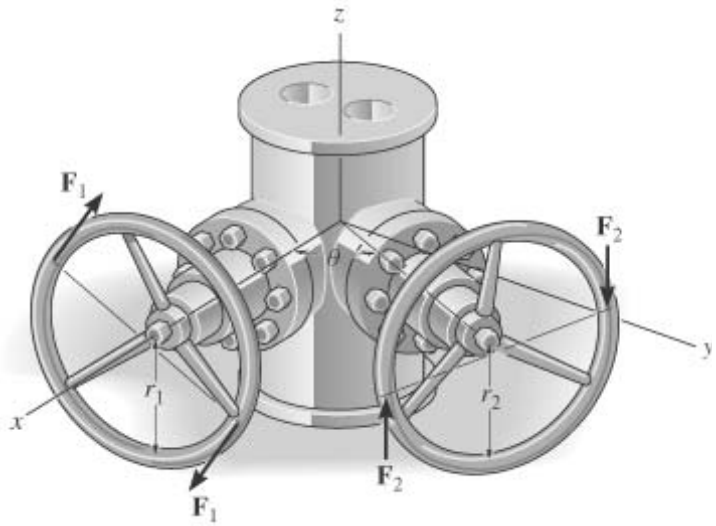
(a) $\mathbf{M} = \begin{pmatrix} 0 \\ c \\ -b \end{pmatrix} \times \mathbf{F}$ $\mathbf{M} = \begin{pmatrix} 2.6 \\ 2.4 \\ 3.6 \end{pmatrix}$ N·m

(b) $\mathbf{M} = \begin{pmatrix} 0 \\ 0 \\ -a \end{pmatrix} \times (-\mathbf{F}) + \begin{pmatrix} 0 \\ c \\ -a-b \end{pmatrix} \times \mathbf{F}$ $\mathbf{M} = \begin{pmatrix} 2.6 \\ 2.4 \\ 3.6 \end{pmatrix}$ N·m

---

Problem 4-95

A couple acts on each of the handles of the minidual valve. Determine the magnitude and coordinate direction angles of the resultant couple moment.



Given:

$F_1 = 35$ N $\quad \theta = 60$ deg

$F_2 = 25$ N

$r_1 = 175$ mm

$r_2 = 175$ mm

Solution:

$$\mathbf{M} = \begin{pmatrix} -F_1\, 2\, r_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -F_2\, 2 r_2 \cos(\theta) \\ -F_2\, 2\, r_2 \sin(\theta) \\ 0 \end{pmatrix} \qquad \mathbf{M} = \begin{pmatrix} -16.63 \\ -7.58 \\ 0 \end{pmatrix} \text{N·m} \qquad |\mathbf{M}| = 18.3\,\text{N·m}$$

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$$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{M}}{|\mathbf{M}|}\right) \qquad \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}155.496\\ 114.504\\ 90\end{pmatrix}\text{deg}$$

Problem 4-96

Express the moment of the couple acting on the pipe in Cartesian vector form. What is the magnitude of the couple moment?

Given:

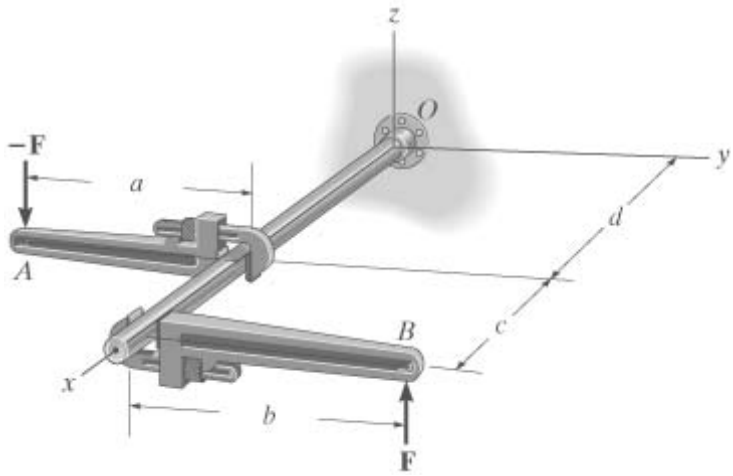
$F = 125$ N

$a = 150$ mm

$b = 150$ mm

$c = 200$ mm

$d = 600$ mm



Solution:

$$\mathbf{M} = \begin{pmatrix}c\\ a+b\\ 0\end{pmatrix} \times \begin{pmatrix}0\\ 0\\ F\end{pmatrix} \qquad \mathbf{M} = \begin{pmatrix}37.5\\ -25\\ 0\end{pmatrix}\text{N}\cdot\text{m} \qquad |\mathbf{M}| = 45.1\,\text{N}\cdot\text{m}$$

Problem 4-97

If the couple moment acting on the pipe has a magnitude M, determine the magnitude F of the forces applied to the wrenches.

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Given:

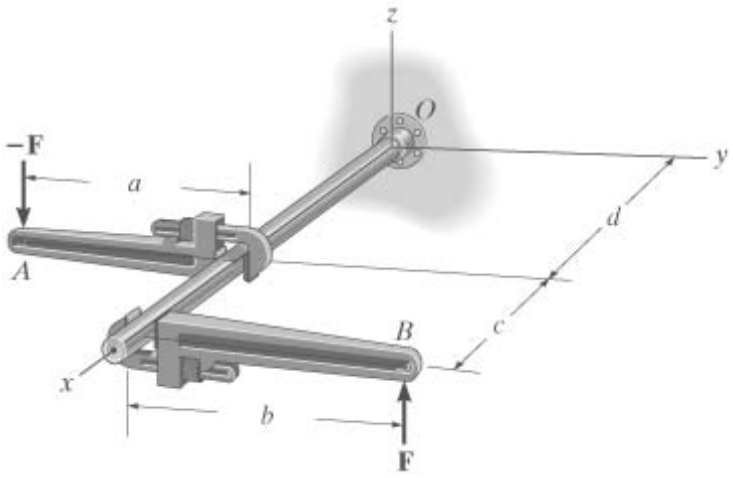
$M = 300\ \text{N}\cdot\text{m}$

$a = 150\ \text{mm}$

$b = 150\ \text{mm}$

$c = 200\ \text{mm}$

$d = 600\ \text{mm}$



Solution:

Initial guess: $F = 1\ \text{N}$

Given $\left\| \begin{pmatrix} c \\ a+b \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ F \end{pmatrix} \right\| = M$ $F = \text{Find}(F)$ $F = 832.1\ \text{N}$

Problem 4-98

Replace the force at A by an equivalent force and couple moment at point O.

Given:

$F = 375\ \text{N}$

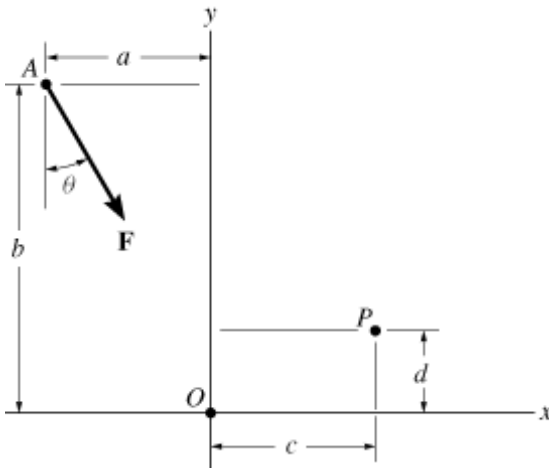
$a = 2\ \text{m}$

$b = 4\ \text{m}$

$c = 2\ \text{m}$

$d = 1\ \text{m}$

$\theta = 30\ \text{deg}$



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Solution:

$$\mathbf{F_V} = F\begin{pmatrix}\sin(\theta)\\ -\cos(\theta)\\ 0\end{pmatrix} \qquad \mathbf{F_V} = \begin{pmatrix}187.5\\ -324.76\\ 0\end{pmatrix}\ \text{N}$$

$$\mathbf{M_O} = \begin{pmatrix}-a\\ b\\ 0\end{pmatrix} \times \mathbf{F_V} \qquad \mathbf{M_O} = \begin{pmatrix}0\\ 0\\ -100.481\end{pmatrix}\ \text{N}\cdot\text{m}$$

Problem 4-99

Replace the force at A by an equivalent force and couple moment at point P.

Given:

$F = 375$ N

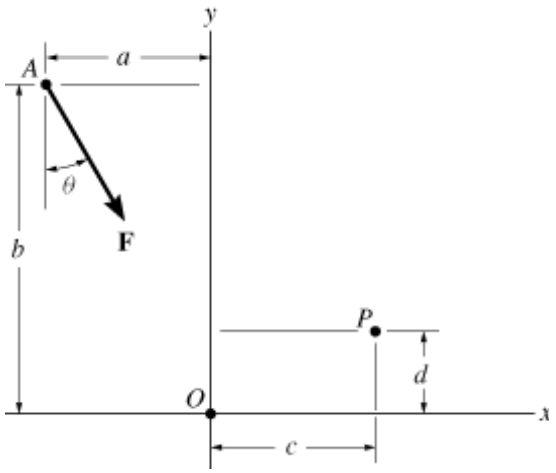
$a = 2$ m

$b = 4$ m

$c = 2$ m

$d = 1$ m

$\theta = 30$ deg



Solution:

$$\mathbf{F_V} = F\begin{pmatrix}\sin(\theta)\\ -\cos(\theta)\\ 0\end{pmatrix} \qquad \mathbf{F_V} = \begin{pmatrix}187.5\\ -324.76\\ 0\end{pmatrix}\ \text{N}$$

$$\mathbf{M_P} = \begin{pmatrix}-a-c\\ b-d\\ 0\end{pmatrix} \times \mathbf{F_V} \qquad \mathbf{M_P} = \begin{pmatrix}0\\ 0\\ 736.538\end{pmatrix}\ \text{N}\cdot\text{m}$$

Problem 4-100

Replace the force system by an equivalent resultant force and couple moment at point O.

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Given:

$F_1 = 60$ lb $\quad a = 2$ ft

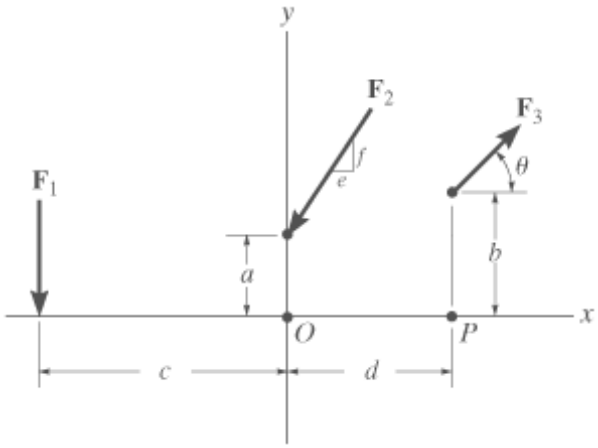
$F_2 = 85$ lb $\quad b = 3$ ft

$F_3 = 25$ lb $\quad c = 6$ ft

$\theta = 45$ deg $\quad d = 4$ ft

$e = 3$

$f = 4$



Solution:

$$\mathbf{F} = \begin{pmatrix} 0 \\ -F_1 \\ 0 \end{pmatrix} + \frac{F_2}{\sqrt{e^2 + f^2}} \begin{pmatrix} -e \\ -f \\ 0 \end{pmatrix} + F_3 \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix}$$

$$\mathbf{F} = \begin{pmatrix} -33.322 \\ -110.322 \\ 0 \end{pmatrix} \text{lb} \qquad |\mathbf{F}| = 115.245 \text{ lb}$$

$$\mathbf{M_O} = \begin{pmatrix} -c \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -F_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \left[\frac{F_2}{\sqrt{e^2 + f^2}} \begin{pmatrix} -e \\ -f \\ 0 \end{pmatrix} \right] + \begin{pmatrix} d \\ b \\ 0 \end{pmatrix} \times \left[F_3 \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix} \right]$$

$$\mathbf{M_O} = \begin{pmatrix} 0 \\ 0 \\ 480 \end{pmatrix} \text{lb} \cdot \text{ft} \qquad |\mathbf{M_O}| = 480 \text{ lb} \cdot \text{ft}$$

Problem 4-101

Replace the force system by an equivalent resultant force and couple moment at point P.

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Given:

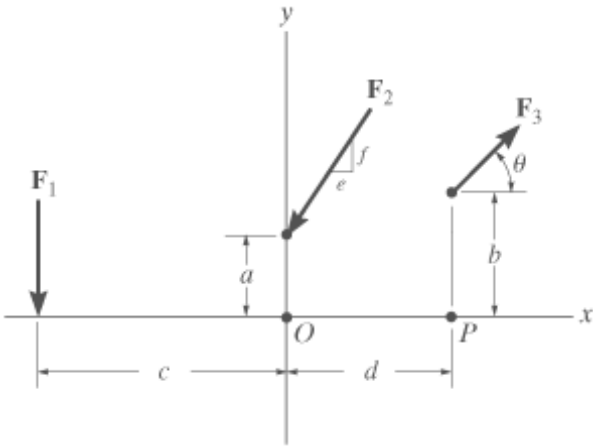
$F_1 = 60$ lb $\quad a = 2$ ft

$F_2 = 85$ lb $\quad b = 3$ ft

$F_3 = 25$ lb $\quad c = 6$ ft

$\theta = 45$ deg $\quad d = 4$ ft

$e = 3 \quad f = 4$



Solution:

$$\mathbf{F} = \begin{pmatrix} 0 \\ -F_1 \\ 0 \end{pmatrix} + \frac{F_2}{\sqrt{e^2 + f^2}} \begin{pmatrix} -e \\ -f \\ 0 \end{pmatrix} + F_3 \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix}$$

$$\mathbf{F} = \begin{pmatrix} -33.322 \\ -110.322 \\ 0 \end{pmatrix} \text{lb} \qquad |\mathbf{F}| = 115.245\,\text{lb}$$

$$\mathbf{M_P} = \begin{pmatrix} -c-d \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -F_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -d \\ a \\ 0 \end{pmatrix} \times \left[\frac{F_2}{\sqrt{e^2 + f^2}} \begin{pmatrix} -e \\ -f \\ 0 \end{pmatrix} \right] + \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \times \left[F_3 \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix} \right]$$

$$\mathbf{M_P} = \begin{pmatrix} 0 \\ 0 \\ 921 \end{pmatrix} \text{lb}\cdot\text{ft} \qquad |\mathbf{M_P}| = 921\,\text{lb}\cdot\text{ft}$$

Problem 4-102

Replace the force system by an equivalent force and couple moment at point *O*.

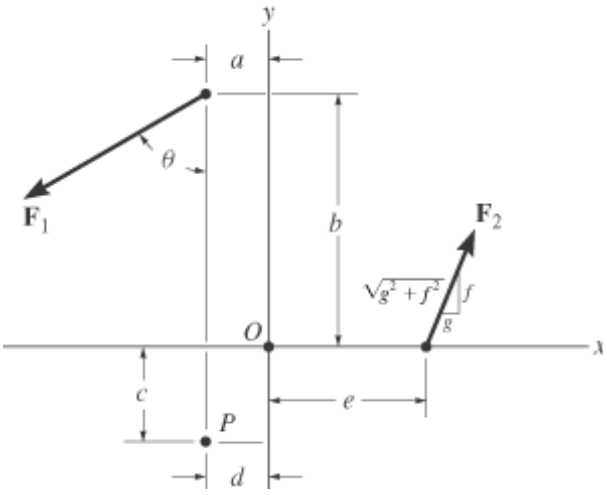
Units Used:

kip $= 10^3$ lb

Given:

$F_1 = 430$ lb $\quad F_2 = 260$ lb

$a = 2$ ft $\quad e = 5$ ft



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$b = 8$ ft $f = 12$

$c = 3$ ft $g = 5$

$d = a$ $\theta = 60$ deg

Solution:

$$\mathbf{F_R} = F_1\begin{pmatrix}-\sin(\theta)\\-\cos(\theta)\\0\end{pmatrix} + \frac{F_2}{\sqrt{g^2+f^2}}\begin{pmatrix}g\\f\\0\end{pmatrix} \qquad \mathbf{F_R} = \begin{pmatrix}-272\\25\\0\end{pmatrix}\text{lb} \qquad |\mathbf{F_R}| = 274\,\text{lb}$$

$$\mathbf{M_O} = \begin{pmatrix}-d\\b\\0\end{pmatrix}\times\left[F_1\begin{pmatrix}-\sin(\theta)\\-\cos(\theta)\\0\end{pmatrix}\right] + \begin{pmatrix}e\\0\\0\end{pmatrix}\times\left[\frac{F_2}{\sqrt{g^2+f^2}}\begin{pmatrix}g\\f\\0\end{pmatrix}\right] \qquad \mathbf{M_O} = \begin{pmatrix}0\\0\\4.609\end{pmatrix}\text{kip}\cdot\text{ft}$$

Problem 4-103

Replace the force system by an equivalent force and couple moment at point *P*.

Units Used:

kip $= 10^3$ lb

Given:

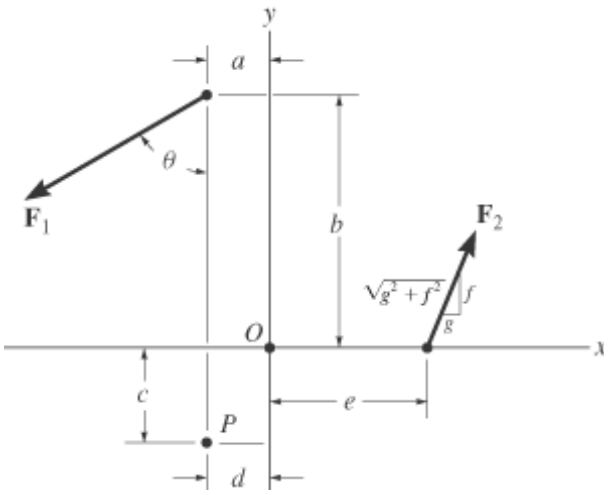
$F_1 = 430$ lb $F_2 = 260$ lb

$a = 2$ ft $e = 5$ ft

$b = 8$ ft $f = 12$

$c = 3$ ft $g = 5$

$d = a$ $\theta = 60$ deg



Solution:

$$\mathbf{F_R} = F_1\begin{pmatrix}-\sin(\theta)\\-\cos(\theta)\\0\end{pmatrix} + \frac{F_2}{\sqrt{g^2+f^2}}\begin{pmatrix}g\\f\\0\end{pmatrix} \qquad \mathbf{F_R} = \begin{pmatrix}-272\\25\\0\end{pmatrix}\text{lb} \qquad |\mathbf{F_R}| = 274\,\text{lb}$$

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$$\mathbf{M_P} = \begin{pmatrix} 0 \\ b+c \\ 0 \end{pmatrix} \times \left[F_1 \begin{pmatrix} -\sin(\theta) \\ -\cos(\theta) \\ 0 \end{pmatrix} \right] + \begin{pmatrix} d+e \\ c \\ 0 \end{pmatrix} \times \left[\frac{F_2}{\sqrt{g^2+f^2}} \begin{pmatrix} g \\ f \\ 0 \end{pmatrix} \right] \qquad \mathbf{M_P} = \begin{pmatrix} 0 \\ 0 \\ 5.476 \end{pmatrix} \text{kip}\cdot\text{ft}$$

Problem 4-104

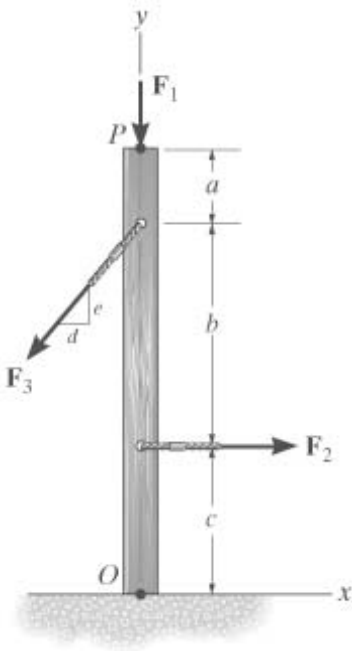
Replace the loading system acting on the post by an equivalent resultant force and couple moment at point *O*.

Given:

$F_1 = 30$ lb $\quad a = 1$ ft $\quad d = 3$

$F_2 = 40$ lb $\quad b = 3$ ft $\quad e = 4$

$F_3 = 60$ lb $\quad c = 2$ ft



Solution:

$$\mathbf{F_R} = F_1 \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} + F_2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{F_3}{\sqrt{d^2+e^2}} \begin{pmatrix} -d \\ -e \\ 0 \end{pmatrix}$$

$$\mathbf{F_R} = \begin{pmatrix} 4 \\ -78 \\ 0 \end{pmatrix} \text{lb} \qquad |\mathbf{F_R}| = 78.1 \text{ lb}$$

$$\mathbf{M_O} = \begin{pmatrix} 0 \\ a+b+c \\ 0 \end{pmatrix} \times \left[F_1 \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} \right] + \begin{pmatrix} 0 \\ c \\ 0 \end{pmatrix} \times \left[F_2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right] + \begin{pmatrix} 0 \\ b+c \\ 0 \end{pmatrix} \times \left[\frac{F_3}{\sqrt{d^2+e^2}} \begin{pmatrix} -d \\ -e \\ 0 \end{pmatrix} \right]$$

$$\mathbf{M_O} = \begin{pmatrix} 0 \\ 0 \\ 100 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 4-105

Replace the loading system acting on the post by an equivalent resultant force and couple moment at point *P*.

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Given:

$F_1 = 30$ lb

$F_2 = 40$ lb

$F_3 = 60$ lb

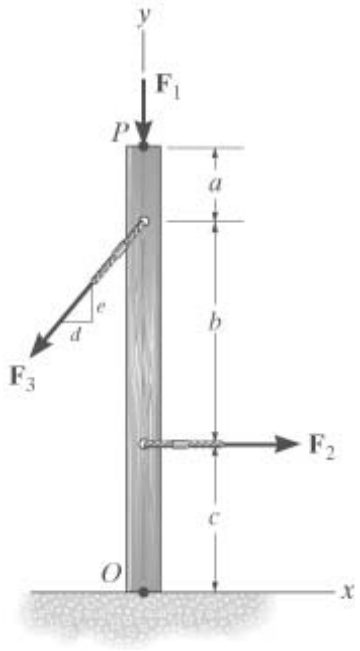
$a = 1$ ft

$b = 3$ ft

$c = 2$ ft

$d = 3$

$e = 4$



Solution:

$$\mathbf{F_R} = F_1\begin{pmatrix}0\\-1\\0\end{pmatrix} + F_2\begin{pmatrix}1\\0\\0\end{pmatrix} + \frac{F_3}{\sqrt{d^2+e^2}}\begin{pmatrix}-d\\-e\\0\end{pmatrix}$$

$$\mathbf{F_R} = \begin{pmatrix}4\\-78\\0\end{pmatrix}\text{ lb} \qquad \left|\mathbf{F_R}\right| = 78.1\text{ lb}$$

$$\mathbf{M_P} = \left[\begin{pmatrix}0\\0\\0\end{pmatrix}\text{ft}\right] \times \left[F_1\begin{pmatrix}0\\-1\\0\end{pmatrix}\right] + \begin{pmatrix}0\\-a-b\\0\end{pmatrix} \times \left[F_2\begin{pmatrix}1\\0\\0\end{pmatrix}\right] + \begin{pmatrix}0\\-a\\0\end{pmatrix} \times \left[\frac{F_3}{\sqrt{d^2+e^2}}\begin{pmatrix}-d\\-e\\0\end{pmatrix}\right]$$

$$\mathbf{M_P} = \begin{pmatrix}0\\0\\124\end{pmatrix}\text{ lb}\cdot\text{ft}$$

Problem 4-106

Replace the force and couple system by an equivalent force and couple moment at point O.

Units Used:

$\text{kN} = 10^3$ N

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Given:

$M = 8\ \text{kN m}$ $\theta = 60\ \text{deg}$

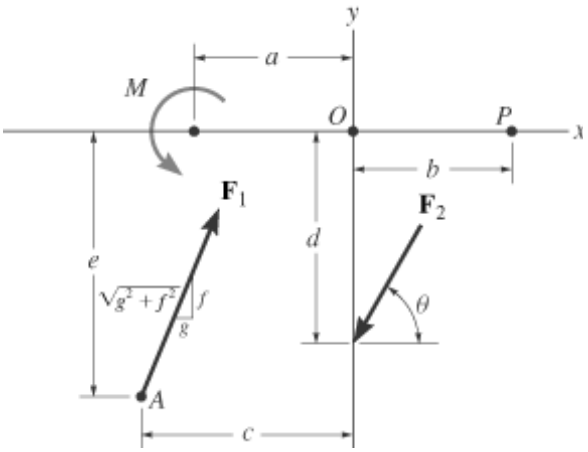
$a = 3\ \text{m}$ $f = 12$

$b = 3\ \text{m}$ $g = 5$

$c = 4\ \text{m}$ $F_1 = 6\ \text{kN}$

$d = 4\text{m}$ $F_2 = 4\ \text{kN}$

$e = 5\ \text{m}$



Solution:

$$\mathbf{F_R} = \frac{F_1}{\sqrt{f^2 + g^2}}\begin{pmatrix} g \\ f \\ 0 \end{pmatrix} + F_2\begin{pmatrix} -\cos(\theta) \\ -\sin(\theta) \\ 0 \end{pmatrix}$$

$$\mathbf{F_R} = \begin{pmatrix} 0.308 \\ 2.074 \\ 0 \end{pmatrix}\text{kN} \qquad \left|\mathbf{F_R}\right| = 2.097\,\text{kN}$$

$$\mathbf{M_O} = \begin{pmatrix} 0 \\ 0 \\ M \end{pmatrix} + \begin{pmatrix} -c \\ -e \\ 0 \end{pmatrix} \times \left[\frac{F_1}{\sqrt{f^2 + g^2}}\begin{pmatrix} g \\ f \\ 0 \end{pmatrix}\right] + \begin{pmatrix} 0 \\ -d \\ 0 \end{pmatrix} \times \left[F_2\begin{pmatrix} -\cos(\theta) \\ -\sin(\theta) \\ 0 \end{pmatrix}\right]$$

$$\mathbf{M_O} = \begin{pmatrix} 0 \\ 0 \\ -10.615 \end{pmatrix}\text{kN}\cdot\text{m}$$

Problem 4-107

Replace the force and couple system by an equivalent force and couple moment at point P.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

$M = 8\ \text{kN}\cdot\text{m}$ $\theta = 60\ \text{deg}$

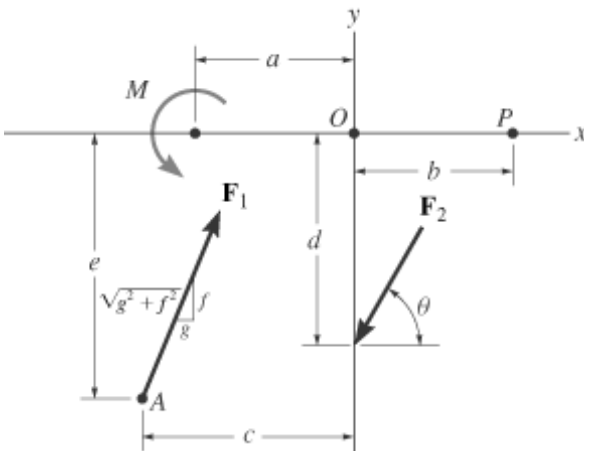
$a = 3\ \text{m}$ $f = 12$

$b = 3\ \text{m}$ $g = 5$

$c = 4\ \text{m}$ $F_1 = 6\ \text{kN}$

$d = 4\ \text{m}$ $F_2 = 4\ \text{kN}$

$e = 5\ \text{m}$



Solution:

$$\mathbf{F_R} = \frac{F_1}{\sqrt{f^2 + g^2}}\begin{pmatrix} g \\ f \\ 0 \end{pmatrix} + F_2\begin{pmatrix} -\cos(\theta) \\ -\sin(\theta) \\ 0 \end{pmatrix}$$

$$\mathbf{F_R} = \begin{pmatrix} 0.308 \\ 2.074 \\ 0 \end{pmatrix}\text{kN} \qquad \left|\mathbf{F_R}\right| = 2.097\,\text{kN}$$

$$\mathbf{M_P} = \begin{pmatrix} 0 \\ 0 \\ M \end{pmatrix} + \begin{pmatrix} -c-b \\ -e \\ 0 \end{pmatrix} \times \left[\frac{F_1}{\sqrt{f^2 + g^2}}\begin{pmatrix} g \\ f \\ 0 \end{pmatrix}\right] + \begin{pmatrix} -b \\ -d \\ 0 \end{pmatrix} \times \left[F_2\begin{pmatrix} -\cos(\theta) \\ -\sin(\theta) \\ 0 \end{pmatrix}\right]$$

$$\mathbf{M_P} = \begin{pmatrix} 0 \\ 0 \\ -16.838 \end{pmatrix}\text{kN}\cdot\text{m}$$

Problem 4-108

Replace the force system by a single force resultant and specify its point of application, measured along the *x* axis from point *O*.

Given:

$F_1 = 125\ \text{lb}$

$F_2 = 350\ \text{lb}$

$F_3 = 850\ \text{lb}$

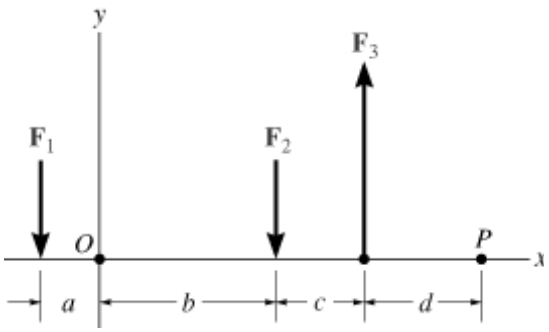
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$a = 2$ ft

$b = 6$ ft

$c = 3$ ft

$d = 4$ ft



Solution:

$$F_{Ry} = F_3 - F_2 - F_1 \qquad F_{Ry} = 375\,\text{lb}$$

$$F_{Ry}\,x = F_3(b + c) - F_2(b) + F_1(a)$$

$$x = \frac{F_3(b + c) - F_2(b) + F_1\,a}{F_{Ry}} \qquad x = 15.5\,\text{ft}$$

Problem 4-109

Replace the force system by a single force resultant and specify its point of application, measured along the x axis from point P.

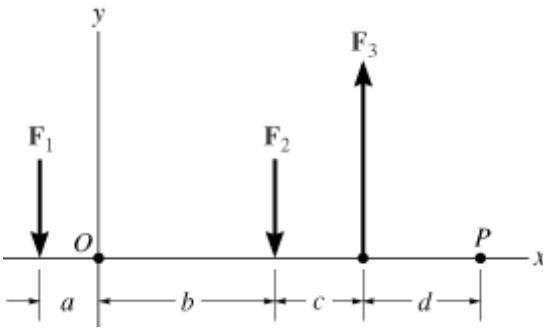
Given:

$F_1 = 125$ lb $\quad a = 2$ ft

$F_2 = 350$ lb $\quad b = 6$ ft

$F_3 = 850$ lb $\quad c = 3$ ft

$d = 4$ ft



Solution:

$$F_{Ry} = F_3 - F_2 - F_1 \qquad F_{Ry} = 375\,\text{lb}$$

$$F_{Ry}\,x = F_2(d + c) - F_3(d) + F_1(a + b + c + d)$$

$$x = \frac{F_2\,d + F_2\,c - F_3\,d + F_1\,a + F_1\,b + F_1\,c + F_1\,d}{F_{Ry}}$$

$x = 2.47$ ft (to the right of P)

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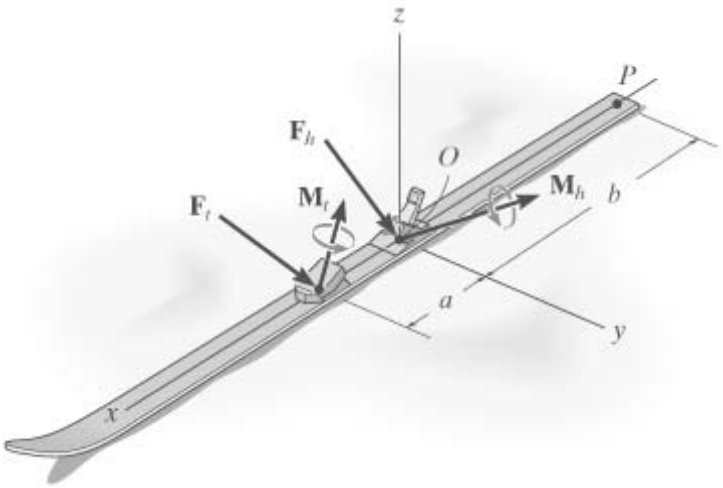
Problem 4-110

The forces and couple moments which are exerted on the toe and heel plates of a snow ski are $\mathbf{F}_t$, $\mathbf{M}_t$, and $\mathbf{F}_h$, $\mathbf{M}_h$, respectively. Replace this system by an equivalent force and couple moment acting at point *O*. Express the results in Cartesian vector form.

Given:

$a = 120$ mm

$b = 800$ mm



Solution:

$$\mathbf{F_t} = \begin{pmatrix} -50 \\ 80 \\ -158 \end{pmatrix} \text{N} \qquad \mathbf{F_h} = \begin{pmatrix} -20 \\ 60 \\ -250 \end{pmatrix} \text{N} \qquad \mathbf{M_t} = \begin{pmatrix} -6 \\ 4 \\ 2 \end{pmatrix} \text{N}\cdot\text{m} \qquad \mathbf{M_h} = \begin{pmatrix} -20 \\ 8 \\ 3 \end{pmatrix} \text{N}\cdot\text{m}$$

$$\mathbf{F_R} = \mathbf{F_t} + \mathbf{F_h}$$

$$\mathbf{F_R} = \begin{pmatrix} -70 \\ 140 \\ -408 \end{pmatrix} \text{N}$$

$$\mathbf{r_{0Ft}} = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{M_{RP}} = \left(\mathbf{r_{0Ft}} \times \mathbf{F_t}\right) + \mathbf{M_t} + \mathbf{M_h}$$

$$\mathbf{M_{RP}} = \begin{pmatrix} -26 \\ 31 \\ 14.6 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-111

The forces and couple moments which are exerted on the toe and heel plates of a snow ski are $\mathbf{F}_t$, $\mathbf{M}_t$, and $\mathbf{F}_h$, $\mathbf{M}_h$, respectively. Replace this system by an equivalent force and couple moment

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acting at point *P*. Express the results in Cartesian vector form.

Given:

$a = 120$ mm

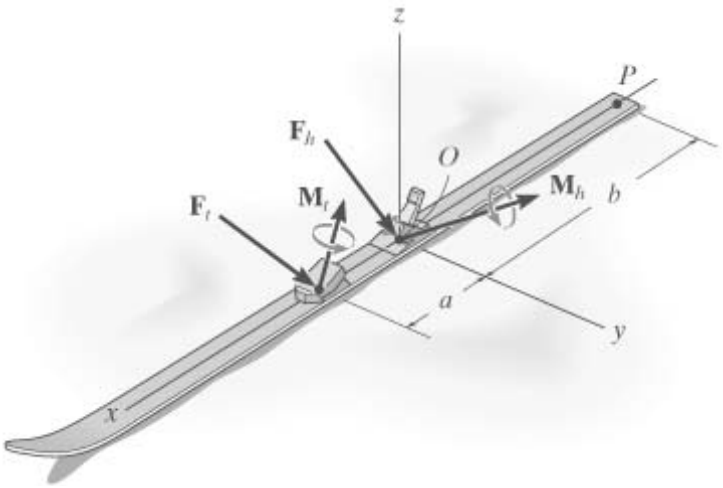
$b = 800$ mm

$$\mathbf{F_t} = \begin{pmatrix} -50 \\ 80 \\ -158 \end{pmatrix} \text{N}$$

$$\mathbf{M_t} = \begin{pmatrix} -6 \\ 4 \\ 2 \end{pmatrix} \text{N}\cdot\text{m}$$

$$\mathbf{F_h} = \begin{pmatrix} -20 \\ 60 \\ -250 \end{pmatrix} \text{N}$$

$$\mathbf{M_h} = \begin{pmatrix} -20 \\ 8 \\ 3 \end{pmatrix} \text{N}\cdot\text{m}$$



Solution:

$$\mathbf{F_R} = \mathbf{F_t} + \mathbf{F_h} \qquad \mathbf{F_R} = \begin{pmatrix} -70 \\ 140 \\ -408 \end{pmatrix} \text{N}$$

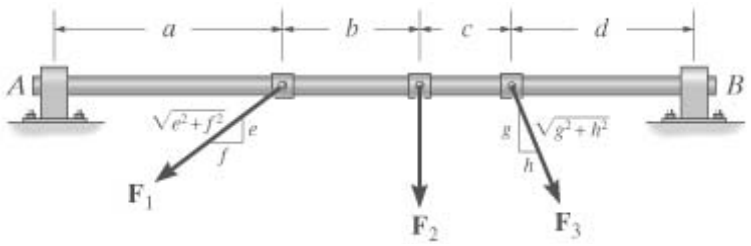
$$\mathbf{M_P} = \mathbf{M_t} + \mathbf{M_h} + \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix} \times \mathbf{F_h} + \begin{pmatrix} a+b \\ 0 \\ 0 \end{pmatrix} \times \mathbf{F_t} \qquad \mathbf{M_P} = \begin{pmatrix} -26 \\ 357.4 \\ 126.6 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-112

Replace the three forces acting on the shaft by a single resultant force. Specify where the force acts, measured from end *B*.

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Given:



$F_1 = 500$ lb

$F_2 = 200$ lb

$F_3 = 260$ lb

$a = 5$ ft $\quad e = 3$

$b = 3$ ft $\quad f = 4$

$c = 2$ ft $\quad g = 12$

$d = 4$ ft $\quad h = 5$

Solution:

$$\mathbf{F_R} = \frac{F_1}{\sqrt{e^2+f^2}}\begin{pmatrix}-f\\-e\\0\end{pmatrix} + F_2\begin{pmatrix}0\\-1\\0\end{pmatrix} + \frac{F_3}{\sqrt{g^2+h^2}}\begin{pmatrix}h\\-g\\0\end{pmatrix} \qquad \mathbf{F_R} = \begin{pmatrix}-300\\-740\\0\end{pmatrix}\text{lb} \qquad |\mathbf{F_R}| = 798\,\text{lb}$$

Initial guess: $x = 1$ ft

Given

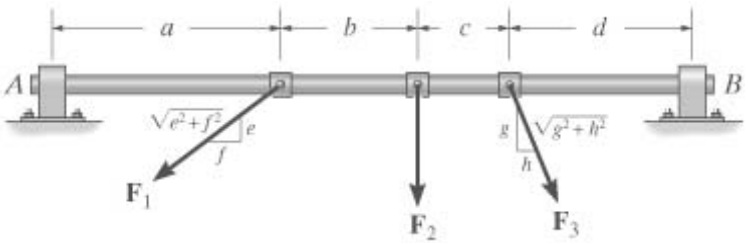
$$\begin{pmatrix}a\\0\\0\end{pmatrix} \times \left[\frac{F_1}{\sqrt{e^2+f^2}}\begin{pmatrix}-f\\-e\\0\end{pmatrix}\right] + \begin{pmatrix}a+b\\0\\0\end{pmatrix} \times \left[F_2\begin{pmatrix}0\\-1\\0\end{pmatrix}\right] + \begin{pmatrix}a+b+c\\0\\0\end{pmatrix} \times \left[\frac{F_3}{\sqrt{g^2+h^2}}\begin{pmatrix}h\\-g\\0\end{pmatrix}\right] = \begin{pmatrix}-x\\0\\0\end{pmatrix} \times \mathbf{F_R}$$

$x = \text{Find}(x) \qquad x = -7.432\,\text{ft}$

Problem 4-113

Replace the three forces acting on the shaft by a single resultant force. Specify where the force acts, measured from end *B*.

Given:



$F_1 = 500$ lb

$F_2 = 200$ lb

$F_3 = 260$ lb

$a = 5$ ft $\quad e = 3$

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$b = 3$ ft $f = 4$

$c = 2$ ft $g = 12$

$d = 4$ ft $h = 5$

Solution:

$$\mathbf{F_R} = \frac{F_1}{\sqrt{e^2 + f^2}}\begin{pmatrix} -f \\ -e \\ 0 \end{pmatrix} + F_2\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} + \frac{F_3}{\sqrt{g^2 + h^2}}\begin{pmatrix} h \\ -g \\ 0 \end{pmatrix} \qquad \mathbf{F_R} = \begin{pmatrix} -300 \\ -740 \\ 0 \end{pmatrix} \text{lb} \qquad |\mathbf{F_R}| = 798\,\text{lb}$$

Initial guess: $x = 1$ft

Given

$$\begin{pmatrix} -b-c-d \\ 0 \\ 0 \end{pmatrix} \times \left[\frac{F_1}{\sqrt{e^2 + f^2}}\begin{pmatrix} -f \\ -e \\ 0 \end{pmatrix}\right] + \begin{pmatrix} -c-d \\ 0 \\ 0 \end{pmatrix} \times \left[F_2\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}\right] + \begin{pmatrix} -d \\ 0 \\ 0 \end{pmatrix} \times \left[\frac{F_3}{\sqrt{g^2 + h^2}}\begin{pmatrix} h \\ -g \\ 0 \end{pmatrix}\right] = \begin{pmatrix} -x \\ 0 \\ 0 \end{pmatrix} \times \mathbf{F}$$

$x = \text{Find}(x)$ $x = 6.568$ ft measured to the left of B

Problem 4-114

Replace the loading on the frame by a single resultant force. Specify where its line of action intersects member AB, measured from A.

Given:

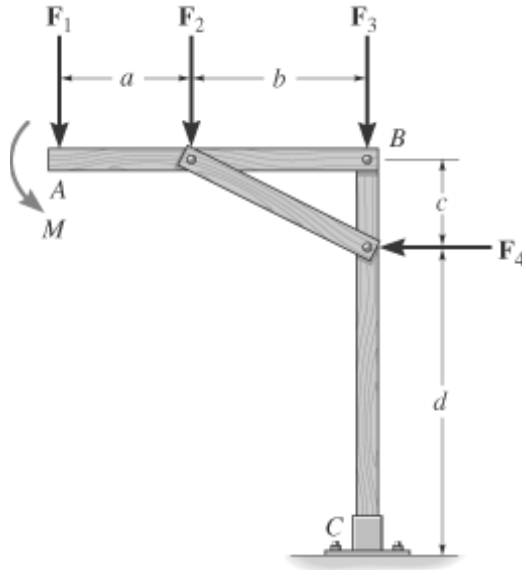
$F_1 = 300$ lb $M = 600$ lb·ft

$F_2 = 200$ lb $a = 3$ ft

$F_3 = 400$ lb $b = 4$ ft

$F_4 = 200$ lb $c = 2$ ft

$d = 7$ ft



Solution:

$F_{Rx} = -F_4$ $F_{Rx} = -200$ lb

$F_{Ry} = -F_1 - F_2 - F_3$ $F_{Ry} = -900$ lb

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$$F = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F = 922\,\text{lb} \qquad \theta = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta = 77.5\,\text{deg}$$

$$F_{Ry}\,x = -F_2\,a - F_3(a+b) - F_4\,c + M$$

$$x = -\frac{F_2(a) + F_3(a+b) + F_4\,c - M}{F_{Ry}} \qquad x = 3.556\,\text{ft}$$

Problem 4-115

Replace the loading on the frame by a single resultant force. Specify where the force acts, measured from end *A*.

Given:

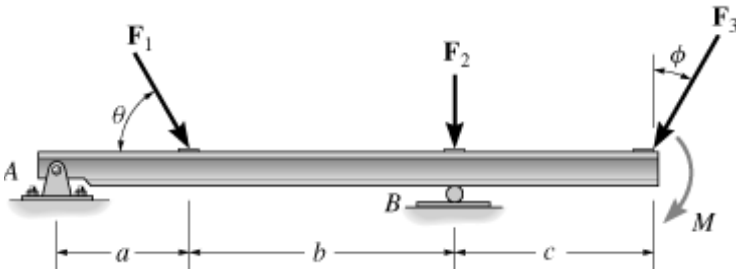
$F_1 = 450\ \text{N}$ $a = 2\ \text{m}$

$F_2 = 300\ \text{N}$ $b = 4\ \text{m}$

$F_3 = 700\ \text{N}$ $c = 3\ \text{m}$

$\theta = 60\ \text{deg}$ $M = 1500\ \text{N}\cdot\text{m}$

$\phi = 30\ \text{deg}$



Solution:

$$F_{Rx} = F_1\cos(\theta) - F_3\sin(\phi) \qquad F_{Rx} = -125\ \text{N}$$

$$F_{Ry} = -F_1\sin(\theta) - F_3\cos(\phi) - F_2 \qquad F_{Ry} = -1.296\times10^3\ \text{N}$$

$$F = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F = 1.302\times10^3\ \text{N}$$

$$\theta_1 = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = 84.5\,\text{deg}$$

$$F_{Ry}(x) = -F_1\sin(\theta)a - F_2(a+b) - F_3\cos(\phi)(a+b+c) - M$$

$$x = \frac{-F_1\sin(\theta)a - F_2(a+b) - F_3\cos(\phi)(a+b+c) - M}{F_{Ry}} \qquad x = 7.36\ \text{m}$$

Problem 4-116

Replace the loading on the frame by a single resultant force. Specify where the force acts, measured from end *B*.

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Given:

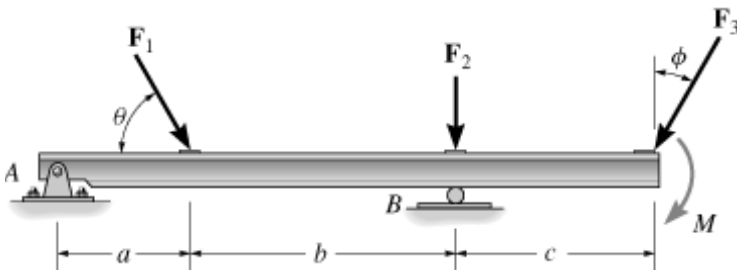
$F_1 = 450$ N $a = 2$ m

$F_2 = 300$ N $b = 4$ m

$F_3 = 700$ N $c = 3$ m

$\theta = 60$ deg $M = 1500$ N·m

$\phi = 30$deg



Solution:

$F_{Rx} = F_1 \cos(\theta) - F_3 \sin(\phi)$ $F_{Rx} = -125$ N

$F_{Ry} = -F_1 \sin(\theta) - F_3 \cos(\phi) - F_2$ $F_{Ry} = -1.296 \times 10^3$ N

$F = \sqrt{F_{Rx}^2 + F_{Ry}^2}$ $F = 1.302 \times 10^3$ N

$\theta_1 = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right)$ $\theta_1 = 84.5$ deg

$F_{Ry} x = F_1 \sin(\theta) b - F_3 \cos(\phi) c - M$

$x = \frac{F_1 \sin(\theta) b - F_3 \cos(\phi) c - M}{F_{Ry}}$ $x = 1.36$ m (to the right)

Problem 4-117

Replace the loading system acting on the beam by an equivalent resultant force and couple moment at point O.

Given:

$F_1 = 200$ N

$F_2 = 450$ N

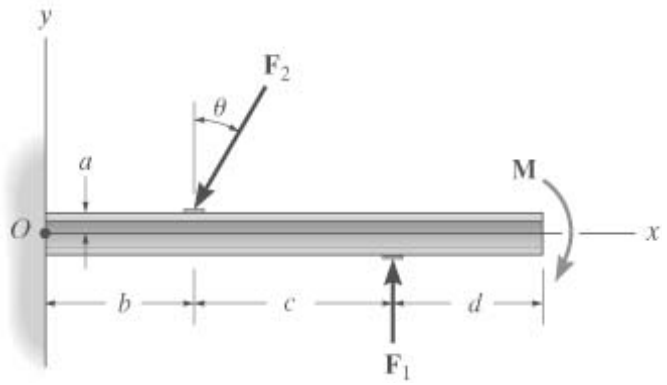
$M = 200$ N·m

$a = 0.2$ m

$b = 1.5$ m

$c = 2$ m

$d = 1.5$ m



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$\theta = 30 \text{ deg}$

Solution:

$$\mathbf{F_R} = F_1\begin{pmatrix}0\\1\\0\end{pmatrix} + F_2\begin{pmatrix}-\sin(\theta)\\-\cos(\theta)\\0\end{pmatrix}$$

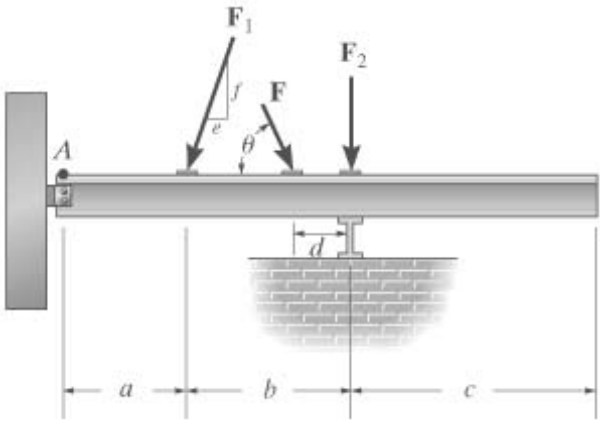
$$\mathbf{F_R} = \begin{pmatrix}-225\\-190\\0\end{pmatrix}\text{ N} \qquad |\mathbf{F_R}| = 294 \text{ N}$$

$$\mathbf{M_O} = \begin{pmatrix}b+c\\a\\0\end{pmatrix} \times \left[F_1\begin{pmatrix}0\\1\\0\end{pmatrix}\right] + \begin{pmatrix}b\\a\\0\end{pmatrix} \times \left[F_2\begin{pmatrix}-\sin(\theta)\\-\cos(\theta)\\0\end{pmatrix}\right] + M\begin{pmatrix}0\\0\\-1\end{pmatrix}$$

$$\mathbf{M_O} = \begin{pmatrix}0\\0\\-39.6\end{pmatrix}\text{ N·m}$$

Problem 4-118

Determine the magnitude and direction θ of force **F** and its placement d on the beam so that the loading system is equivalent to a resultant force $\mathbf{F}_R$ acting vertically downward at point A and a clockwise couple moment M.



Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$F_1 = 5 \text{ kN}$ $a = 3 \text{ m}$

$F_2 = 3 \text{ kN}$ $b = 4 \text{ m}$

$F_R = 12 \text{ kN}$ $c = 6 \text{ m}$

$M = 50 \text{ kN·m}$ $e = 7$ $f = 24$

Solution:

Initial guesses: $F = 1 \text{ kN}$ $\theta = 30 \text{ deg}$ $d = 2 \text{ m}$

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Given $\left(\frac{-e}{\sqrt{e^2+f^2}}\right)F_1 + F\cos(\theta) = 0$

$\left(\frac{-f}{\sqrt{e^2+f^2}}\right)F_1 - F\sin(\theta) - F_2 = -F_R$

$\left(\frac{f}{\sqrt{e^2+f^2}}\right)F_1\,a + F\sin(\theta)(a+b-d) + F_2(a+b) = M$

$\begin{pmatrix} F \\ \theta \\ d \end{pmatrix} = \text{Find}(F,\theta,d)$ $F = 4.427\,\text{kN}$ $\theta = 71.565\,\text{deg}$ $d = 3.524\ \text{m}$

Problem 4-119

Determine the magnitude and direction θ of force $\mathbf{F}$ and its placement d on the beam so that the loading system is equivalent to a resultant force $\mathbf{F}_R$ acting vertically downward at point A and a clockwise couple moment M.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

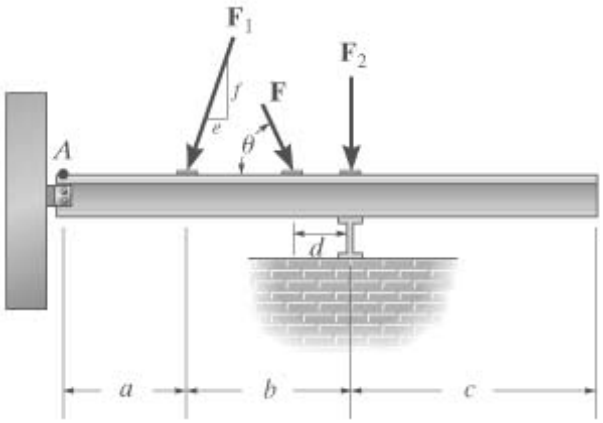
$F_1 = 5\ \text{kN}$ $a = 3\ \text{m}$

$F_2 = 3\ \text{kN}$ $b = 4\ \text{m}$

$F_R = 10\ \text{kN}$ $c = 6\ \text{m}$

$M = 45\ \text{kN·m}$ $e = 7$

$f = 24$



Solution:

Initial guesses: $F = 1\ \text{kN}$ $\theta = 30\ \text{deg}$ $d = 1\ \text{m}$

Given $\left(\frac{-e}{\sqrt{e^2+f^2}}\right)F_1 + F\cos(\theta) = 0$

$\left(\frac{-f}{\sqrt{e^2+f^2}}\right)F_1 - F\sin(\theta) - F_2 = -F_R$

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$$\left(\frac{f}{\sqrt{e^2 + f^2}}\right) F_1\, a + F \sin(\theta)(a + b - d) + F_2(a + b) = M$$

$$\begin{pmatrix} F \\ \theta \\ d \end{pmatrix} = \text{Find}(F, \theta, d) \qquad F = 2.608\,\text{kN} \qquad \theta = 57.529\,\text{deg} \qquad d = 2.636\ \text{m}$$

Problem 4-120

Replace the loading on the frame by a single resultant force. Specify where its line of action intersects member *AB*, measured from *A*.

Given:

$F_1 = 500$ N $\quad a = 3$ m

$F_2 = 300$ N $\quad b = 2$ m

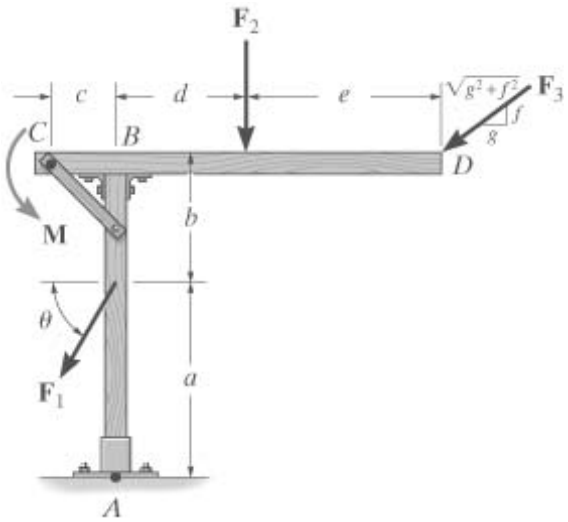
$c = 1$ m

$F_3 = 250$ N $\quad d = 2$ m

$M = 400$ N·m $\quad e = 3$ m

$\theta = 60$ deg $\quad f = 3$

$g = 4$



Solution:

$$F_{Rx} = -F_3\left(\frac{g}{\sqrt{g^2 + f^2}}\right) - F_1(\cos(\theta)) \qquad F_{Rx} = -450\ \text{N}$$

$$F_{Ry} = -F_2 - F_3\left(\frac{f}{\sqrt{f^2 + g^2}}\right) - F_1 \sin(\theta) \qquad F_{Ry} = -883.0127\ \text{N}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 991.066\ \text{N}$$

$$\theta_1 = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = 62.996\,\text{deg}$$

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$$-F_{Rx}(y) = M + F_1\cos(\theta)a + F_3\frac{g}{\sqrt{g^2+f^2}}(b+a) - F_2(d) - F_3\left(\frac{f}{\sqrt{g^2+f^2}}\right)(d+e)$$

$$y = \frac{M + F_1\cos(\theta)a + F_3\frac{g}{\sqrt{g^2+f^2}}(b+a) - F_2(d) - F_3\left(\frac{f}{\sqrt{g^2+f^2}}\right)(d+e)}{-F_{Rx}}$$

$y = 1.78$ m

Problem 4-121

Replace the loading on the frame by a single resultant force. Specify where its line of action intersects member *CD*, measured from end *C*.

Given:

$F_1 = 500$ N $\quad a = 3$ m

$F_2 = 300$ N $\quad b = 2$ m

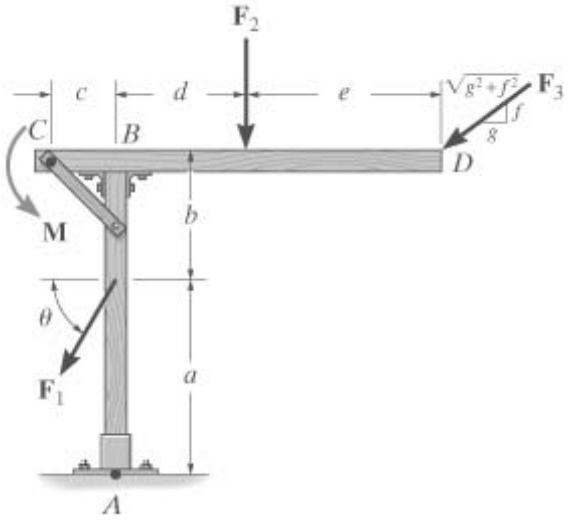
$F_3 = 250$ N $\quad c = 1$ m

$M = 400$ N·m $\quad d = 2$ m

$\theta = 60$ deg $\quad e = 3$ m

$f = 3$

$g = 4$



Solution:

$$F_{Rx} = -F_3\left(\frac{g}{\sqrt{g^2+f^2}}\right) - F_1(\cos(\theta)) \qquad F_{Rx} = -450 \text{ N}$$

$$F_{Ry} = -F_2 - F_3\left(\frac{f}{\sqrt{f^2+g^2}}\right) - F_1\sin(\theta) \qquad F_{Ry} = -883.0127 \text{ N}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 991.066 \text{ N}$$

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$$\theta_1 = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = 62.996\,\text{deg}$$

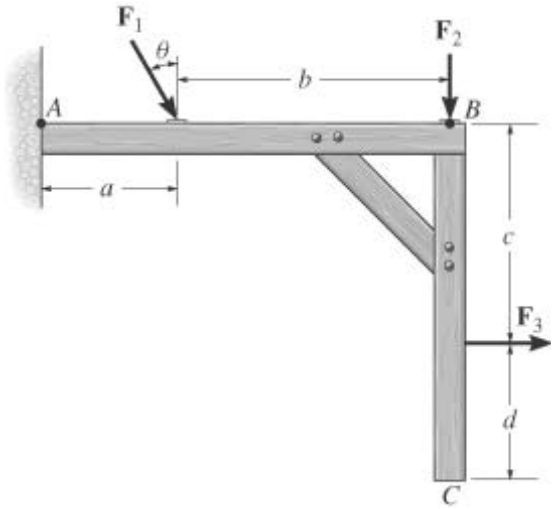
$$F_{Ry}(x) = M - F_2(d+c) - F_3\left(\frac{f}{\sqrt{g^2+f^2}}\right)(c+d+e) - F_1(b)\cos(\theta) - F_1\,c\sin(\theta)$$

$$x = \frac{M - F_2(d+c) - F_3\left(\frac{f}{\sqrt{g^2+f^2}}\right)(c+d+e) - F_1(b)\cos(\theta) - F_1\,c\sin(\theta)}{F_{Ry}}$$

$$x = 2.64\ \text{m}$$

Problem 4-122

Replace the force system acting on the frame by an equivalent resultant force and specify where the resultant's line of action intersects member *AB*, measured from point *A*.



Given:

$F_1 = 35\ \text{lb} \quad a = 2\ \text{ft}$

$F_2 = 20\ \text{lb} \quad b = 4\ \text{ft}$

$F_3 = 25\ \text{lb} \quad c = 3\ \text{ft}$

$\theta = 30\ \text{deg} \quad d = 2\ \text{ft}$

Solution:

$F_{Rx} = F_1\sin(\theta) + F_3 \qquad F_{Rx} = 42.5\,\text{lb}$

$F_{Ry} = -F_1\cos(\theta) - F_2 \qquad F_{Ry} = -50.311\,\text{lb}$

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 65.9\,\text{lb}$

$\theta_1 = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = -49.8\,\text{deg}$

$F_{Ry}\,x = -F_1\cos(\theta)a - F_2(a+b) + F_3(c)$

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$$x = \frac{-F_1 \cos(\theta) a - F_2(a + b) + F_3(c)}{F_{Ry}} \qquad x = 2.099 \text{ ft}$$

Problem 4-123

Replace the force system acting on the frame by an equivalent resultant force and specify where the resultant's line of action intersects member *BC*, measured from point *B*.

Given:

$F_1 = 35$ lb

$F_2 = 20$ lb

$F_3 = 25$ lb

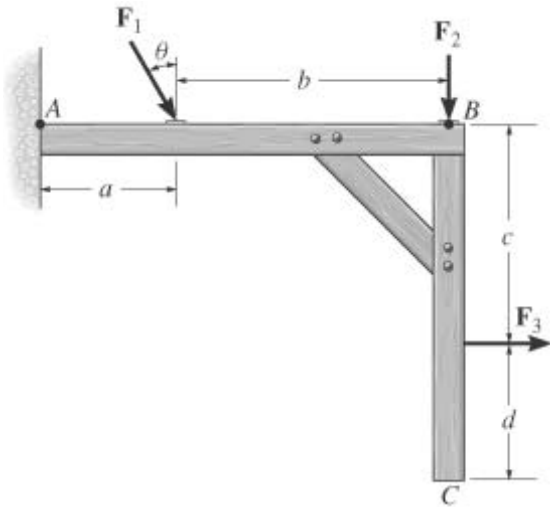
$\theta = 30$ deg

$a = 2$ ft

$b = 4$ ft

$c = 3$ ft

$d = 2$ ft



Solution:

$$F_{Rx} = F_1 \sin(\theta) + F_3 \qquad F_{Rx} = 42.5 \text{ lb}$$

$$F_{Ry} = -F_1 \cos(\theta) - F_2 \qquad F_{Ry} = -50.311 \text{ lb}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 65.9 \text{ lb}$$

$$\theta_1 = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = -49.8 \text{ deg}$$

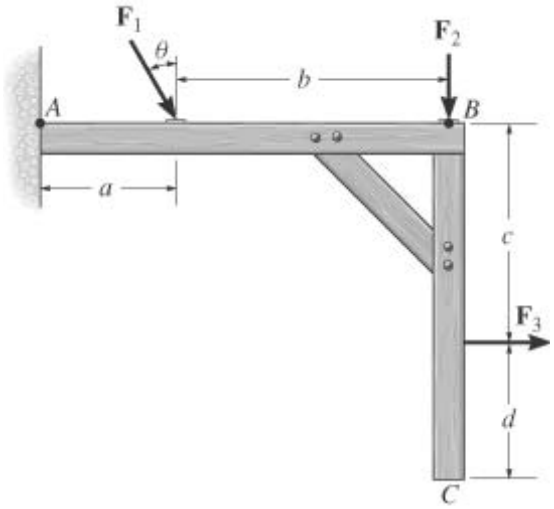
$$F_{Rx}\, y = F_1 \cos(\theta) b + F_3(c)$$

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$$y = \frac{F_1 \cos(\theta) b + F_3 (c)}{F_{Rx}} \qquad y = 4.617\,\text{ft} \quad \text{(Below point } B\text{)}$$

Problem 4-124

Replace the force system acting on the frame by an equivalent resultant force and couple moment acting at point *A*.



Given:

$F_1 = 35\ \text{lb}$ $a = 2\ \text{ft}$

$F_2 = 20\ \text{lb}$ $b = 4\ \text{ft}$

$F_3 = 25\ \text{lb}$ $c = 3\ \text{ft}$

$\theta = 30\ \text{deg}$ $d = 2\ \text{ft}$

Solution:

$$F_{Rx} = F_1 \sin(\theta) + F_3 \qquad F_{Rx} = 42.5\,\text{lb}$$

$$F_{Ry} = F_1 \cos(\theta) + F_2 \qquad F_{Ry} = 50.311\,\text{lb}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 65.9\,\text{lb}$$

$$\theta_1 = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta_1 = 49.8\,\text{deg}$$

$$M_{RA} = -F_1 \cos(\theta) a - F_2 (a + b) + F_3 (c) \qquad M_{RA} = -106\,\text{lb}\cdot\text{ft}$$

Problem 4-125

Replace the force and couple-moment system by an equivalent resultant force and couple moment at point *O*. Express the results in Cartesian vector form.

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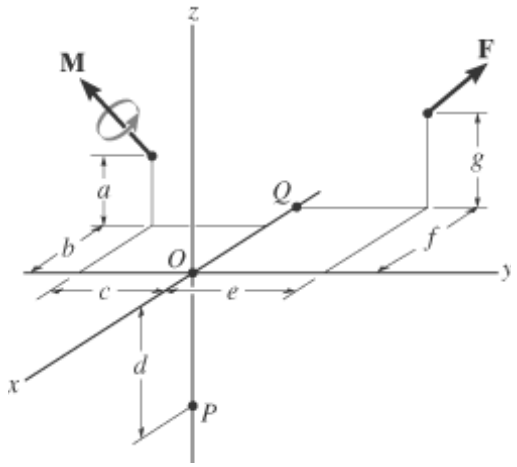
Units Used:

$$\text{kN} = 10^3 \text{ N}$$

Given:

$$\mathbf{F} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN} \qquad \mathbf{M} = \begin{pmatrix} -20 \\ -70 \\ 20 \end{pmatrix} \text{kN·m}$$

$a = 3$ m, $b = 3$ m, $c = 4$ m, $d = 6$ m, $e = 5$ m, $f = 6$ m, $g = 5$ m



Solution:

$$\mathbf{F_R} = \mathbf{F} \qquad \mathbf{M_R} = \mathbf{M} + \begin{pmatrix} -f \\ e \\ g \end{pmatrix} \times \mathbf{F} \qquad \mathbf{F_R} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN} \qquad \mathbf{M_R} = \begin{pmatrix} -10 \\ 18 \\ -56 \end{pmatrix} \text{kN·m}$$

Problem 4-126

Replace the force and couple-moment system by an equivalent resultant force and couple moment at point *P*. Express the results in Cartesian vector form.

Units Used:

$$\text{kN} = 10^3 \text{ N}$$

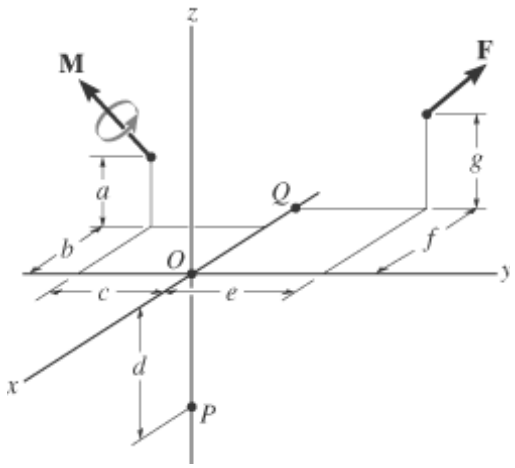
Given:

$$\mathbf{F} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN} \qquad \mathbf{M} = \begin{pmatrix} -20 \\ -70 \\ 20 \end{pmatrix} \text{kN·m}$$

$a = 3$ m

$b = 3$ m $\quad e = 5$ m

$c = 4$ m $\quad f = 6$ m



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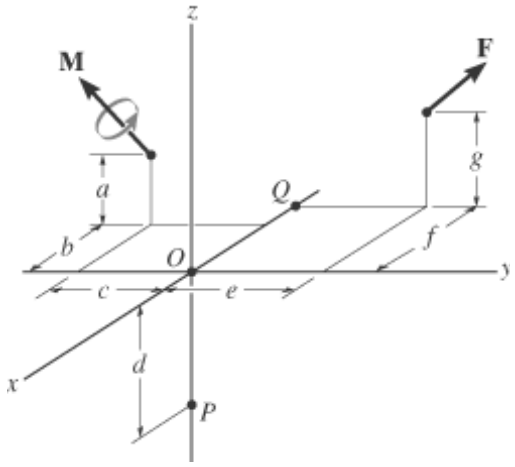
$d = 6\ \text{m}$ $g = 5\ \text{m}$

Solution:

$$\mathbf{F_R} = \mathbf{F} \qquad \mathbf{M_R} = \mathbf{M} + \begin{pmatrix} -f \\ e \\ d+g \end{pmatrix} \times \mathbf{F} \qquad \mathbf{F_R} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN} \qquad \mathbf{M_R} = \begin{pmatrix} -46 \\ 66 \\ -56 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-127

Replace the force and couple-moment system by an equivalent resultant force and couple moment at point *Q*. Express the results in Cartesian vector form.



Units Used:

$1\ \text{kN} = 10^3\ \text{N}$

Given:

$$\mathbf{F} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN}$$

$$\mathbf{M} = \begin{pmatrix} -20 \\ -70 \\ 20 \end{pmatrix} \text{kN}\cdot\text{m}$$

$a = 3\ \text{m}$

$b = 3\ \text{m}$ $e = 5\ \text{m}$

$c = 4\ \text{m}$ $f = 6\ \text{m}$

$d = 6\ \text{m}$ $g = 5\ \text{m}$

Solution:

$$\mathbf{F_R} = \mathbf{F} \qquad \mathbf{M_R} = \mathbf{M} + \begin{pmatrix} 0 \\ e \\ g \end{pmatrix} \times \mathbf{F} \qquad \mathbf{F_R} = \begin{pmatrix} 8 \\ 6 \\ 8 \end{pmatrix} \text{kN} \qquad \mathbf{M_R} = \begin{pmatrix} -10 \\ -30 \\ -20 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-128

The belt passing over the pulley is subjected to forces $\mathbf{F_1}$ and $\mathbf{F_2}$. $\mathbf{F_1}$ acts in the $-\mathbf{k}$ direction.

Replace these forces by an equivalent force and couple moment at point *A*. Express the result in

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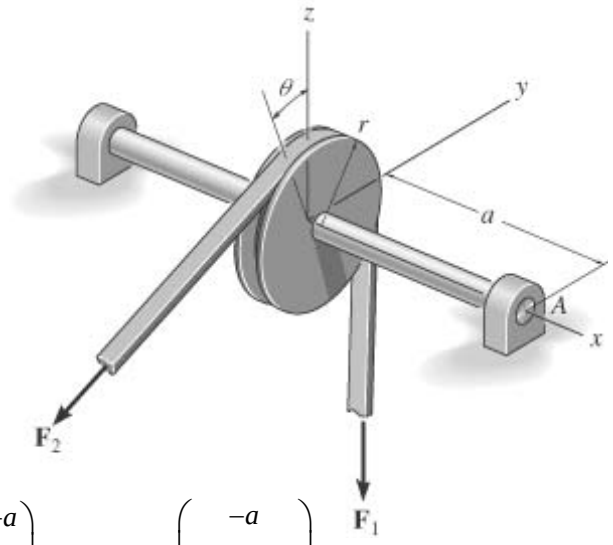
Cartesian vector form.

Given:

$F_1 = 40$ N $\quad r = 80$ mm

$F_2 = 40$ N $\quad a = 300$ mm

$\theta = 0$ deg



Solution:

$$\mathbf{F_{1v}} = F_1 \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$$

$$\mathbf{F_{2v}} = F_2 \begin{pmatrix} 0 \\ -\cos(\theta) \\ -\sin(\theta) \end{pmatrix} \qquad \mathbf{r_1} = \begin{pmatrix} -a \\ r \\ 0 \end{pmatrix} \qquad \mathbf{r_2} = \begin{pmatrix} -a \\ -r\sin(\theta) \\ r\cos(\theta) \end{pmatrix}$$

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad \mathbf{M_A} = \mathbf{r_1} \times \mathbf{F_{1v}} + \mathbf{r_2} \times \mathbf{F_{2v}}$$

$$\mathbf{F_R} = \begin{pmatrix} 0 \\ -40 \\ -40 \end{pmatrix} \text{N} \qquad \mathbf{M_A} = \begin{pmatrix} 0 \\ -12 \\ 12 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-129

The belt passing over the pulley is subjected to forces $\mathbf{F_1}$ and $\mathbf{F_2}$. $\mathbf{F_1}$ acts in the $-\mathbf{k}$ direction.

Replace these forces by an equivalent force and couple moment at point *A*. Express the result in Cartesian vector form.

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Given:

$F_1 = 40$ N

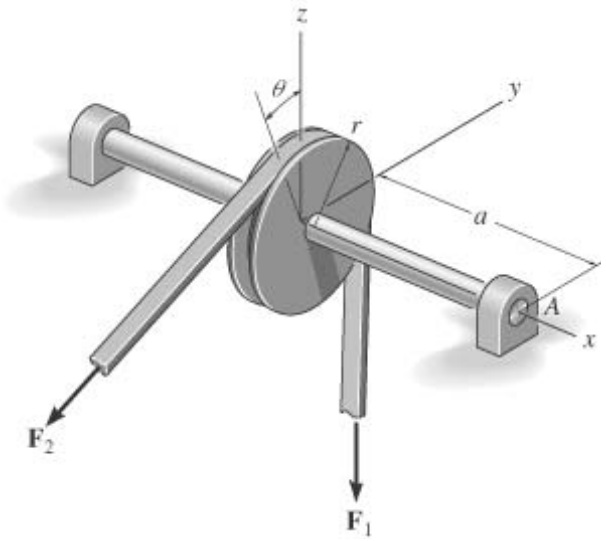
$F_2 = 40$ N

$\theta = 0$ deg

$r = 80$ mm

$a = 300$ mm

$\theta = 45$ deg



Solution:

$$\mathbf{F_{1v}} = F_1\begin{pmatrix}0\\0\\-1\end{pmatrix} \qquad \mathbf{F_{2v}} = F_2\begin{pmatrix}0\\-\cos(\theta)\\-\sin(\theta)\end{pmatrix} \qquad \mathbf{r_1} = \begin{pmatrix}-a\\r\\0\end{pmatrix} \qquad \mathbf{r_2} = \begin{pmatrix}-a\\-r\sin(\theta)\\r\cos(\theta)\end{pmatrix}$$

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad \mathbf{M_A} = \mathbf{r_1} \times \mathbf{F_{1v}} + \mathbf{r_2} \times \mathbf{F_{2v}}$$

$$\mathbf{F_R} = \begin{pmatrix}0\\-28.28\\-68.28\end{pmatrix}\ \text{N} \qquad \mathbf{M_A} = \begin{pmatrix}0\\-20.49\\8.49\end{pmatrix}\ \text{N}\cdot\text{m}$$

Problem 4-130

Replace this system by an equivalent resultant force and couple moment acting at O. Express the results in Cartesian vector form.

Given:

$F_1 = 50$ N

$F_2 = 80$ N

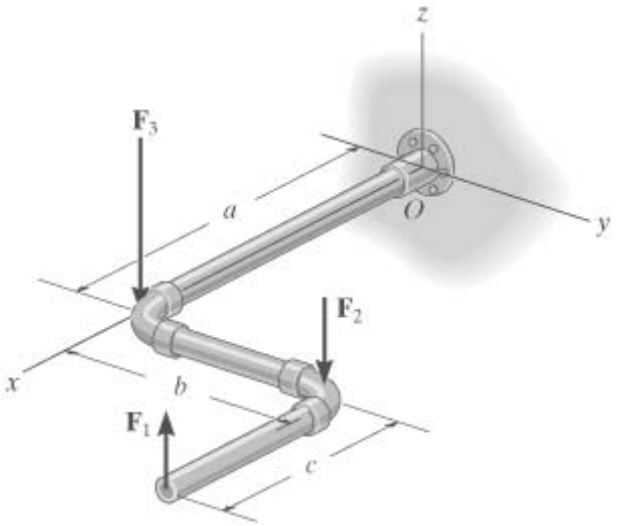
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$F_3 = 180$ N

$a = 1.25$ m

$b = 0.5$ m

$c = 0.75$ m



Solution:

$$\mathbf{F_R} = \begin{pmatrix} 0 \\ 0 \\ F_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -F_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -F_3 \end{pmatrix}$$

$$\mathbf{F_R} = \begin{pmatrix} 0 \\ 0 \\ -210 \end{pmatrix} \text{ N}$$

$$\mathbf{M_O} = \begin{pmatrix} a+c \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ F_1 \end{pmatrix} + \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F_2 \end{pmatrix} + \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F_3 \end{pmatrix} \qquad \mathbf{M_O} = \begin{pmatrix} -15 \\ 225 \\ 0 \end{pmatrix} \text{ N}\cdot\text{m}$$

Problem 4-131

Handle forces $\mathbf{F_1}$ and $\mathbf{F_2}$ are applied to the electric drill. Replace this system by an equivalent resultant force and couple moment acting at point *O*. Express the results in Cartesian vector form.

Given:

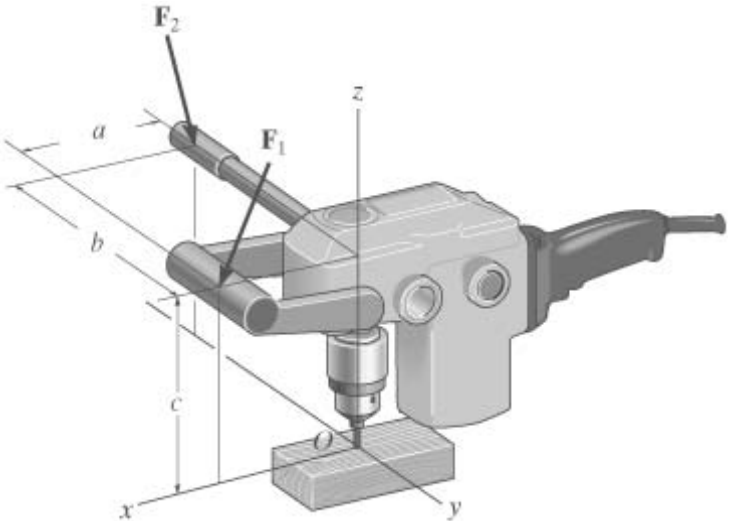
$a = 0.15$ m

$b = 0.25$ m

$c = 0.3$ m

$$\mathbf{F_1} = \begin{pmatrix} 6 \\ -3 \\ -10 \end{pmatrix} \text{ N}$$

$$\mathbf{F_2} = \begin{pmatrix} 0 \\ 2 \\ -4 \end{pmatrix} \text{ N}$$



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Solution:

$$\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2} \qquad \mathbf{F_R} = \begin{pmatrix} 6 \\ -1 \\ -14 \end{pmatrix} \text{N}$$

$$\mathbf{M_O} = \begin{pmatrix} a \\ 0 \\ c \end{pmatrix} \times \mathbf{F_1} + \begin{pmatrix} 0 \\ -b \\ c \end{pmatrix} \times \mathbf{F_2} \qquad \mathbf{M_O} = \begin{pmatrix} 1.3 \\ 3.3 \\ -0.45 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-132

A biomechanical model of the lumbar region of the human trunk is shown. The forces acting in the four muscle groups consist of $\mathbf{F_R}$ for the rectus, $\mathbf{F_O}$ for the oblique, $\mathbf{F_L}$ for the lumbar latissimus dorsi, and $\mathbf{F_E}$ for the erector spinae. These loadings are symmetric with respect to the *y - z* plane. Replace this system of parallel forces by an equivalent force and couple moment acting at the spine, point *O.* Express the results in Cartesian vector form.

Given:

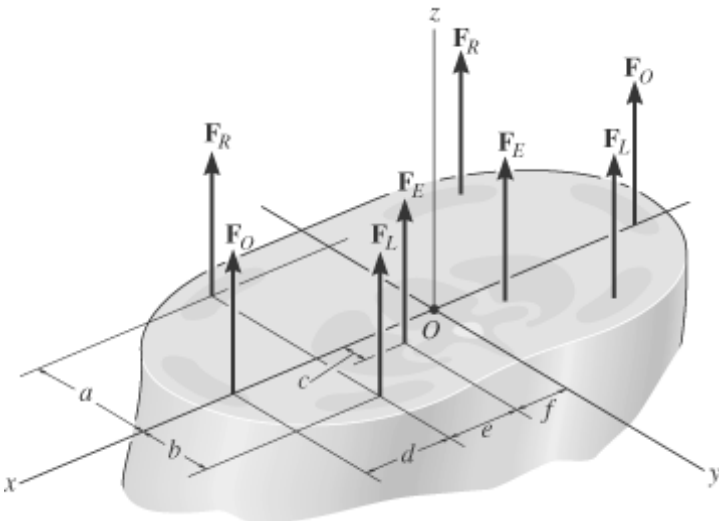
$F_R = 35$ N $\quad a = 75$ mm

$F_O = 45$ N $\quad b = 45$ mm

$F_L = 23$ N $\quad c = 15$ mm

$F_E = 32$ N $\quad d = 50$ mm

$e = 40$ mm $\quad f = 30$ mm



Solution:

$F_{Res} = \Sigma F_i;$ $\qquad F_{Res} = 2\left(F_R + F_O + F_L + F_E\right)$ $\qquad F_{Res} = 270$ N

$M_{ROx} = \Sigma M_{Ox};$ $\qquad M_{RO} = -2F_R a + 2F_E c + 2F_L b$ $\qquad M_{RO} = -2.22$ N·m

---

Problem 4-133

The building slab is subjected to four parallel column loadings.Determine the equivalent resultant force and specify its location (*x, y*) on the slab.

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Units Used:

$kN = 10^3\ N$

Given:

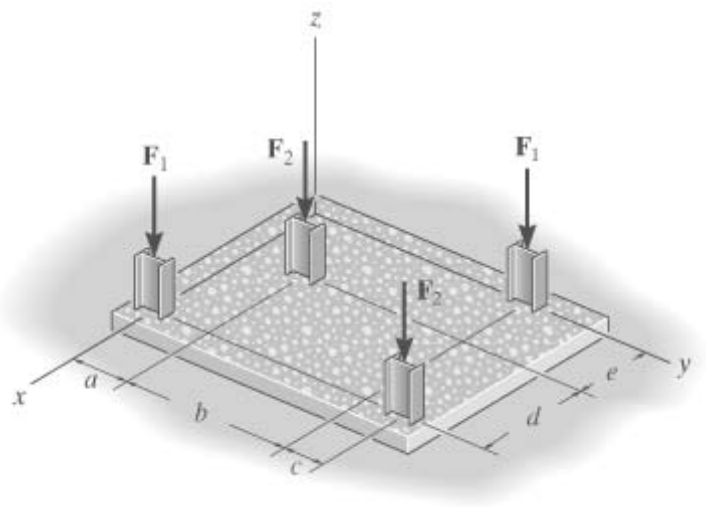
$F_1 = 30\ kN$ $a = 3\ m$

$F_2 = 40\ kN$ $b = 8\ m$

$F_3 = 20\ kN$ $c = 2\ m$

$F_4 = 50\ kN$ $d = 6\ m$

$e = 4\ m$



Solution:

$+\uparrow F_R = \Sigma F_x;\quad F_R = F_1 + F_2 + F_3 + F_4$

$F_R = 140\,kN$

$M_{Rx} = \Sigma M_x;\quad -F_R(y) = -(F_4)(a) - [(F_1)(a+b)] - [(F_2)(a+b+c)]$

$$y = \frac{F_4\ a + F_1\ a + F_1\ b + F_2\ a + F_2\ b + F_2\ c}{F_R}$$

$y = 7.14\ m$

$M_{Ry} = \Sigma M_y;\quad (F_R)x = (F_4)(e) + (F_3)(d+e) + (F_2)(b+c)$

$$x = \frac{F_4\ e + F_3\ d + F_3\ e + F_2\ b + F_2\ c}{F_R}$$

$x = 5.71\ m$

Problem 4-134

The building slab is subjected to four parallel column loadings. Determine the equivalent resultant force and specify its location (x, y) on the slab.

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Units Used:

$kN = 10^3 N$

Given:

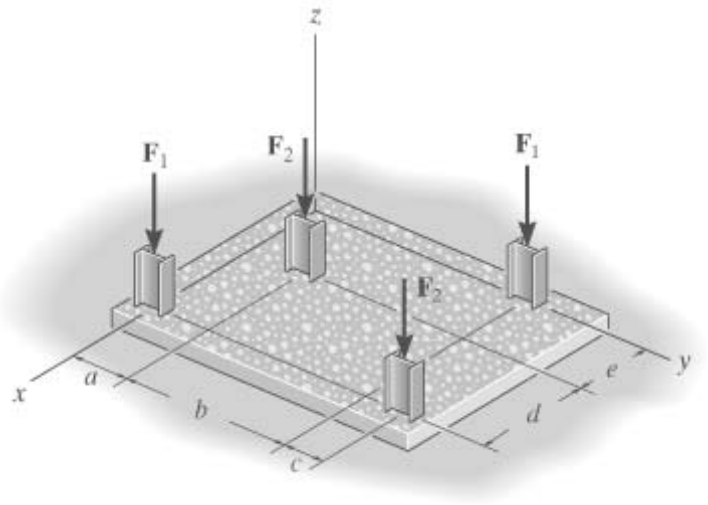
$F_1 = 20\ kN \quad a = 3\ m$

$F_2 = 50\ kN \quad b = 8\ m$

$F_3 = 20\ kN \quad c = 2\ m$

$F_4 = 50\ kN \quad d = 6\ m$

$e = 4\ m$



Solution:

$$F_R = F_1 + F_2 + F_3 + F_4 \qquad F_R = 140\,kN$$

$$F_R x = F_2 e + F_1(d + e) + F_2(d + e)$$

$$x = \frac{2\,F_2\,e + F_1\,d + F_1\,e + F_2\,d}{F_R} \qquad x = 6.43\,m$$

$$-F_R y = -F_2 a - F_3(a + b) - F_2(a + b + c)$$

$$y = \frac{2\,F_2\,a + F_3\,a + F_3\,b + F_2\,b + F_2\,c}{F_R} \qquad y = 7.29\ m$$

Problem 4-135

The pipe assembly is subjected to the action of a wrench at B and a couple at A. Determine the magnitude F of the couple forces so that the system can be simplified to a wrench acting at point C.

Given:

$a = 0.6\ m$

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$b = 0.8$ m

$c = 0.25$ m

$d = 0.7$ m

$e = 0.3$ m

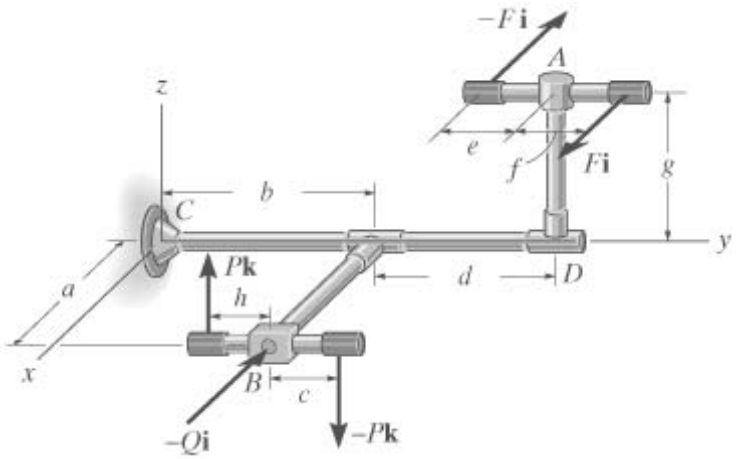
$f = 0.3$ m

$g = 0.5$ m

$h = 0.25$ m

$P = 60$ N

$Q = 40$ N



Solution:

Initial Guess $F = 1$ N $M_C = 1$ N·m

Given

$$\begin{pmatrix} -M_C \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} -P(c+h) \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -F(e+f) \end{bmatrix} + \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} -Q \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} F \\ M_C \end{pmatrix} = \text{Find}(F, M_C) \qquad M_C = 30\,\text{N·m} \qquad F = 53.3 \text{ N}$$

Problem 4-136

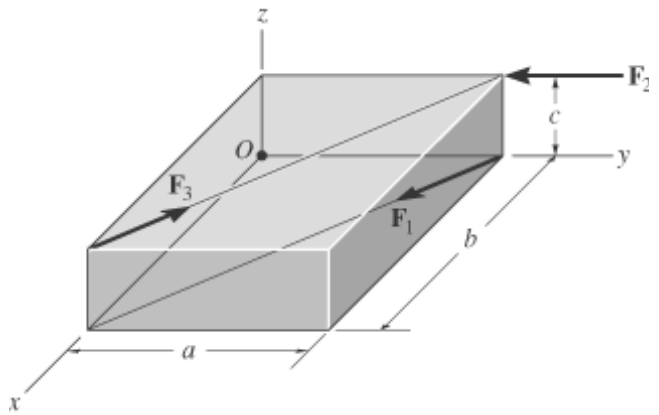
The three forces acting on the block each have a magnitude $F_1 = F_2 = F_3$. Replace this system by a wrench and specify the point where the wrench intersects the z axis, measured from point O.

Given:

$F_1 = 10$ lb $a = 6$ ft

$F_2 = F_1$ $b = 6$ ft

$F_3 = F_1$ $c = 2$ ft



Solution:

The vectors

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$$\mathbf{F_{1v}} = \frac{F_1}{\sqrt{b^2 + a^2}}\begin{pmatrix} b \\ -a \\ 0 \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ -F_2 \\ 0 \end{pmatrix} \qquad \mathbf{F_{3v}} = \frac{F_3}{\sqrt{b^2 + a^2}}\begin{pmatrix} -b \\ a \\ 0 \end{pmatrix}$$

Place the wrench in the x - z plane.

Guesses $x = 1\text{ft}$ $z = 1\text{ft}$ $M = 1\ \text{lb}\cdot\text{ft}$ $R_x = 1\ \text{lb}$ $R_y = 1\ \text{lb}$ $R_z = 1\ \text{lb}$

Given $$\begin{pmatrix} R_x \\ R_y \\ R_z \end{pmatrix} = \mathbf{F_{1v}} + \mathbf{F_{2v}} + \mathbf{F_{3v}}$$

$$\begin{pmatrix} x \\ 0 \\ z \end{pmatrix} \times \begin{pmatrix} R_x \\ R_y \\ R_z \end{pmatrix} + \frac{M}{\sqrt{R_x{}^2 + R_y{}^2 + R_z{}^2}}\begin{pmatrix} R_x \\ R_y \\ R_z \end{pmatrix} = \begin{pmatrix} 0 \\ a \\ c \end{pmatrix} \times \mathbf{F_{2v}} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \mathbf{F_{1v}} + \begin{pmatrix} b \\ 0 \\ c \end{pmatrix} \times \mathbf{F_{3v}}$$

$$\begin{pmatrix} x \\ z \\ M \\ R_x \\ R_y \\ R_z \end{pmatrix} = \text{Find}\left(x, z, M, R_x, R_y, R_z\right) \qquad \mathbf{M_v} = \frac{M}{\sqrt{R_x{}^2 + R_y{}^2 + R_z{}^2}}\begin{pmatrix} R_x \\ R_y \\ R_z \end{pmatrix}$$

$$\begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0.586 \end{pmatrix}\text{ft} \qquad \mathbf{M_v} = \begin{pmatrix} 0 \\ -14.142 \\ 0 \end{pmatrix}\text{lb}\cdot\text{ft} \qquad \begin{pmatrix} R_x \\ R_y \\ R_z \end{pmatrix} = \begin{pmatrix} 0 \\ -10 \\ 0 \end{pmatrix}\text{lb}$$

Problem 4-137

Replace the three forces acting on the plate by a wrench. Specify the magnitude of the force and couple moment for the wrench and the point $P(x, y)$ where its line of action intersects the plate.

Units Used:

$\text{kN} = 10^3\text{N}$

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Given:

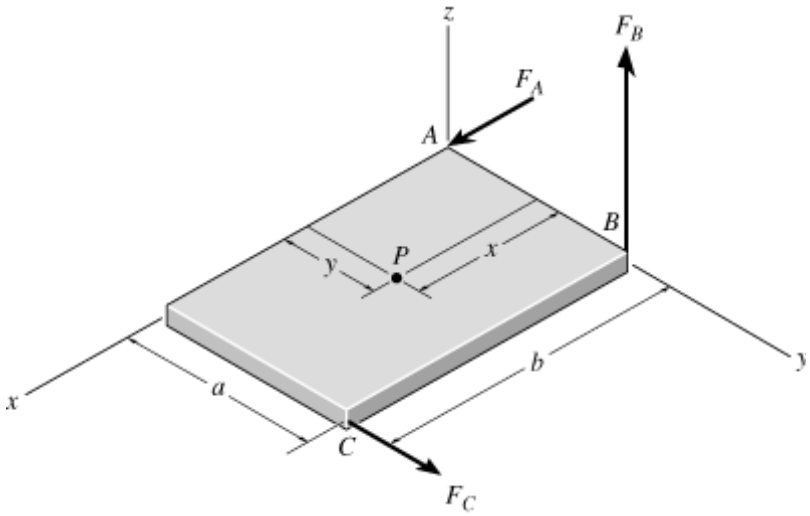
$F_A = 500$ N

$F_B = 800$ N

$F_C = 300$ N

$a = 4$ m

$b = 6$ m



Solution:

$$\mathbf{F_R} = \begin{pmatrix} F_A \\ F_C \\ F_B \end{pmatrix} \qquad |\mathbf{F_R}| = 0.9899\,\text{kN}$$

Guesses $x = 1$ m $y = 1$ m $M = 100$ N·m

Given $$M\frac{\mathbf{F_R}}{|\mathbf{F_R}|} + \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \times \mathbf{F_R} = \begin{pmatrix} b \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ F_C \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ F_B \end{pmatrix}$$

$$\begin{pmatrix} M \\ x \\ y \end{pmatrix} = \text{Find}(M, x, y) \qquad M = 3.07\,\text{kN·m} \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1.163 \\ 2.061 \end{pmatrix} \text{m}$$

Problem 4-138

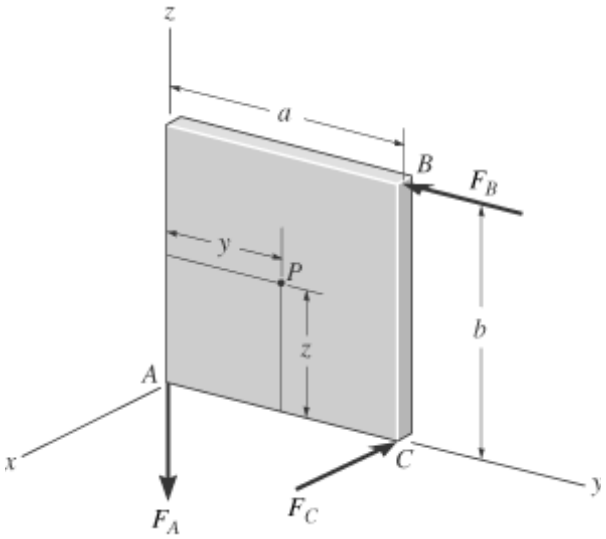
Replace the three forces acting on the plate by a wrench. Specify the magnitude of the force and couple moment for the wrench and the point $P(y, z)$ where its line of action intersects the plate.

Given:

$F_A = 80$ lb $a = 12$ ft

$F_B = 60$ lb $b = 12$ ft

$F_C = 40$ lb



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Solution:

$$\mathbf{F_R} = \begin{pmatrix} -F_C \\ -F_B \\ -F_A \end{pmatrix} \qquad |\mathbf{F_R}| = 108\,\text{lb}$$

Guesses $y = 1\ \text{ft}$ $z = 1\ \text{ft}$ $M = 1\ \text{lb}\cdot\text{ft}$

Given $$M\frac{\mathbf{F_R}}{|\mathbf{F_R}|} + \begin{pmatrix} 0 \\ y \\ z \end{pmatrix} \times \mathbf{F_R} = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} -F_C \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ b \end{pmatrix} \times \begin{pmatrix} 0 \\ -F_B \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} M \\ y \\ z \end{pmatrix} = \text{Find}(M, y, z) \qquad M = -624\,\text{lb}\cdot\text{ft} \qquad \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} 0.414 \\ 8.69 \end{pmatrix}\text{ft}$$

Problem 4-139

The loading on the bookshelf is distributed as shown. Determine the magnitude of the equivalent resultant location, measured from point *O*.

Given:

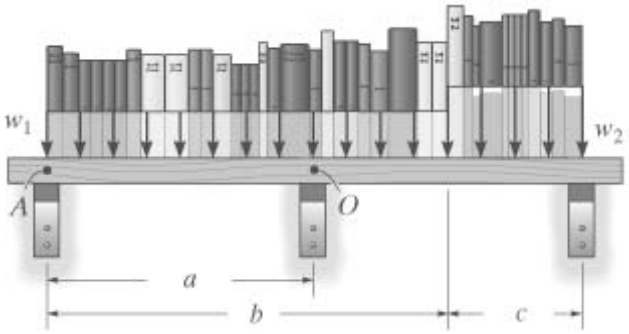
$w_1 = 2\,\frac{\text{lb}}{\text{ft}}$

$w_2 = 3.5\,\frac{\text{lb}}{\text{ft}}$

$a = 2.75\ \text{ft}$

$b = 4\ \text{ft}$

$c = 1.5\ \text{ft}$



Solution: Guesses $R = 1\ \text{lb}$ $d = 1\text{ft}$

Given $w_1 b + w_2 c = R$

$$w_1 b\left(a - \frac{b}{2}\right) - w_2 c\left(\frac{c}{2} + b - a\right) = -d\,R$$

$$\begin{pmatrix} R \\ d \end{pmatrix} = \text{Find}(R, d) \qquad R = 13.25\,\text{lb} \qquad d = 0.34\,\text{ft}$$

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Problem 4-140

Replace the loading by an equivalent resultant force and couple moment acting at point A.

Units Used:

$\text{kN} = 10^3\ \text{N}$

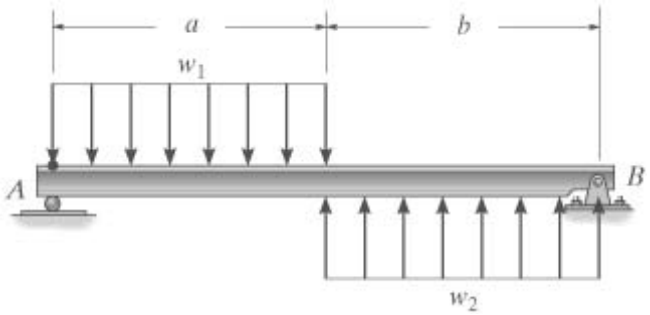
Given:

$w_1 = 600\ \frac{\text{N}}{\text{m}}$

$w_2 = 600\ \frac{\text{N}}{\text{m}}$

$a = 2.5\ \text{m}$

$b = 2.5\ \text{m}$



Solution:

$F_R = w_1 a - w_2 b$ $\quad F_R = 0\ \text{N}$

$M_{RA} = w_1 a\left(\frac{a+b}{2}\right)$ $\quad M_{RA} = 3.75\,\text{kN}\cdot\text{m}$

Problem 4-141

Replace the loading by an equivalent force and couple moment acting at point O.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

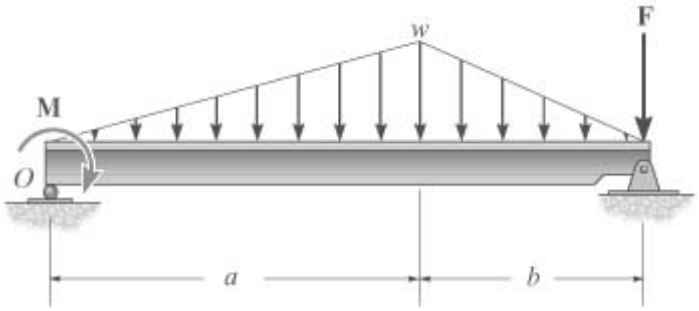
$w = 6\ \frac{\text{kN}}{\text{m}}$

$F = 15\ \text{kN}$

$M = 500\ \text{kN}\cdot\text{m}$

$a = 7.5\ \text{m}$

$b = 4.5\ \text{m}$



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Solution:

$$F_R = \frac{1}{2}w(a+b) + F \qquad F_R = 51.0\,\text{kN}$$

$$M_R = -M - \left(\frac{1}{2}wa\right)\left(\frac{2}{3}a\right) - \left(\frac{1}{2}wb\right)\left(a + \frac{b}{3}\right) - F(a+b) \qquad M_R = -914\,\text{kN}\cdot\text{m}$$

Problem 4-142

Replace the loading by a single resultant force, and specify the location of the force on the beam measured from point *O*.

Units Used:

$$\text{kN} = 10^3\ \text{N}$$

Given:

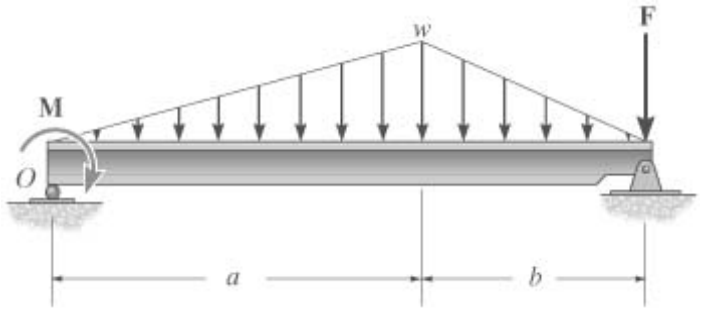
$$w = 6\ \frac{\text{kN}}{\text{m}}$$

$$F = 15\ \text{kN}$$

$$M = 500\ \text{kN}\cdot\text{m}$$

$$a = 7.5\ \text{m}$$

$$b = 4.5\ \text{m}$$



Solution:

Initial Guesses: $F_R = 1\ \text{kN}$ $d = 1\ \text{m}$

Given

$$F_R = \frac{1}{2}w(a+b) + F$$

$$-F_R d = -M - \left(\frac{1}{2}\ wa\right)\left(\frac{2}{3}a\right) - \left(\frac{1}{2}w\ b\right)\left(a + \frac{b}{3}\right) - F(a+b)$$

$$\begin{pmatrix} F_R \\ d \end{pmatrix} = \text{Find}(F_R, d) \qquad F_R = 51\,\text{kN} \qquad d = 17.922\ \text{m}$$

Problem 4-143

The column is used to support the floor which exerts a force P on the top of the column. The effect of soil pressure along its side is distributed as shown. Replace this loading by an

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equivalent resultant force and specify where it acts along the column, measured from its base *A*.

Units Used: $\text{kip} = 10^3\ \text{lb}$

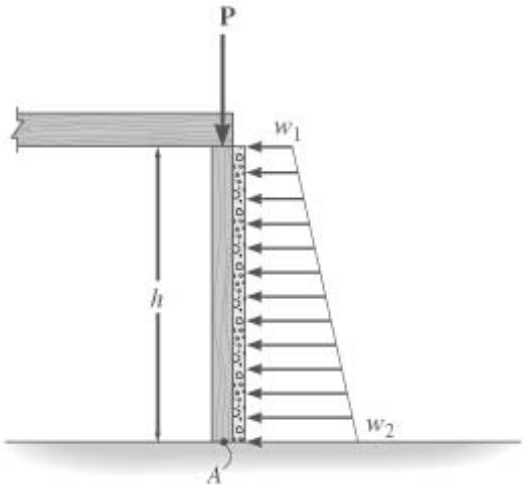
Given:

$P = 3000\ \text{lb}$

$w_1 = 80\ \frac{\text{lb}}{\text{ft}}$

$w_2 = 200\ \frac{\text{lb}}{\text{ft}}$

$h = 9\ \text{ft}$



Solution:

$$F_{Rx} = w_1 h + \frac{1}{2}(w_2 - w_1)h$$

$F_{Rx} = 1260\,\text{lb}$ $F_{Ry} = P$

$F_R = \sqrt{F_{Rx}^2 + P^2}$ $F_R = 3.25\,\text{kip}$

$\theta = \text{atan}\left(\frac{P}{F_{Rx}}\right)$ $\theta = 67.2\,\text{deg}$

$$F_{Rx}\, y = \frac{1}{2}(w_2 - w_1)h\frac{h}{3} + w_1 h\frac{h}{2}$$

$y = \frac{1}{6} h^2 \frac{w_2 + 2\, w_1}{F_{Rx}}$ $y = 3.86\,\text{ft}$

Problem 4-144

Replace the loading by an equivalent force and couple moment at point *O*.

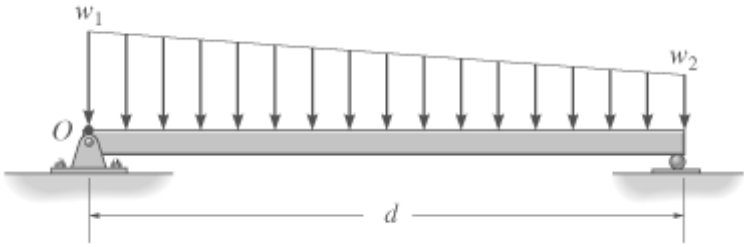
Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$w_1 = 15\ \frac{\text{kN}}{\text{m}}$

$w_2 = 5\ \frac{\text{kN}}{\text{m}}$



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$d = 9\ \text{m}$

Solution:

$$F_R = \frac{1}{2}(w_1 + w_2)d \qquad F_R = 90\,\text{kN}$$

$$M_{RO} = w_2\, d\frac{d}{2} + \frac{1}{2}(w_1 - w_2)d\frac{d}{3} \qquad M_{RO} = 338\,\text{kN}\cdot\text{m}$$

Problem 4-145

Replace the distributed loading by an equivalent resultant force, and specify its location on the beam, measured from the pin at *C*.

Units Used:

$\text{kip} = 10^3\ \text{lb}$

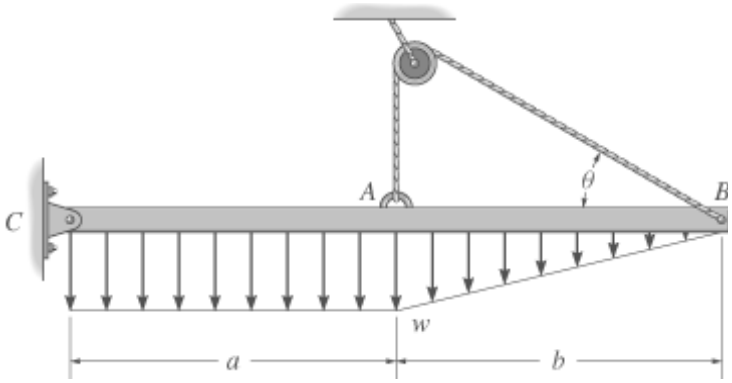
Given:

$w = 800\ \frac{\text{lb}}{\text{ft}}$

$a = 15\ \text{ft}$

$b = 15\ \text{ft}$

$\theta = 30\ \text{deg}$



Solution:

$$F_R = wa + \frac{wb}{2} \qquad F_R = 18\,\text{kip}$$

$$F_R x = wa\frac{a}{2} + \frac{wb}{2}\left(a + \frac{b}{3}\right)$$

$$x = \frac{wa\frac{a}{2} + \frac{wb}{2}\left(a + \frac{b}{3}\right)}{F_R} \qquad x = 11.7\,\text{ft}$$

Problem 4-146

The beam supports the distributed load caused by the sandbags. Determine the resultant force on the beam and specify its location measured from point *A*.

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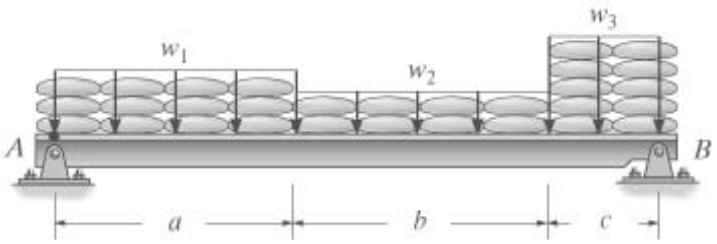
Units Used: $kN = 10^3\ N$

Given:

$w_1 = 1.5\ \frac{kN}{m}$ $a = 3\ m$

$w_2 = 1\ \frac{kN}{m}$ $b = 3\ m$

$w_3 = 2.5\ \frac{kN}{m}$ $c = 1.5\ m$



Solution:

$F_R = w_1 a + w_2 b + w_3 c$ $F_R = 11.25\ kN$

$M_A = w_1 a \frac{a}{2} + w_2 b\left(a + \frac{b}{2}\right) + w_3 c\left(a + b + \frac{c}{2}\right)$

$M_A = 45.563\ kN\cdot m$ $d = \frac{M_A}{F_R}$ $d = 4.05\ m$

Problem 4-147

Determine the length b of the triangular load and its position a on the beam such that the equivalent resultant force is zero and the resultant couple moment is M clockwise.

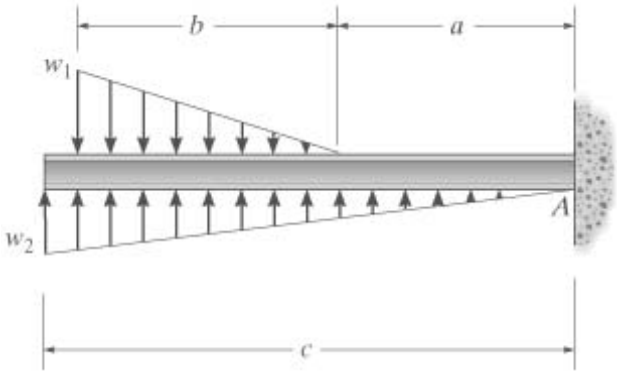
Units Used:

$kN = 10^3\ N$

Given:

$w_1 = 4\ \frac{kN}{m}$ $w_2 = 2.5\ \frac{kN}{m}$

$M = 8\ kN\cdot m$ $c = 9\ m$



Solution:

Initial Guesses: $a = 1\ m$ $b = 1\ m$

Given $\frac{-1}{2} w_1 b + \frac{1}{2} w_2 c = 0$

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$$\frac{1}{2}w_1 b\left(a+\frac{2b}{3}\right)-\frac{1}{2}w_2 c\frac{2c}{3}=-M$$

$$\begin{pmatrix}a\\b\end{pmatrix}=\text{Find}(a,b) \qquad \begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}1.539\\5.625\end{pmatrix}\text{m}$$

Problem 4-148

Replace the distributed loading by an equivalent resultant force and specify its location, measured from point *A*.

Units Used:

$$\text{kN}=10^3\text{N}$$

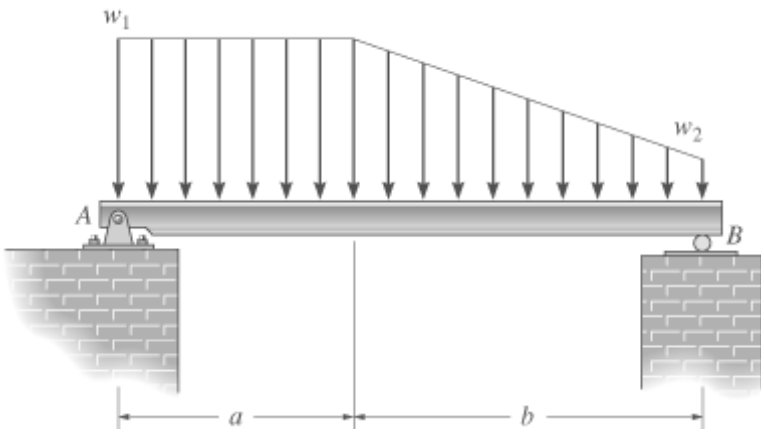
Given:

$$w_1=800\,\frac{\text{N}}{\text{m}}$$

$$w_2=200\,\frac{\text{N}}{\text{m}}$$

$$a=2\text{ m}$$

$$b=3\text{ m}$$



Solution:

$$F_R=w_2 b+w_1 a+\frac{1}{2}(w_1-w_2)b \qquad F_R=3.10\,\text{kN}$$

$$xF_R=w_1 a\frac{a}{2}+\frac{1}{2}(w_1-w_2)b\left(a+\frac{b}{3}\right)+w_2 b\left(a+\frac{b}{2}\right)$$

$$x=\frac{w_1 a\frac{a}{2}+\frac{1}{2}(w_1-w_2)b\left(a+\frac{b}{3}\right)+w_2 b\left(a+\frac{b}{2}\right)}{F_R} \qquad x=2.06\text{ m}$$

Problem 4-149

The distribution of soil loading on the bottom of a building slab is shown. Replace this loading by an equivalent resultant force and specify its location, measured from point *O*.

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$\text{kip} = 10^3 \text{lb}$

Given:

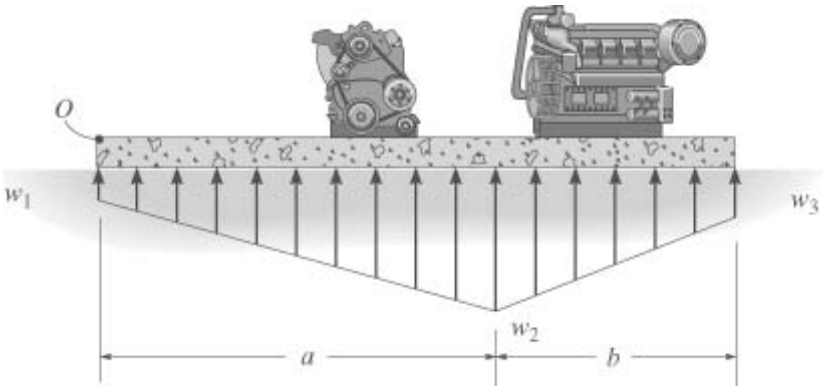
$w_1 = 50 \frac{\text{lb}}{\text{ft}}$

$w_2 = 300 \frac{\text{lb}}{\text{ft}}$

$w_3 = 100 \frac{\text{lb}}{\text{ft}}$

$a = 12 \text{ ft}$

$b = 9 \text{ ft}$



Solution:

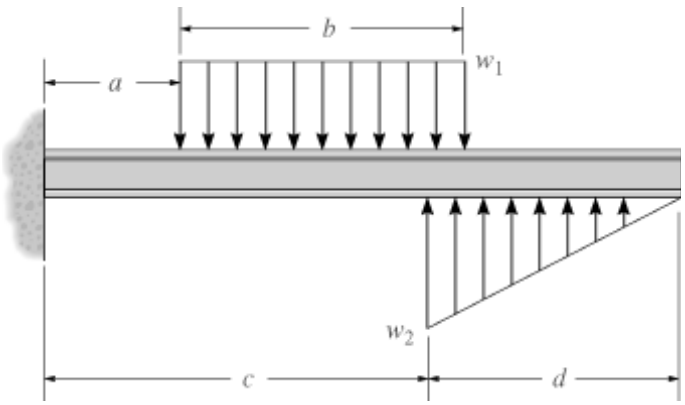
$$F_R = w_1 a + \frac{1}{2}(w_2 - w_1)a + \frac{1}{2}(w_2 - w_3)b + w_3 b \qquad F_R = 3.9 \text{ kip}$$

$$F_R \, d = w_1 a \frac{a}{2} + \frac{1}{2}(w_2 - w_1)a \frac{2a}{3} + \frac{1}{2}(w_2 - w_3)b\left(a + \frac{b}{3}\right) + w_3 b\left(a + \frac{b}{2}\right)$$

$$d = \frac{3\, w_3\, b\, a + 2\, w_3\, b^2 + w_1\, a^2 + 2\, a^2\, w_2 + 3\, b\, w_2\, a + w_2\, b^2}{6 F_R} \qquad d = 11.3 \text{ ft}$$

Problem 4-150

The beam is subjected to the distributed loading. Determine the length b of the uniform load and its position a on the beam such that the resultant force and couple moment acting on the beam are zero.



Given:

$w_1 = 40 \frac{\text{lb}}{\text{ft}}$ $\quad c = 10 \text{ft}$

$w_2 = 60 \frac{\text{lb}}{\text{ft}}$ $\quad d = 6 \text{ ft}$

Solution:

Initial Guesses: $\quad a = 1 \text{ ft} \quad b = 1 \text{ft}$

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Given

$$\frac{1}{2}w_2\,d - w_1\,b = 0 \qquad \frac{1}{2}w_2\,d\left(c + \frac{d}{3}\right) - w_1\,b\left(a + \frac{b}{2}\right) = 0$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \text{Find}(a, b) \qquad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 9.75 \\ 4.5 \end{pmatrix} \text{ft}$$

Problem 4-151

Replace the loading by an equivalent resultant force and specify its location on the beam, measured from point *B*.

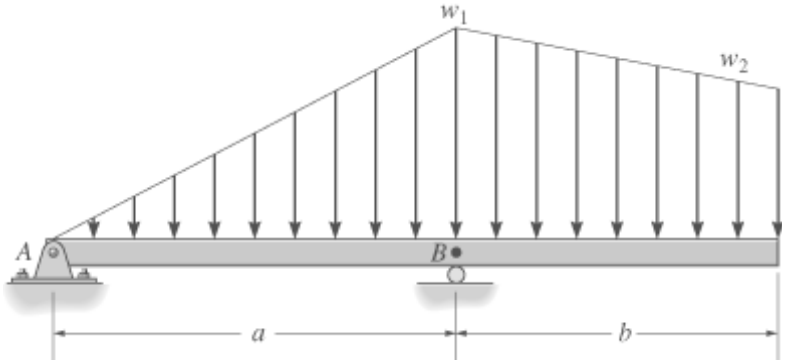
Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

$$w_1 = 800\ \frac{\text{lb}}{\text{ft}}$$

$$w_2 = 500\ \frac{\text{lb}}{\text{ft}}$$



$$a = 12 \text{ ft}$$

$$b = 9 \text{ ft}$$

Solution:

$$F_R = \frac{1}{2}a\,w_1 + \frac{1}{2}(w_1 - w_2)b + w_2\,b \qquad F_R = 10.65 \text{ kip}$$

$$F_R x = -\frac{1}{2}\,a\,w_1\,\frac{a}{3} + \frac{1}{2}(w_1 - w_2)b\,\frac{b}{3} + w_2\,b\,\frac{b}{2}$$

$$x = \frac{-\frac{1}{2}\,a\,w_1\,\frac{a}{3} + \frac{1}{2}(w_1 - w_2)b\,\frac{b}{3} + w_2\,b\,\frac{b}{2}}{F_R} \qquad x = 0.479 \text{ ft}$$

(to the right of B)

Problem 4-152

Replace the distributed loading by an equivalent resultant force and specify where its line of action intersects member *AB*, measured from *A*.

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Given:

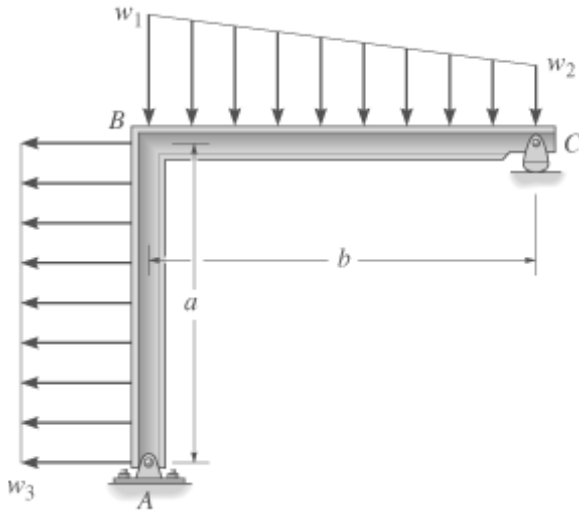
$w_1 = 200 \frac{\text{N}}{\text{m}}$

$w_2 = 100 \frac{\text{N}}{\text{m}}$

$w_3 = 200 \frac{\text{N}}{\text{m}}$

$a = 5\ \text{m}$

$b = 6\ \text{m}$



Solution:

$F_{Rx} = -w_3\, a$ $\quad F_{Rx} = -1000\,\text{N}$

$F_{Ry} = \frac{-1}{2}\left(w_1 + w_2\right) b$ $\quad F_{Ry} = -900\ \text{N}$

$-y F_{Rx} = w_3\, a \frac{a}{2} - w_2\, b \frac{b}{2} - \frac{1}{2}\left(w_1 - w_2\right) b \frac{b}{3}$

$y = \frac{w_3\, a \frac{a}{2} - w_2\, b \frac{b}{2} - \frac{1}{2}\left(w_1 - w_2\right) b \frac{b}{3}}{-F_{Rx}}$ $\quad y = 0.1\ \text{m}$

Problem 4-153

Replace the distributed loading by an equivalent resultant force and specify where its line of action intersects member *BC*, measured from *C*.

Units Used:

$\text{kN} = 10^3 \text{N}$

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Given:

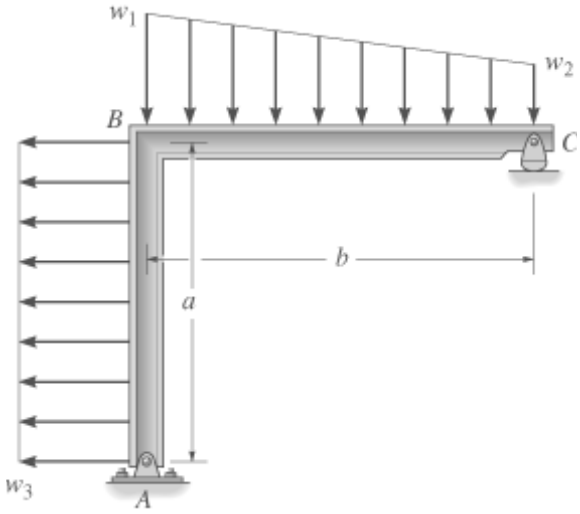
$w_1 = 200 \frac{\text{N}}{\text{m}}$

$w_2 = 100 \frac{\text{N}}{\text{m}}$

$w_3 = 200 \frac{\text{N}}{\text{m}}$

$a = 5\ \text{m}$

$b = 6\ \text{m}$



Solution:

$F_{Rx} = -w_3\, a$ $\qquad F_{Rx} = -1000\ \text{N}$

$F_{Ry} = \frac{-1}{2}\left(w_1 + w_2\right) b$ $\qquad F_{Ry} = -900\ \text{N}$

$-x F_{Ry} = -w_3\, a \frac{a}{2} + w_2\, b \frac{b}{2} + \frac{1}{2}\left(w_1 - w_2\right) b \frac{2b}{3}$

$$x = \frac{-w_3\, a \frac{a}{2} + w_2\, b \frac{b}{2} + \frac{1}{2}\left(w_1 - w_2\right) b \frac{2b}{3}}{-F_{Ry}} \qquad x = 0.556\ \text{m}$$

$$\left| \begin{pmatrix} F_{Rx} \\ F_{Ry} \end{pmatrix} \right| = 1.345\,\text{kN}$$

Problem 4-154

Replace the loading by an equivalent resultant force and couple moment acting at point O.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$w_1 = 7.5 \frac{\text{kN}}{\text{m}}$

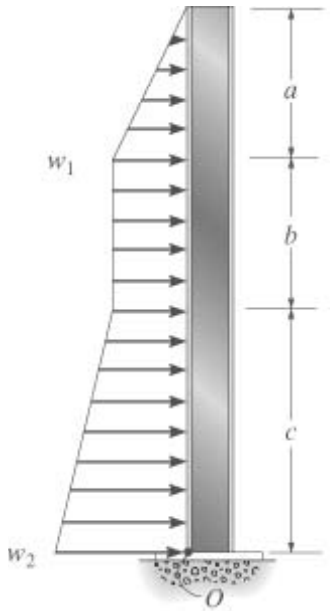
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$w_2 = 20 \frac{\text{kN}}{\text{m}}$

$a = 3 \text{ m}$

$b = 3 \text{ m}$

$c = 4.5 \text{ m}$



Solution:

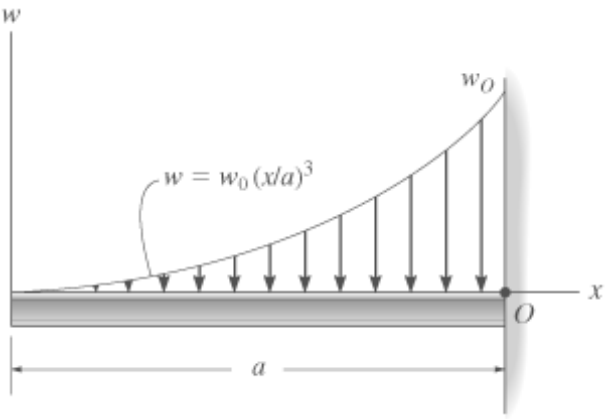
$$F_R = \frac{1}{2}(w_2 - w_1)c + w_1 c + w_1 b + \frac{1}{2} w_1 a$$

$F_R = 95.6 \text{ kN}$

$$M_{Ro} = -\frac{1}{2}(w_2 - w_1)c\frac{c}{3} - w_1 c\frac{c}{2} - w_1 b\left(c + \frac{b}{2}\right) - \frac{1}{2} w_1 a\left(b + c + \frac{a}{3}\right) \qquad M_{Ro} = -349 \text{ kN}\cdot\text{m}$$

Problem 4-155

Determine the equivalent resultant force and couple moment at point *O*.



Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$a = 3 \text{ m}$

$w_O = 3 \frac{\text{kN}}{\text{m}}$

$w(x) = w_O\left(\frac{x}{a}\right)^2$

Solution:

$$F_R = \int_0^a w(x)\, dx \qquad F_R = 3 \text{ kN}$$

$$M_O = \int_0^a w(x)(a - x)\, dx \qquad M_O = 2.25 \text{ kN}\cdot\text{m}$$

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Problem 4-156

Wind has blown sand over a platform such that the intensity of the load can be approximated by the function $w = w_0\left(\frac{x}{d}\right)^3$. Simplify this distributed loading to an equivalent resultant force and specify the magnitude and location of the force, measured from A.

Units Used:

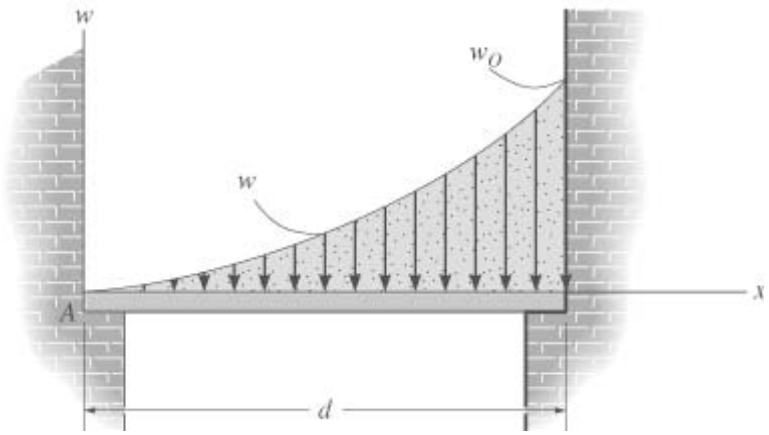
$\text{kN} = 10^3\text{N}$

Given:

$w_0 = 500\,\frac{\text{N}}{\text{m}}$

$d = 10\ \text{m}$

$w(x) = w_0\left(\frac{x}{d}\right)^3$



Solution:

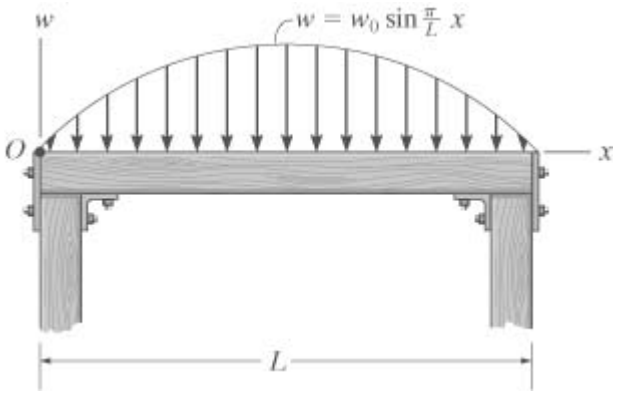
$F_R = \int_0^d w(x)\ \mathrm{d}x$ $\quad F_R = 1.25\,\text{kN}$

$d = \dfrac{\int_0^d x\,w(x)\ \mathrm{d}x}{F_R}$ $\quad d = 8\ \text{m}$

Problem 4-157

Determine the equivalent resultant force and its location, measured from point *O*.

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Solution:

$$F_R = \int_0^L w_0 \sin\left(\frac{\pi x}{L}\right) dx = \frac{2 w_0 L}{\pi}$$

$$d = \frac{\int_0^L x\, w_0 \sin\left(\frac{\pi x}{L}\right) dx}{F_R} = \frac{L}{2}$$

Problem 4-158

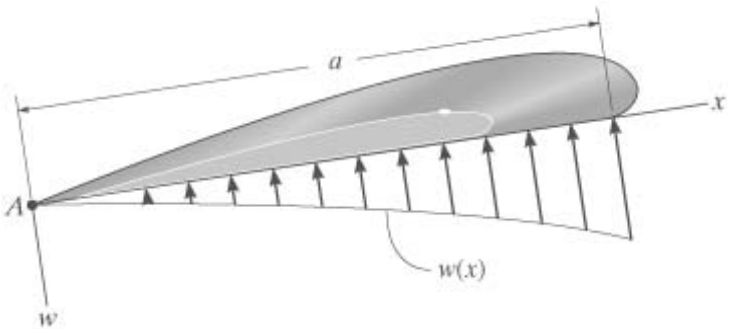
Determine the equivalent resultant force acting on the bottom of the wing due to air pressure and specify where it acts, measured from point *A*.

Given:

$a = 3$ ft

$k = 86 \frac{\text{lb}}{\text{ft}^3}$

$w(x) = k x^2$



Solution:

$$F_R = \int_0^a w(x)\, dx \qquad F_R = 774 \text{ lb}$$

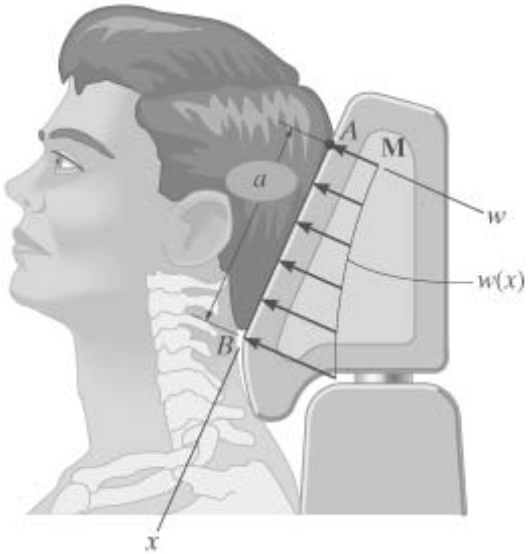
$$x = \frac{\int_0^a x\, w(x)\, dx}{F_R} \qquad x = 2.25 \text{ ft}$$

Problem 4-159

Currently eighty-five percent of all neck injuries are caused by rear-end car collisions. To

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alleviate this problem, an automobile seat restraint has been developed that provides additional pressure contact with the cranium. During dynamic tests the distribution of load on the cranium has been plotted and shown to be parabolic. Determine the equivalent resultant force and its location, measured from point *A*.



Given:

$a = 0.5\ \text{ft}$

$w_0 = 12\ \frac{\text{lb}}{\text{ft}}$

$k = 24\ \frac{\text{lb}}{\text{ft}^3}$

$w(x) = w_0 + kx^2$

Solution:

$$F_R = \int_0^a w(x)\ \mathrm{d}x \qquad F_R = 7\,\text{lb}$$

$$x = \frac{\int_0^a x\,w(x)\ \mathrm{d}x}{F_R} \qquad x = 0.268\,\text{ft}$$

Problem 4-160

Determine the equivalent resultant force of the distributed loading and its location, measured from point A. Evaluate the integrals using Simpson's rule.

Units Used:

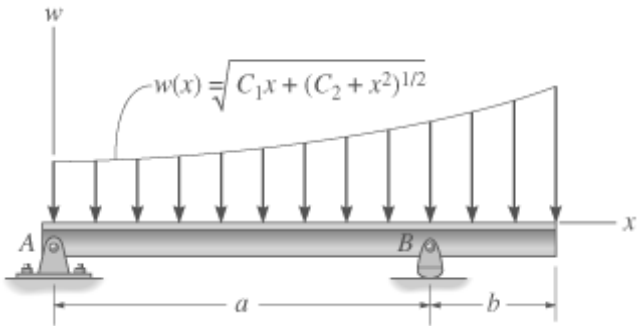
$\text{kN} = 10^3\ \text{N}$

Given:

$c_1 = 5$

$c_2 = 16$

$a = 3$



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$b = 1$

Solution:

$$F_R = \int_0^{a+b} \sqrt{c_1 x + \sqrt{c_2 + x^2}}\, dx \qquad F_R = 14.9$$

$$d = \frac{\int_0^{a+b} x\sqrt{c_1 x + \sqrt{c_2 + x^2}}\, dx}{F_R} \qquad d = 2.27$$

Problem 4-161

Determine the coordinate direction angles of F, which is applied to the end A of the pipe assembly, so that the moment of F about O is zero.

Given:

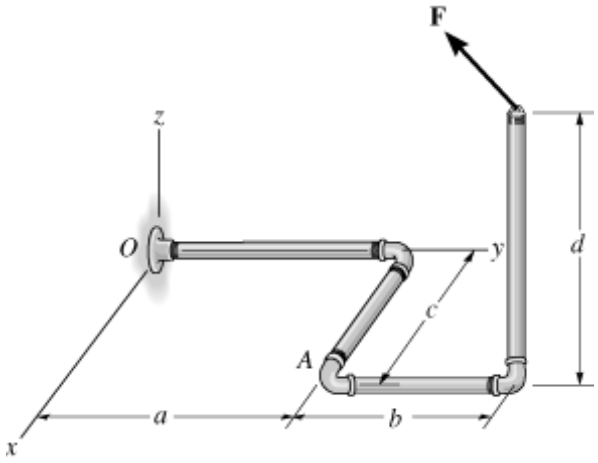
$F = 20$ lb

$a = 8$ in

$b = 6$ in

$c = 6$ in

$d = 10$ in



Solution:

Require $M_o = 0$. This happens when force F is directed either towards or away from point O.

$$\mathbf{r} = \begin{pmatrix} c \\ a+b \\ d \end{pmatrix} \qquad \mathbf{u} = \frac{\mathbf{r}}{|\mathbf{r}|} \qquad \mathbf{u} = \begin{pmatrix} 0.329 \\ 0.768 \\ 0.549 \end{pmatrix}$$

If the force points away from O, then

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}(\mathbf{u}) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 70.774 \\ 39.794 \\ 56.714 \end{pmatrix} \text{deg}$$

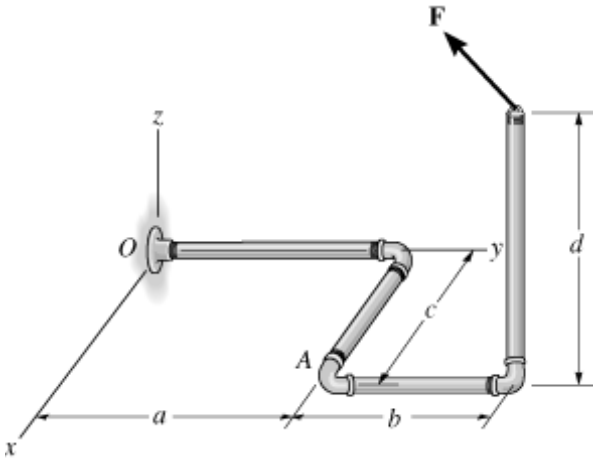
If the force points towards O, then

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$$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \operatorname{acos}(-\mathbf{u}) \qquad \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}109.226\\ 140.206\\ 123.286\end{pmatrix}\text{deg}$$

Problem 4-162

Determine the moment of the force F about point O. The force has coordinate direction angles α, β, γ. Express the result as a Cartesian vector.



Given:

$F = 20$ lb $\quad a = 8$ in

$\alpha = 60$ deg $\quad b = 6$ in

$\beta = 120$ deg $\quad c = 6$ in

$\gamma = 45$ deg $\quad d = 10$ in

Solution:

$$\mathbf{r} = \begin{pmatrix}c\\ a+b\\ d\end{pmatrix} \qquad \mathbf{F_v} = F\begin{pmatrix}\cos(\alpha)\\ \cos(\beta)\\ \cos(\gamma)\end{pmatrix} \qquad \mathbf{M} = \mathbf{r}\times\mathbf{F_v} \qquad \mathbf{M} = \begin{pmatrix}297.99\\ 15.147\\ -200\end{pmatrix}\text{lb}\cdot\text{in}$$

Problem 4-163

Replace the force at A by an equivalent resultant force and couple moment at point P. Express the results in Cartesian vector form.

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$$\text{kN} = 10^3 \text{ N}$$

Given:

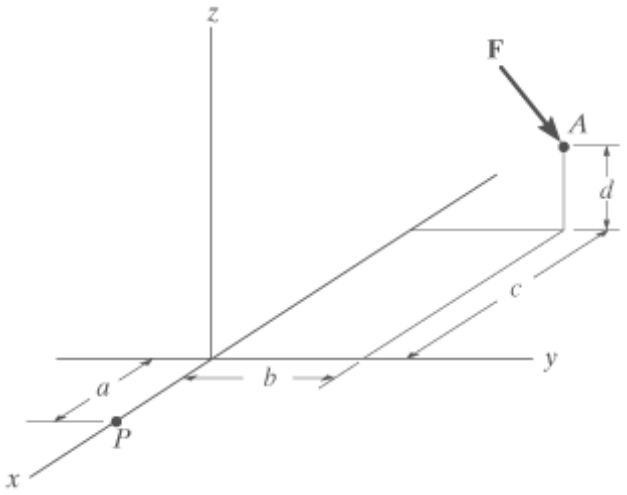
$a = 4$ m

$b = 6$ m

$c = 8$ m

$d = 4$ m

$$\mathbf{F} = \begin{pmatrix} -300 \\ 200 \\ -500 \end{pmatrix} \text{N}$$



Solution:

$$\mathbf{F_R} = \mathbf{F} \qquad \mathbf{F_R} = \begin{pmatrix} -300 \\ 200 \\ -500 \end{pmatrix} \text{N}$$

$$\mathbf{M_P} = \begin{pmatrix} -a - c \\ b \\ d \end{pmatrix} \times \mathbf{F} \qquad \mathbf{M_P} = \begin{pmatrix} -3.8 \\ -7.2 \\ -0.6 \end{pmatrix} \text{kN}\cdot\text{m}$$

Problem 4-164

Determine the moment of the force $\mathbf{F_C}$ about the door hinge at A. Express the result as a Cartesian vector.

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Given:

$F = 250$ N

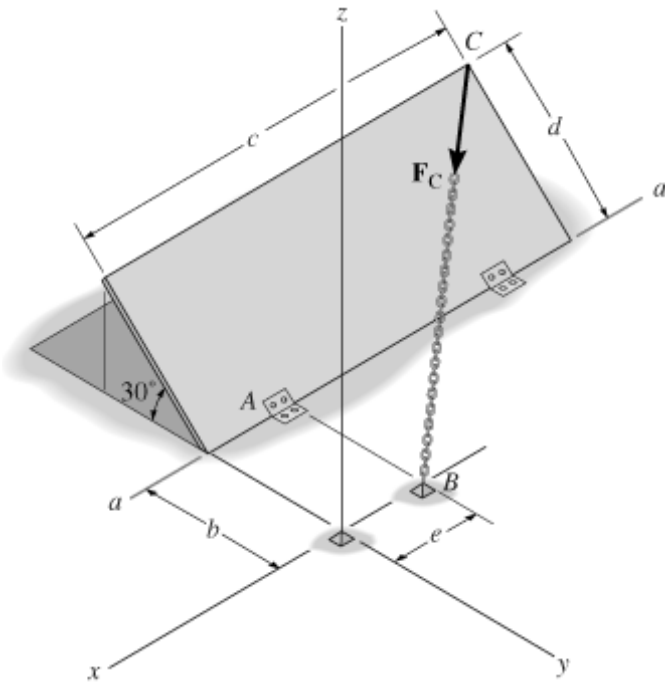
$b = 1$ m

$c = 2.5$ m

$d = 1.5$ m

$e = 0.5$ m

$\theta = 30$ deg



Solution:

$$\mathbf{r_{CB}} = \begin{pmatrix} c - e \\ b + d\cos(\theta) \\ -d\sin(\theta) \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \qquad \mathbf{F_v} = F\frac{\mathbf{r_{CB}}}{|\mathbf{r_{CB}}|}$$

$$\mathbf{M_A} = \mathbf{r_{AB}} \times \mathbf{F_v} \qquad \mathbf{M_A} = \begin{pmatrix} -59.7 \\ 0.0 \\ -159.3 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 4-165

Determine the magnitude of the moment of the force $\mathbf{F_C}$ about the hinged axis *aa* of the door.

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Given:

$F = 250$ N

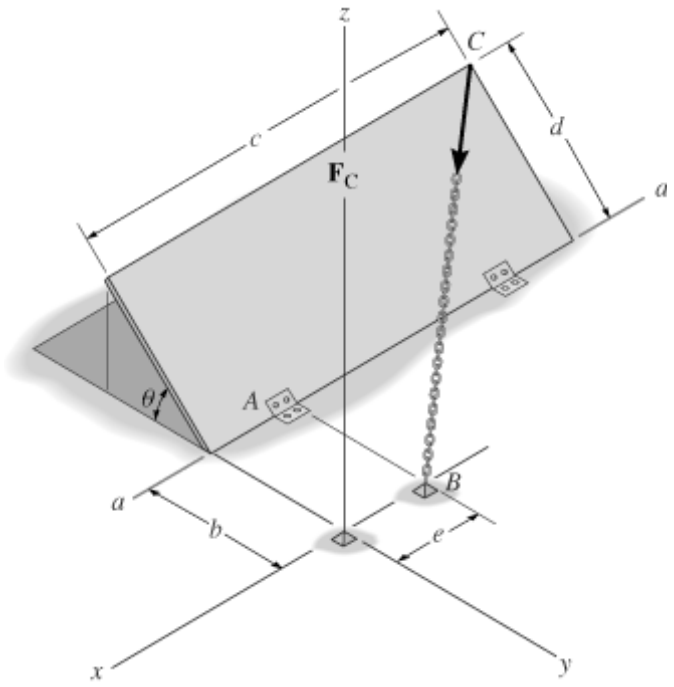
$b = 1$ m

$c = 2.5$ m

$d = 1.5$ m

$e = 0.5$ m

$\theta = 30$ deg



Solution:

$$r_{CB} = \begin{pmatrix} c - e \\ b + d\cos(\theta) \\ -d\sin(\theta) \end{pmatrix} \qquad r_{AB} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \qquad F_v = F\frac{r_{CB}}{|r_{CB}|} \qquad u_a = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$M_{aa} = (r_{AB} \times F_v) \cdot u_a \qquad M_{aa} = -59.7\ \text{N}\cdot\text{m}$$

Problem 4-166

A force F_1 acts vertically downward on the *Z*-bracket. Determine the moment of this force about the bolt axis (*z* axis), which is directed at angle θ from the vertical.

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Given:

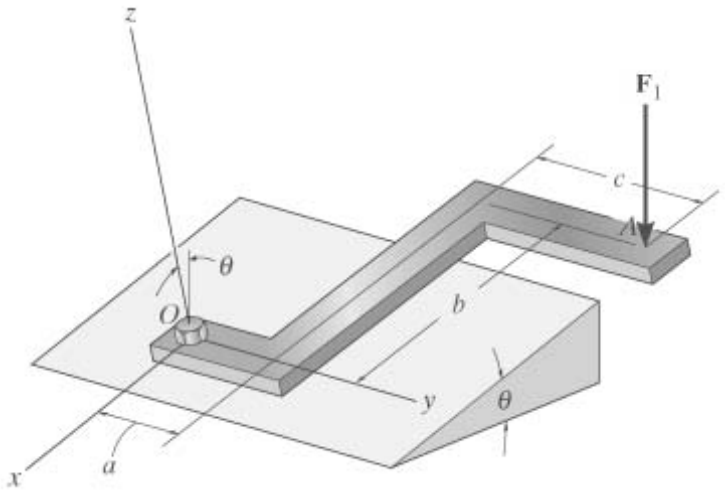
$F_1 = 80\ \text{N}$

$a = 100\ \text{mm}$

$b = 300\ \text{mm}$

$c = 200\ \text{mm}$

$\theta = 15\ \text{deg}$



Solution:

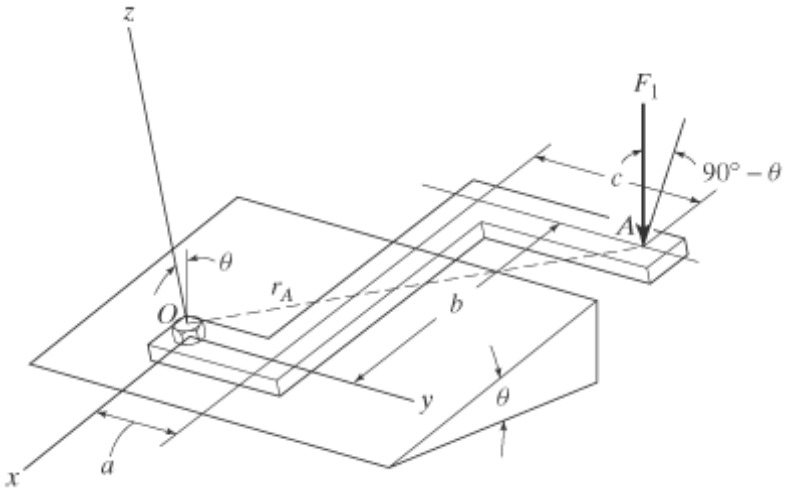
$$\mathbf{r} = \begin{pmatrix} -b \\ a + c \\ 0 \end{pmatrix}$$

$$\mathbf{F} = F_1 \begin{pmatrix} \sin(\theta) \\ 0 \\ -\cos(\theta) \end{pmatrix}$$

$$\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$M_Z = (\mathbf{r} \times \mathbf{F})\mathbf{k}$$

$$M_Z = -6.212\ \text{N}\cdot\text{m}$$



Problem 4-167

Replace the force F having acting at point A by an equivalent force and couple moment at point C.

Units Used: $\text{kip} = 10^3\ \text{lb}$

Given:

$F = 50\ \text{lb}$

$a = 10\ \text{ft}$

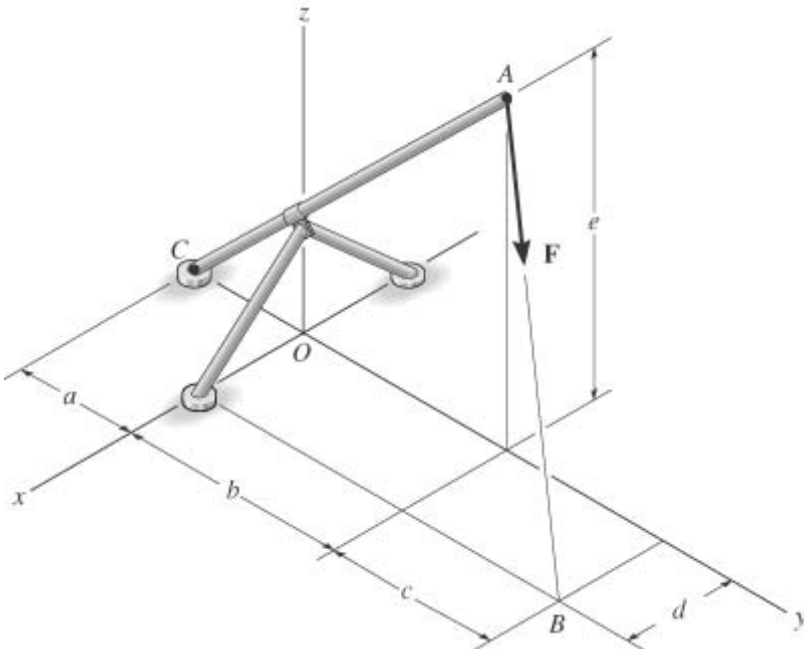
$b = 20\ \text{ft}$

$c = 15\ \text{ft}$

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$d = 10$ ft

$e = 30$ ft



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} d \\ c \\ -e \end{pmatrix}$$

$$\mathbf{F_v} = F\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|}$$

$$\mathbf{r_{CA}} = \begin{pmatrix} 0 \\ a+b \\ e \end{pmatrix}$$

$$\mathbf{F_R} = \mathbf{F_v} \qquad \mathbf{F_R} = \begin{pmatrix} 14.286 \\ 21.429 \\ -42.857 \end{pmatrix} \text{lb}$$

$$\mathbf{M_R} = \mathbf{r_{CA}} \times \mathbf{F_v} \qquad \mathbf{M_R} = \begin{pmatrix} -1.929 \\ 0.429 \\ -0.429 \end{pmatrix} \text{kip}\cdot\text{ft}$$

Problem 4-168

The horizontal force F acts on the handle of the wrench. What is the magnitude of the moment of this force about the z axis?

Given:

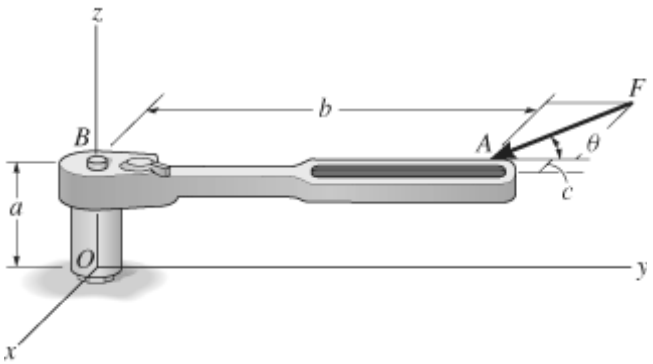
$F = 30$ N

$a = 50$ mm

$b = 200$ mm

$c = 10$ mm

$\theta = 45$ deg



Solution:

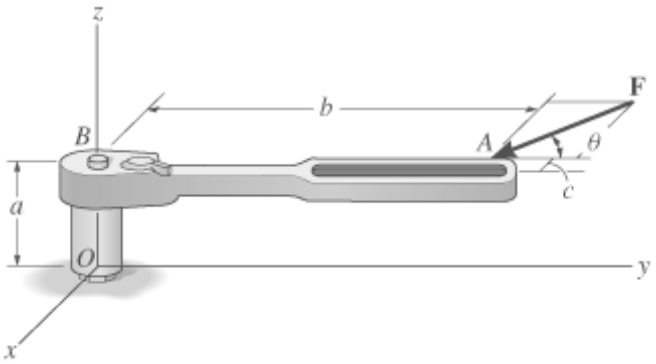
$$\mathbf{F_v} = F\begin{pmatrix} \sin(\theta) \\ -\cos(\theta) \\ 0 \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} -c \\ b \\ a \end{pmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

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$$M_Z = \left(\mathbf{r_{OA}} \times \mathbf{F_V}\right)\mathbf{k} \qquad M_Z = -4.03\,\text{N}\cdot\text{m}$$

Problem 4-169

The horizontal force **F** acts on the handle of the wrench. Determine the moment of this force about point *O*. Specify the coordinate direction angles α, β, γ of the moment axis.



Given:

$F = 30\ \text{N} \quad c = 10\ \text{mm}$

$a = 50\ \text{mm} \quad \theta = 45\ \text{deg}$

$b = 200\ \text{mm}$

Solution:

$$\mathbf{F_V} = F\begin{pmatrix}\sin(\theta)\\ -\cos(\theta)\\ 0\end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix}-c\\ b\\ a\end{pmatrix}$$

$$\mathbf{M_O} = \mathbf{r_{OA}} \times \mathbf{F_V} \qquad \mathbf{M_O} = \begin{pmatrix}1.06\\ 1.06\\ -4.03\end{pmatrix}\text{N}\cdot\text{m}$$

$$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \text{acos}\left(\frac{\mathbf{M_O}}{|\mathbf{M_O}|}\right) \qquad \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}75.7\\ 75.7\\ 159.6\end{pmatrix}\text{deg}$$

Problem 4-170

If the resultant couple moment of the three couples acting on the triangular block is to be zero, determine the magnitudes of forces **F** and **P.**

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Given:

$F_1 = 10$ lb

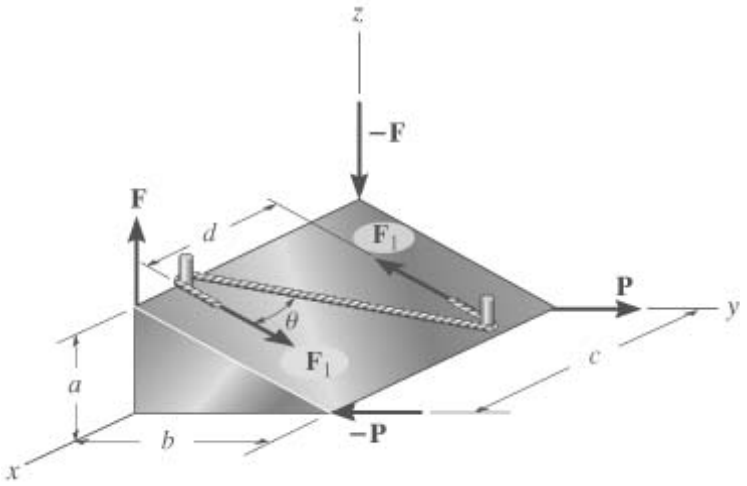
$a = 3$ in

$b = 4$ in

$c = 6$ in

$d = 3$ in

$\theta = 30$ deg



Solution:

Initial Guesses: $F = 1$ lb $P = 1$ lb

Given

$$\begin{pmatrix} 0 \\ -F\,c \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -P\,c \end{pmatrix} + \frac{F_1\,d}{\sqrt{a^2 + b^2}} \begin{pmatrix} 0 \\ a \\ b \end{pmatrix} = 0 \qquad \begin{pmatrix} F \\ P \end{pmatrix} = \text{Find}(F, P) \qquad \begin{pmatrix} F \\ P \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \text{lb}$$

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